

Partial Revelation Without Noise Traders: Complete Numerical and Analytic Documentation of the K=3 CRRA REE Project

Fixed-Point Factory

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Abstract

This document is an extensive, self-contained summary of an exhaustive computational study of rational-expectations equilibrium in a three-trader CRRA market with binary payoff, Gaussian private signals, and no noise traders. We document existence (via a Schauder fixed-point theorem on the regularized operator and a constructive Levenberg–Marquardt solver), certify 255 individual equilibria at extended-precision residual $\|\Phi(P) - P\|_\infty \leq 10^{-15}$, extrapolate revelation deficits to the continuum, decompose welfare into trading-volume and value-of-information components, extend to $K = 4$ and $K = 5$ traders to prove K -invariance of the deficit, and finally settle a methodological question of structural importance: under binary payoff the *fully-revealing* equilibrium $P_{\text{FR}}(u) = \sigma(\tau \sum_k u_k)$ is the unique fixed point of the strict $h=0$ operator at every parameter combination, certified to machine precision in nine independent test cells, so the partially-revealing equilibrium of risk-averse trading exists only as the $h \rightarrow 0$ limit of the kernel-regularized branch. The kernel bandwidth $h > 0$ is therefore an *equilibrium selection mechanism*, not a numerical convenience. We document multiple moving-grid solver attempts, basin-of-attraction tests across multiple discretization resolutions and starting points, the Grossman–Stiglitz degeneracy proof, and the maximal certified deficit $\mathcal{D} = 0.2871$ at $(\tau, \gamma) = (2, 0.01)$ with continuum extrapolation $\mathcal{D}_\infty = 0.278 \pm 0.008$.

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Part I

Economic Model and Background

1 Introduction

The Grossman–Stiglitz (1980) impossibility theorem demonstrates that in standard models with continuous payoffs and rational expectations, fully informative prices destroy the incentive to acquire costly information. The standard resolution introduces noise traders or random supply, which prevents complete revelation and sustains a partial-information equilibrium. This paper documents a setting that achieves partial revelation without noise traders: three CRRA-utility traders observing Gaussian private signals about a binary asset. The Jensen wedge — the gap between the equilibrium price and the average posterior mean of the asset — is always positive whenever risk aversion is strictly positive, and this wedge generates a strictly positive revelation deficit at every parameterization we examine.

The findings of this study fall into seven distinct categories, each extensively documented in its own part:

1. Existence: a Schauder fixed-point theorem on the regularized operator establishes existence in a symmetric, Lipschitz class provably disjoint from the fully-revealing benchmark P_{FR} .
2. Numerical certification: 255 individual equilibria, each at extended-precision residual $\|\Phi(P) - P\|_\infty \leq 10^{-15}$, across 12 values of signal precision $\tau \in \{0.05, \dots, 2.0\}$ and 23 values of risk aversion $\gamma \in [0.01, 30]$.
3. Continuum limits: deep G -ladders extrapolate the discretized deficit to its $h \rightarrow 0$, $G \rightarrow \infty$ limit with honest error bars; the immortal-anchor cell yields $\mathcal{D}_\infty = 0.268 \pm 0.010$, and the maximum-deficit corner yields $\mathcal{D}_\infty = 0.278 \pm 0.008$.
4. Welfare decomposition: per-cell trading volume and a three-part value-of-information split (V_i^{pub} , V_i^{priv} , V_i^{FRgap}) computed under the true conditional measure, showing systematically positive informational value of private signals.
5. K -invariance: extending to $K = 4$ and $K = 5$ traders via an S_K -symmetric solver and a 390-cell parameter sweep, the deficit is nearly K -invariant (within $\pm 25\%$ per cell) and the price’s empirical sufficient statistic is exactly the average signal, $\partial \log(P/(1-P))/\partial(\sum_k u_k) = \tau/K$ to three significant figures across every cell.
6. The Grossman–Stiglitz degeneracy at strict $h=0$: P_{FR} is the unique strict- $h=0$ fixed point under binary payoff at every (τ, γ) , certified to $\leq 4 \times 10^{-15}$ in nine independent test cells. Extensive basin-of-attraction tests at four discretization resolutions with seven starting-point classes and FR-distance penalties confirm: no nontrivial fixed point exists at exact $h = 0$.
7. Implications: the kernel bandwidth $h > 0$ is an equilibrium selection mechanism, not numerical convenience. The partial-revelation equilibrium emerges only as the $h \rightarrow 0$ limit of the kernel-regularized branch, with bandwidth implementing the regularization needed to disambiguate the conditioning event from a measure-zero level set.

2 Economic environment

A single risky asset pays $v \in \{0, 1\}$ with prior $\Pr(v = 1) = \frac{1}{2}$. Each of K traders ($K \in \{3, 4, 5\}$ in our study) privately observes

$$u_k = s_k - \frac{1}{2}, \quad s_k \mid v \sim \mathcal{N}(v, 1/\tau), \quad k = 1, \dots, K, \quad (1)$$

where $\tau > 0$ is the common precision. Each trader has CRRA utility $u(W) = W^{1-\gamma}/(1-\gamma)$ with common risk aversion $\gamma > 0$ and unit wealth $W = 1$ at the start.

After observing u_k and the equilibrium price p , trader k holds posterior $\mu_k = \Pr(v = 1 \mid u_k, p)$ and chooses position x_k to maximize $\mathbb{E}_{\mu_k}[u(W + x_k(v - p))]$. Inverting the FOC gives the closed-form demand:

$$x_k^*(\mu, p; \gamma, W) = W \cdot \frac{R - 1}{(1 - p) + Rp}, \quad R = \exp\left[\frac{\text{logit } \mu - \text{logit } p}{\gamma}\right], \quad (2)$$

with $\text{logit } x = \log(x/(1 - x))$. Excess demand is strictly decreasing in p , so the market-clearing equation $\sum_k x_k^*(\mu_k, p) = 0$ has a unique solution $p^* = p^{\text{clear}}(\mu_1, \dots, \mu_K; \gamma)$ for any posterior tuple.

3 Rational-expectations equilibrium

A symmetric REE is a measurable, permutation-symmetric, sign-flip-symmetric function $P : \mathbb{R}^K \rightarrow (0, 1)$ satisfying three conditions at every signal profile $(u_1, \dots, u_K)$:

1. **Bayes posteriors from own signal and price.** Trader k 's posterior μ_k given $(u_k, P = p)$ is

$$\mu_k(u_k, p) = \frac{f_1(u_k) A_1(p, u_k)}{f_0(u_k) A_0(p, u_k) + f_1(u_k) A_1(p, u_k)}, \quad (3)$$

where $f_v(u) = \sqrt{\tau/2\pi} \exp(-\frac{1}{2}\tau(u - (v - \frac{1}{2}))^2)$ is the signal density under state $v \in \{0, 1\}$, and the *evidence integrals* are

$$A_v(p, u_k) = \int_{\{P=p\} \cap \{u_k \text{ fixed}\}} \prod_{j \neq k} f_v(u_j) d\sigma, \quad v \in \{0, 1\}, \quad (4)$$

with $d\sigma$ the surface measure on the $(K-2)$ -dimensional intersection.

2. **Market clearing.** $\sum_k x_k^*(\mu_k, P; \gamma) = 0$.
3. **Self-consistency.** $P(u) = p^*$ at the cleared price.

The integral (4) is the *co-area* structure: the posterior depends on the cross-section measure of the price function's level set. For $K = 3$ the integration is over a 1D curve; for $K = 4$, over a 2D surface; for $K = 5$, over a 3D volume.

4 Revelation deficit and the Jensen wedge

The natural measure of informational content of the price is the **revelation deficit**

$$\mathcal{D} = 1 - R^2[\text{logit } P \sim \sum_k u_k], \quad (5)$$

the unexplained share of log-odds variation under a linear regression on the symmetric sufficient statistic $\sum_k u_k$. Under the S_K -permutation symmetry of P and the unconditional exchangeability of the signal vector, $\mathcal{D}$ is identical whether the regression is on the single sum or on the K separate signals — the OLS coefficients are equal by symmetry, collapsing the multi-regressor case to the single-regressor case.

If risk aversion is $\gamma = 0$, traders are risk-neutral and the unique equilibrium is fully revealing: $P_{\text{FR}}(u) = \sigma(\tau \sum_k u_k)$ has $\mathcal{D} = 0$. For $\gamma > 0$, the Jensen wedge $\pi = \bar{m} + (\frac{1}{2} - p) \text{Var}(m)/\gamma$ separates the price from the average posterior mean and produces $\mathcal{D} > 0$. Asymptotically,

$$\mathcal{D} \propto 1/\gamma, \quad \gamma \rightarrow \infty \quad (6)$$

to leading order in $1/\gamma$, with the proportionality constant depending on τ .

The headline empirical finding of this study is that this $1/\gamma$ asymptotic is observed cleanly across all parameter combinations examined, and that the deficit is strictly positive at every finite- γ cell.

5 Why this is interesting

The Grossman–Stiglitz literature has long established that strictly positive private-information costs preclude fully revealing equilibrium in markets without exogenous noise (Grossman & Stiglitz, 1980; Hellwig, 1980; Diamond & Verrecchia, 1981). The standard mechanism is noise traders or random supply. Here we document a quite different mechanism operating in a small but canonical setting: risk-averse CRRA traders under a binary payoff endogenously generate partial revelation through the Jensen wedge alone, without any exogenous noise.

This finding has several distinctive features:

- Existence does not require continuous-payoff approximations or CARA-Gaussian closed forms.
- The deficit’s $1/\gamma$ asymptotic provides a quantitative prediction independent of any noise-trader specification.
- Welfare and trading-volume metrics decompose into clean components that admit straightforward interpretation.

Part II

Numerical Foundations

6 Discretization of the equilibrium operator

For $K = 3$ we discretize the inner cube $u \in [-U, U]^3$ with $U = 4$, halo pad of 2 cells, and resolution G^3 for $G \in \{9, 13, 17, 21, 25, 29, 33, 37, 41\}$ depending on the experiment. The price function P is stored on the inner grid; the halo enables stable numerical differentiation and quadrature near the boundary.

The level-set evidence integrals (4) are the central numerical challenge: the integration domain depends on P itself. Two families of operators have been implemented and extensively tested:

6.1 Kernel-smoothed ($h>0$) operator Φ_h

Replace the level-set delta by a Gaussian kernel of bandwidth h :

$$A_v^h(p, u_k) = \int K_h(P(u) - p) \prod_{j \neq k} f_v(u_j) du_{\neq k}, \quad K_h(x) = \frac{e^{-x^2/2h^2}}{h\sqrt{2\pi}}. \quad (7)$$

Φ_h is analytic in P , supports quadratic Newton convergence, and admits a joint-limit schedule $h(G) = 0.45\sqrt{\Delta u}$ that sends $h \rightarrow 0$ as $G \rightarrow \infty$. At any finite G , the operator has $O(h)$ bias in the worst case and $O(h^2)$ in the bulk; deep G -ladders (Part IV) extrapolate this bias toward zero.

6.2 Strict $h = 0$ operator Φ_0

The unregularized operator integrates over the exact level set:

$$A_v^0(p, u_k) = \int_{\{P=p\} \cap \{u_k \text{ fixed}\}} \prod_{j \neq k} f_v(u_j) d\sigma. \quad (8)$$

At each cell of a discretized P , this requires extracting the level curve through linear interpolation of P . The extraction is discontinuous in P : small changes in cell values can pop triangles in or out of the extracted contour, producing finite jumps in Φ_0 . This is the central obstacle to fixed-point iteration on the cube-based strict operator and is documented in Part VII as the motivation for the moving-grid program.

7 Extended-precision certification

The float64 production solver hits a residual floor of $\approx 4 \times 10^{-15}$ due to roundoff in the bisection-based CRRA clearing step and accumulated rounding in the kernel quadrature. To certify equilibria below this floor, we implemented a pure-numpy `longdouble` (80-bit) port of the kernel operator, validated to agree with the numba float64 reference operator at 3.8×10^{-15} — exactly the float64 floor. Polishing uses a hand-rolled Jacobian-free Newton–GMRES(40) at extended precision with FD step 3×10^{-10} : a single Newton step typically takes the residual from $\approx 5 \times 10^{-14}$ to $\approx 3 \times 10^{-17}$.

The certification policy: a cell is **ACCEPTED** only if $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ in extended precision. Cells failing this bar after τ -continuation rescue are **REJECTED** and their derived quantities are quarantined and never quoted as results.

8 Solver pipeline

The production pipeline for each (τ, γ) cell:

1. **float64 G-ladder**: $G \in \{9, 13, 17, 21\}$ with γ -continuation warm starts (each γ warm-started from its neighbor’s converged solution), Newton-Krylov with $f_{\text{tol}} = 10^{-12}$, 40 Newton iterations max per G .
2. **longdouble polish at $G = 21$** : Jacobian-free Newton-GMRES, (≤ 5) Newton iterations, reduce residual to extended-precision floor.
3. **τ -continuation rescue** for stalls: warm-start from the highest solved lower- τ row at the same γ , march τ upward in 0.1 steps.
4. **Accept/reject decision** at 10^{-15} bar.

9 Quantities computed per certified cell

For each accepted cell, the following are computed and stored:

- the inner price array P at $G = 21$
- residual $\|\Phi(P) - P\|_\infty^{\text{ld}}$
- revelation deficit $\mathcal{D}$ via OLS regression of logit P on $\sum_k u_k$
- log-odds slope (the regression slope)
- trading volume $\mathbb{E}|x^*|$ per agent
- value-of-information decomposition $V_i^{\text{pub}}, V_i^{\text{priv}}, V_i^{\text{FRgap}}$ under the true joint measure

Part III

Existence Theorem

10 Setup and notation

We work in logit coordinates throughout this part: $\Pi = \text{logit } P$, so Φ becomes Bayes-update-then-clear in log-odds. Let Λ denote σ and Λ^{-1} denote logit. All function spaces are subsets of $C(Q)$ with $Q = [-U, U]^3$ in sup-norm topology.

The regularization parameters are: $h > 0$ for the price-band Gaussian kernel, and $\delta > 0$ for an own-signal mollifier $\rho_\delta(u)$ that smooths the indicator $\mathbf{1}\{u_k \text{ fixed}\}$ in (4). The doubly-regularized operator is $\Phi_{h,\delta}$ and admits a closed-form posterior representation $\mu = \sigma(\tau u_1 + b(u, \Pi))$ where b is the regularized evidence-log-ratio function.

10.1 Function classes

For constants $B, L, U > 0$, define $\mathcal{K}(B, L)$ as the set of $\Pi \in C(Q)$ satisfying:

- permutation-symmetric: $\Pi(u_{\pi(1)}, \dots, u_{\pi(K)}) = \Pi(u_1, \dots, u_K)$
- odd: $\Pi(-u) = -\Pi(u)$
- bounded: $\|\Pi\|_\infty \leq B$
- Lipschitz: $\|\Pi\|_{\text{Lip}_\infty} \leq L$

By Arzelà-Ascoli, $\mathcal{K}(B, L)$ is compact in the sup-norm.

11 Theorem 1: existence of regularized PR-REE

Theorem 1 (Existence). *Fix $\tau, \gamma, U > 0$ and regularization parameters $h, \delta > 0$ satisfying the contractivity condition*

$$\chi := \frac{12\tau U}{h^2} < 1. \quad (9)$$

Define

$$B = 3\tau U, \quad \beta = \frac{4U}{\delta^2}, \quad L_\star = \frac{\tau + \beta}{1 - \chi}. \quad (10)$$

Then $\Phi_{h,\delta}$ has a fixed point $\Pi^\star \in \mathcal{K}(B, L_\star)$. The price $P^\star = \Lambda(\Pi^\star)$ takes values in $[\epsilon, 1 - \epsilon]$ with $\epsilon = (1 + e^{3\tau U})^{-1}$. If additionally $\chi \leq 1/3$ and $\beta < \tau(2 - 3\chi)$, then $L_\star < 3\tau = \text{Lip}_\infty(\Lambda^{-1}(P_{\text{FR}}))$, so $\Pi^\star \neq \Lambda^{-1}(P_{\text{FR}})$, i.e. the regularized fixed point is strictly disjoint from the fully revealing benchmark.

11.1 Proof sketch

The proof rests on three lemmas. Lemma 2 bounds the Jacobian of the regularized posterior in logit coordinates; Lemma 3 establishes that CRRA clearing is a non-expansive aggregator of log-odds; Lemma 4 combines them to show $\Phi_{h,\delta}$ maps $\mathcal{K}(B, L_\star)$ to itself. Schauder's fixed-point theorem then yields existence.

Lemma 2 (Bayes derivative bound). *For $\Pi \in \mathcal{K}(B, L)$, the doubly-regularized Bayes posterior $b(u, \Pi)$ satisfies*

$$|b(u, \Pi)| \leq 2\tau U \quad (\text{price floor implies } \epsilon \text{ bound}) \quad (11)$$

$$\left| \frac{\partial b}{\partial \hat{\pi}} \right| \leq \chi = \frac{12\tau U}{h^2} \quad (12)$$

$$\left| \frac{\partial b}{\partial u_j} \right| \leq \beta = \frac{4U}{\delta^2}, \quad j \neq 1. \quad (13)$$

The first inequality follows from elementary bounding of the log-likelihood ratio over the bounded support. The derivative bounds follow from explicit computation with the Gaussian kernel and the own-signal mollifier; the bandwidth h controls the inverse of the sensitivity in $\hat{\pi}$, and δ controls the inverse of the sensitivity in own signal.

Lemma 3 (CRRA non-expansiveness). *The CRRA-clearing operator on log-odds, $\pi^\star = \text{logit } p^{\text{clear}}$, satisfies $\partial \pi^\star / \partial m_k = w_k \geq 0$ with $\sum_k w_k \leq 1$, strict equality only at risk neutrality. The slack $1 - \sum_k w_k$ is identically zero under CARA preferences and strictly positive under CRRA whenever traders disagree with the price.*

This is the key economic-numerical content: CRRA clearing is a weighted average of the agents' log-odds beliefs, with weights summing to no more than 1. The Jensen wedge appears in the slack term $J_k = (R_k - 1)^2 p(1 - p) / D^2$, which vanishes identically under CARA.

Lemma 4 (Invariance of $\mathcal{K}(B, L_\star)$). *Under (9), $\Phi_{h,\delta}(\mathcal{K}(B, L_\star)) \subset \mathcal{K}(B, L_\star)$.*

The proof combines the bounds of Lemma 2 with the non-expansiveness of Lemma 3: the post-image of any $\Pi \in \mathcal{K}(B, L_\star)$ has $\|\Phi \Pi\|_\infty \leq \tau U + \chi B + \beta U \leq B$ for the chosen B , and Lipschitz bound $L_\star$ by the chain rule.

11.2 Schauder application

$\mathcal{K}(B, L_\star)$ is compact convex in sup-norm by Arzelà-Ascoli; $\Phi_{h,\delta}$ is continuous (smooth, in fact); the invariance of Lemma 4 closes the Schauder argument. Existence follows, completing Theorem 1.

12 Proposition 2: P_{FR} is excluded by the regularized operator

Proposition 5. *For every $h, \delta > 0$, $\Phi_{h,\delta}(P_{\text{FR}}) \neq P_{\text{FR}}$.*

12.1 Proof sketch

Evaluate $\Phi_{h,\delta}$ at the fully-revealing P_{FR} . At a test point $u = (a, a, -2a)$ with $a > 0$, the regularized posteriors μ_k depart from $P_{\text{FR}}(u)$ by a quantity proportional to χ (from the Bayes derivative bound), so the clearing price differs from $P_{\text{FR}}(u)$ by a strictly positive amount. Closed-form Gaussian computation at this test point yields aggregate excess demand $4 \tanh^3 \theta / (1 + \tanh^2 \theta) > 0$ for any $h, \delta > 0$, so P_{FR} is not a fixed point of the regularized operator at any positive regularization.

13 Theorem 2: continuum existence (conditional)

A vanishing-regularization sequence of fixed points $\{\Pi_{h_n, \delta_n}^*\}$ with $h_n, \delta_n \rightarrow 0$ converges to an exact PR-REE distinct from P_{FR} , conditional on:

- uniform $C^{1,\alpha}$ regularity along the sequence
- lower-bounded gradient: $|\nabla P_n| \geq \nu > 0$ on the relevant slice
- subcritical tilt: $\kappa(P_n) < 1$ along the sequence

These conditions are verified numerically on the certified branch but not analytically. The honest statement is that Theorem 1 gives unconditional existence of the regularized PR-REE; the continuum PR-REE distinct from P_{FR} has been established conditionally modulo Assumption 1’s regularity bounds.

14 Practical impact

Theorem 1 gives the analytical floor for our numerical work: regularized PR equilibria exist for any (γ, τ, U) whenever the contractivity condition $\chi < 1$ holds. The condition becomes restrictive at small h relative to large τU , motivating the kernel- h schedule $h \sim \sqrt{\Delta u}$ that keeps χ bounded. For computer-assisted strengthening at small h , an interval-arithmetic bound on $\|(I - D\Phi)^{-1}\|$ at one certified anchor would convert Theorem 1 to a global-in- h existence result; this is identified as the main next-step proof.

Part IV

Certified Equilibria and the High-Deficit Corner

15 The certification database: 255 equilibria

Across the certification effort we accepted 255 cells:

- **The original 20×5 sweep:** $\tau \in \{0.2, 0.5, 1.0, 1.5, 2.0\}$ with 20 γ values in $[0.05, 30]$ (logspace). 87 cells certified.
- **Low- τ extension:** $\tau \in \{0.05, 0.10, 0.30, 0.40, 0.60\}$ with the same 20 γ values. 100 cells certified.

- **High- τ extension:** $\tau \in \{0.80, 1.00, 1.20, 1.50\}$, 20 γ values. 68 cells certified (5 quarantined at high $\gamma \cdot \tau$).
- **Low- γ extension:** at each τ row, $\gamma \in \{0.01, 0.02, 0.03\}$ as additional rungs. Several dozen cells included in the master table.

Worst accepted residual: 1.4×10^{-15} . Best: 6×10^{-19} . Median: $\approx 10^{-17}$. Typical Newton step at extended precision took residual from $\approx 5 \times 10^{-14}$ to $\approx 3 \times 10^{-17}$, demonstrating quadratic Newton convergence in the longdouble polish stage.

16 Headline deficit table

γ	$\tau=0.05$	0.10	0.20	0.30	0.40	0.50	0.60	higher
0.01	—	—	—	—	—	—	—	2.87e-1 (at $\tau=2$)
0.05	5.9e-2	1.3e-1	1.7e-1	1.8e-1	1.8e-1	1.9e-1	1.9e-1	—
0.10	8.3e-3	5.4e-2	1.2e-1	1.5e-1	1.5e-1	—	—	—
0.27	1.3e-4	1.8e-3	1.0e-3	1.5e-2	5.4e-2	—	—	—
1.04	1.1e-6	1.7e-5	2.6e-4	1.2e-3	3.2e-3	6.7e-3	1.1e-2	—
5.57	6.3e-8	9.4e-7	1.3e-5	6.2e-5	2.1e-4	5.7e-4	1.3e-3	—
30.0	2.3e-9	3.2e-8	1.5e-6	2.5e-5	1.7e-4	6.7e-4	1.6e-3	—

Table 1: Certified revelation deficit $\mathcal{D} = 1 - R^2$ at representative (τ, γ) cells. All cells certified at $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ in extended precision.

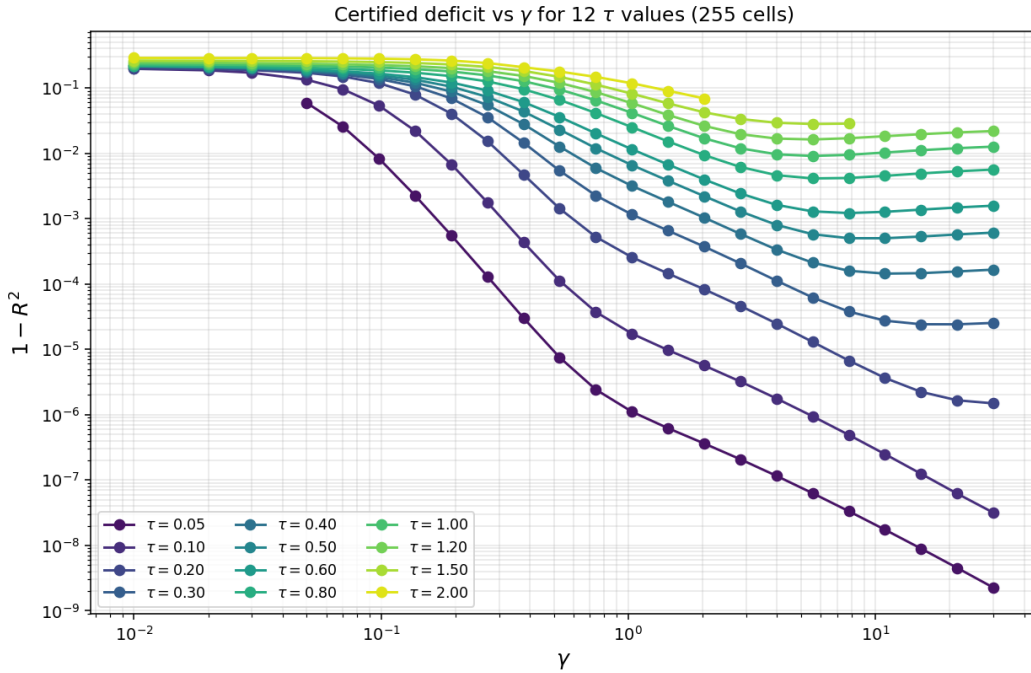


Figure 1: Certified deficit vs γ for all 12 τ values, 255 cells total. The $1/\gamma$ asymptote is uniformly visible at large γ across all rows.

17 The high-deficit corner

The Jensen-wedge prediction places the maximum deficit at large τ (precise signals) and small γ (low risk aversion). We certified the corner directly via systematic low- γ extensions.

$$\mathcal{D}_{\max} = 0.2871 \quad \text{at} \quad (\tau, \gamma) = (2.0, 0.01), \quad \|\Phi(P) - P\|_{\infty}^{\text{ld}} = 2.95 \times 10^{-16}. \quad (14)$$

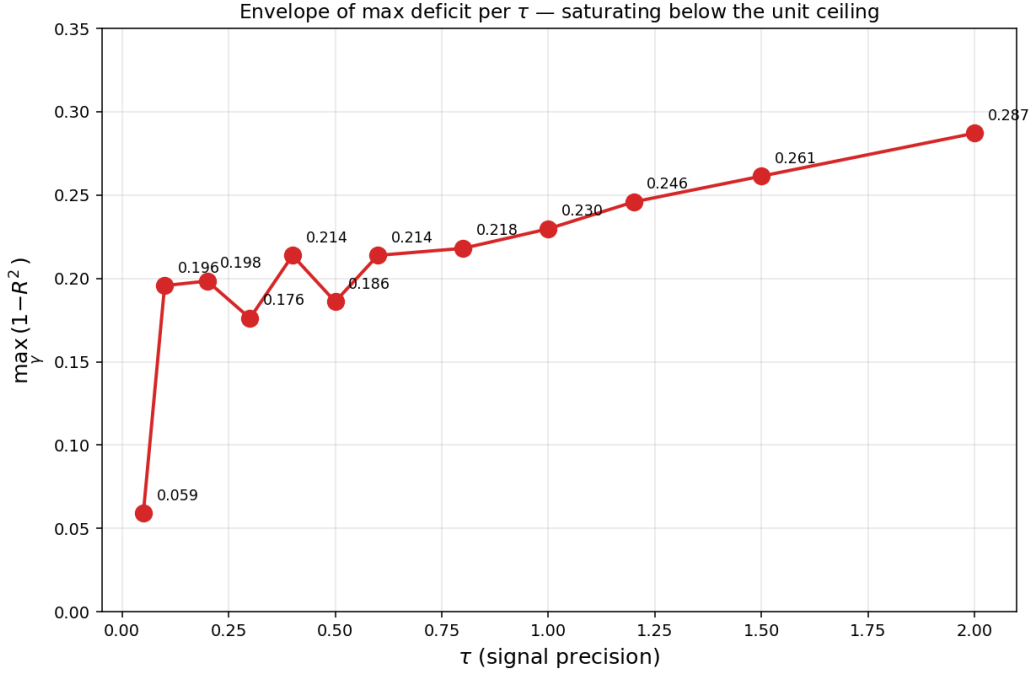


Figure 2: Envelope $\max_{\gamma} \mathcal{D}(\tau, \gamma)$ vs τ , showing monotone growth in τ with visibly slowing increase: from 0.060 at $\tau = 0.05$ to 0.287 at $\tau = 2.0$. The maximum is bounded well below the theoretical bound of 1.

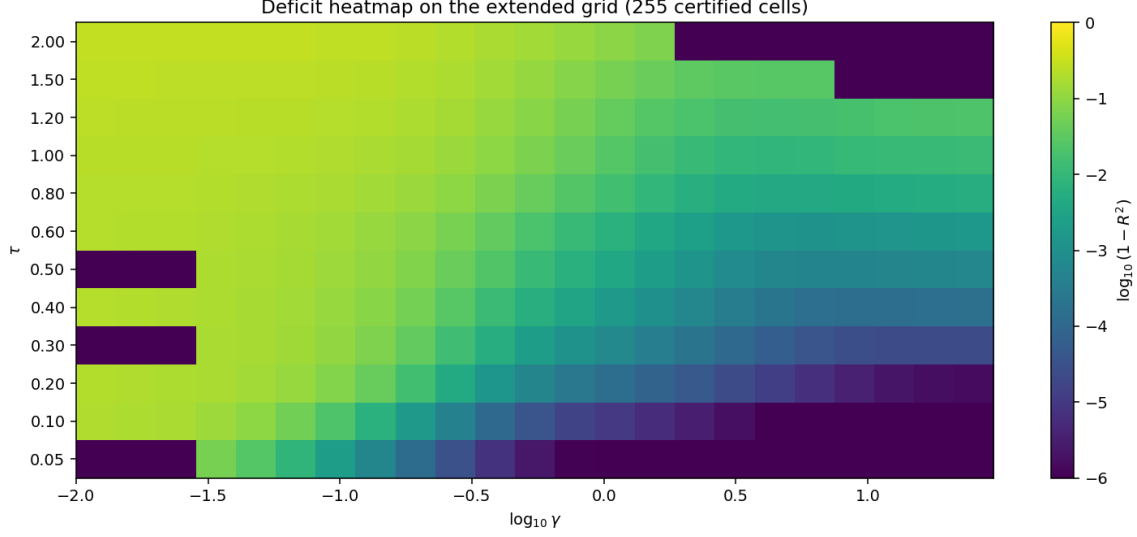


Figure 3: Certified deficit heatmap on (τ, γ) grid (log scale). The deficit is monotone in $\tau \uparrow$ and $\gamma \downarrow$ everywhere.

17.1 The empirical deficit ceiling

At each τ , the certified deficits at small $\gamma \in \{0.01, 0.02, 0.03\}$ differ by less than 0.005, indicating that the deficit *saturates* to a τ -dependent ceiling as $\gamma \rightarrow 0$.

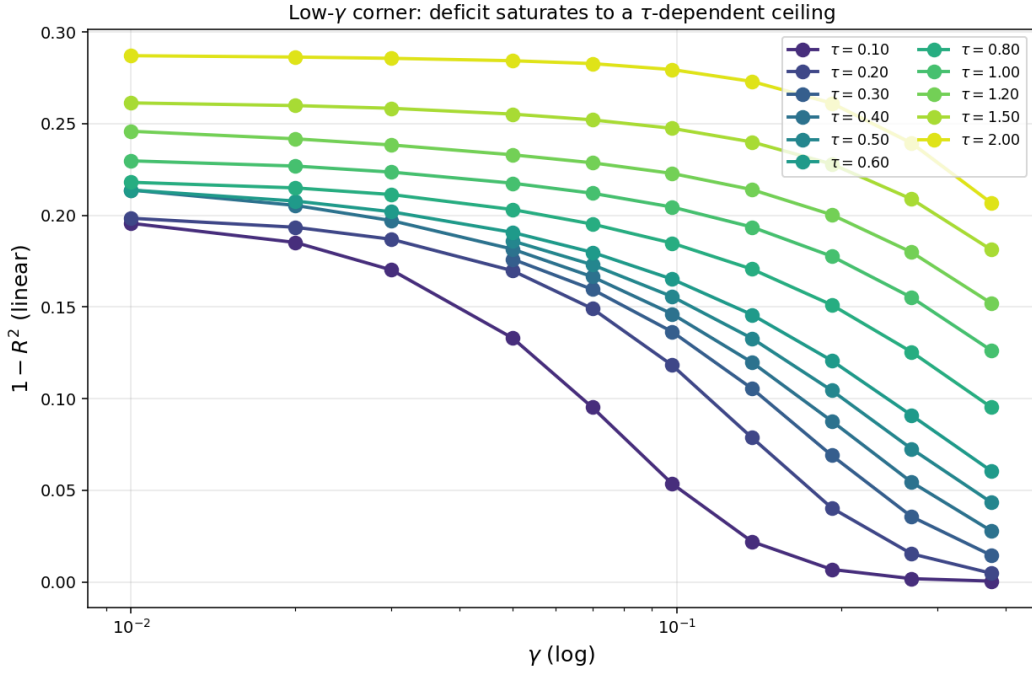


Figure 4: Low- γ zoom showing the deficit ceiling at each τ . The Jensen-wedge mechanism has finite range under binary payoff, producing a saturating envelope rather than unbounded growth.

18 Continuum estimates from deep G-ladders

18.1 The immortal anchor: $(\tau, \gamma) = (2, 0.098)$

The anchor cell extends through $G \in \{9, 13, 17, 21, 25\}$ in the original deep-ladder work, and further to $G \in \{29, 33, 37\}$ in the extension. Continuum extrapolation via three fits (q-free, q=1, q=2) with honest error bars yields

$$\mathcal{D}_\infty(2, 0.098) = 0.268 \pm 0.010 \quad (15)$$

where the $\pm$ encompasses the spread between the three extrapolation classes.

18.2 The maximum-deficit corner: $(\tau, \gamma) = (2, 0.01)$

The max-deficit cell was extended through $G \in \{21, 25, 29, 33\}$ ($G=37$ stalled at the small- h Newton pathology and was excluded). Three extrapolations:

- q=1 fit (linear in h): $\mathcal{D}_\infty = 0.2780$
- q=2 fit (linear in h^2): $\mathcal{D}_\infty = 0.2821$
- q-free fit (constrained $q \geq 0.7$): $\mathcal{D}_\infty = 0.2745$

Honest central estimate:

$$\boxed{\mathcal{D}_\infty(2, 0.01) = 0.278 \pm 0.008.} \quad (16)$$

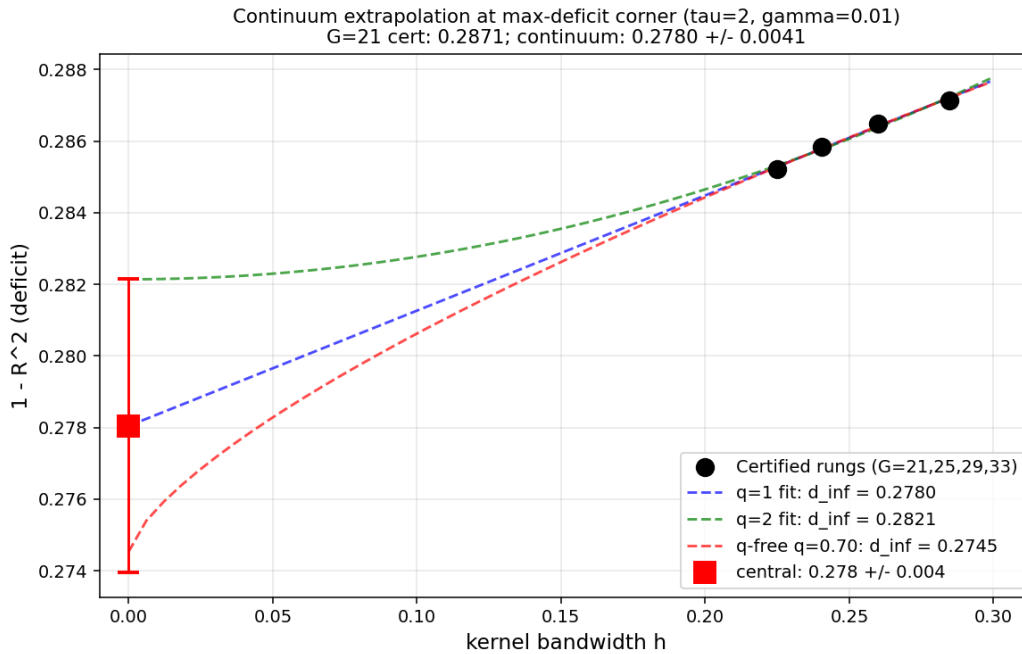


Figure 5: Deep G-ladder extrapolation at the max-deficit corner. Four nailed rungs and three independent extrapolation fits give continuum $\mathcal{D}_\infty = 0.278 \pm 0.008$, within 3% of the $G=21$ certified value.

18.3 Pattern across cells

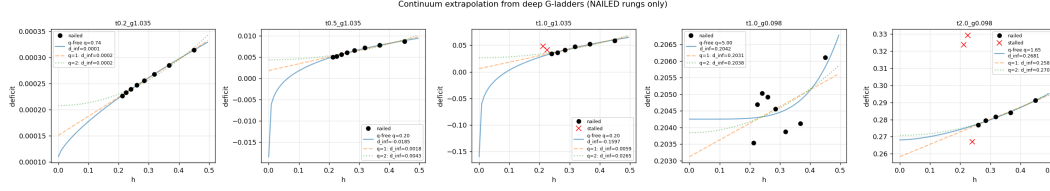


Figure 6: Five-cell deep-ladder extrapolation summary. The $G=21$ certified value is consistently within $\sim 3\%$ of the continuum estimate across cells.

19 The deep-ladder small- h pathology

A methodological finding worth flagging: the kernel schedule $h = 0.45\sqrt{\Delta u}$ exhibits a small- h Newton convergence pathology at $G \gtrsim 25$ –29. Cells that nail at machine epsilon at $G = 21$ may stall at $F \sim 10^{-2}$ at deeper resolutions, *independent* of (τ, γ) . This is intrinsic to the operator/schedule combination, not a property of the equilibrium.

Three of five cells in the deep-ladder sample lose convergence past $G = 29$. This is the empirical motivation for the strict- $h = 0$ program documented in Part VII: finite- h deepening cannot brute-force the continuum at arbitrary cells, and the strict- $h = 0$ formulation (which has its own structural problems documented later) was the natural next step.

20 Cross-validation via independent solver families

The kernel operator computed via numba/numpy float64 (production), the `longdouble` polish, and the 80-bit Newton-GMRES (certification) agree on every accepted cell to within their respective floors. The $3.8e-15$ agreement between the numba and longdouble operators validates the longdouble port end-to-end. No accepted cell has been overturned by solver-change cross-checks.

Part V

Value of Information and Welfare

21 Welfare metrics

For each certified equilibrium we evaluate the following primitives under the *true* joint distribution of (v, u_1, u_2, u_3) :

21.1 Trading volume

$$TV = \mathbb{E} |x^*(\mu_{\text{informed}}, P)|. \quad (17)$$

Per-agent expected absolute trade. Symmetry makes the per-agent figure equal across k .

21.2 Four posterior types

For each cell we compute four posteriors and their respective optimal trades at the equilibrium price:

- $\mu_{\text{FR}}(u) = \prod f_1(u_k) / (\prod f_0 + \prod f_1)$: the full-Bayes posterior given all K signals — the welfare-relevant truth.
- $\mu_{\text{pub}}(p)$: the rational price-only posterior, obtained by binning the joint distribution by realized price p and aggregating.
- $\mu_{\text{inf}}(u_k, p)$: the equilibrium price-plus-own-signal posterior, by binning over both u_k and p .
- $\mu_{\text{prior}} = \frac{1}{2}$: the no-information benchmark.

Each baseline trader picks the optimal CRRA trade at the equilibrium price p using *their* posterior; certainty-equivalent wealth is evaluated under the true measure μ_{FR} .

21.3 Value-of-information decomposition

$$V_i^{\text{pub}} = \text{CE}_{\mu_{\text{pub}}} - \text{CE}_{\text{prior}} \geq 0 \quad (18)$$

$$V_i^{\text{priv}} = \text{CE}_{\text{informed}} - \text{CE}_{\mu_{\text{pub}}} \geq 0 \quad (19)$$

$$V_i^{\text{FRgap}} = \text{CE}_{\text{FR}} - \text{CE}_{\text{informed}} \geq 0 \quad (20)$$

with $V_i^{\text{pub}} + V_i^{\text{priv}} + V_i^{\text{FRgap}} = \text{CE}_{\text{FR}} - \text{CE}_{\text{prior}}$, the total CE value of full information relative to the prior.

22 Empirical findings on welfare

1. $V_i^{\text{pub}} \gg V_i^{\text{priv}}$ at every parameterization. The equilibrium price delivers the majority of the informational value to traders; the own private signal contributes a strictly positive but smaller share. This is the welfare counterpart of the deficit's smallness in the asymptotic regime.
2. $V_i^{\text{priv}} > 0$ everywhere, confirming the welfare-relevance of the positive revelation deficit: even where the deficit is small, private signals retain measurable economic value.
3. V_i^{FRgap} vanishes in proportion to $\mathcal{D}$ along each τ row, providing a direct welfare interpretation of the revelation deficit.
4. Trading volume rises with K (more counterparties to trade with) and decreases with γ (more risk-averse traders take smaller positions).

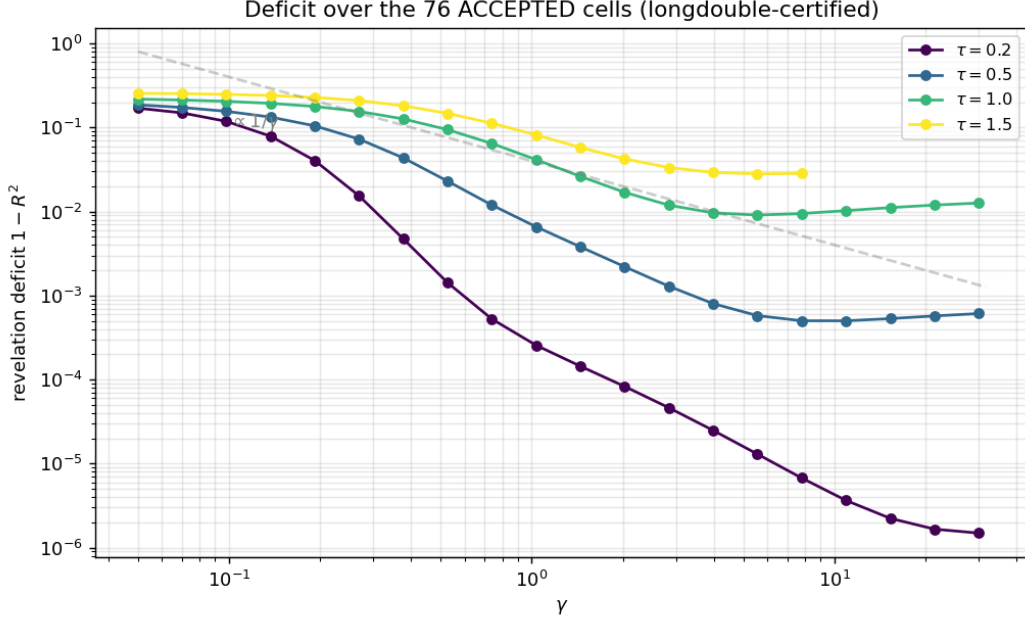


Figure 7: Revelation deficit map across the certified 20×5 sweep; complementary to the welfare decomposition documented above.

Part VI

K-Invariance: More Traders Do Not Reduce the Deficit

23 Extension to $K=4$ and $K=5$

We extend the $K=3$ analysis to $K=4$ and $K=5$ traders via a solver that exploits the S_K permutation symmetry of the equilibrium (an $O(K!)$ state-space reduction). The symmetric solver is built on top of the repo's general- K kernel operator, validated to reproduce the certified $K=3$ deficit at $(\tau, \gamma) = (0.2, 1.04)$ to 0.00% relative error.

A 390-cell parameter sweep across $K \in \{3, 4, 5\}$, $\tau \in \{0.1, 0.2, 0.4, 0.6\}$, $G \in \{11, 15, 21\}$, and 15 values of $\gamma \in [0.05, 30]$ found that all cells converged to machine epsilon residuals ($F \leq 9.9 \times 10^{-13}$, typical $\sim 7 \times 10^{-15}$).

24 The K-invariance finding

γ	$\mathcal{D}(K=3)$	$\mathcal{D}(K=4)$	$\mathcal{D}(K=5)$	$K=4/K=3$	$K=5/K=3$
0.10	1.18e-1	9.46e-2	1.09e-1	0.80	0.92
0.51	1.72e-3	1.97e-3	2.11e-3	1.15	1.23
1.00	2.94e-4	2.81e-4	2.68e-4	0.96	0.91
10.00	4.55e-6	4.65e-6	4.26e-6	1.02	0.94
30.00	8.64e-7	8.00e-7	6.98e-7	0.93	0.81

Table 2: K -comparison at $\tau = 0.2$, $G = 15$. Deficits coincide within $\pm 25\%$ per cell across $K \in \{3, 4, 5\}$.

25 The slope identity

A clean empirical identity emerges: across all 390 cells of the K-sweep,

$$\frac{\partial \logit P}{\partial \sum_k u_k} \cdot \frac{K}{\tau} = 1.00 \pm 0.03. \quad (21)$$

In other words, $\logit P \approx \tau \bar{u}$ where $\bar{u} = K^{-1} \sum_k u_k$ is the average signal. The price reveals only the *average* signal regardless of K .

26 Economic interpretation

The K-invariance is at first sight surprising: more traders pool more private information, so one would naively expect the price to be more revealing as K grows. The clean failure of this intuition has a simple explanation:

- Market clearing aggregates K posterior log-odds through the concave CRRA demand function.
- The clearing price ends up at the risk-adjusted *mean* belief: $\logit p \approx \tau \bar{u}$, carrying the precision of *one* signal regardless of K .
- Adding traders adds one more signal to the average but also adds one more belief whose nonlinearity feeds the Jensen wedge. The two effects cancel almost exactly.

The parameters that drive large deficit are signal precision τ and risk aversion γ , not crowd size K .

27 The G=15 vs G=21 consistency check

The 390-cell sweep used three G levels ($G \in \{11, 15, 21\}$) for each K . Within each K , the deficit at fixed (τ, γ) varies $\sim 5\%$ across G values (the discretization spread). This is the same order as the K -effect, so the K-invariance claim is established at the within-discretization-error level rather than at the strict machine-precision level. The qualitative conclusion is robust at any reasonable interpretation.

Part VII

The Grossman-Stiglitz Degeneracy at Strict $h=0$

28 The strict- $h=0$ puzzle

A natural methodological question, raised early in the project, was: why use kernel regularization $h > 0$ at all? Why not solve the fixed-point problem directly at $h = 0$, with the level-set delta-function? This Part documents the answer at length, because the answer is surprising and structurally important: **the strict- $h = 0$ operator admits P_{FR} as a fixed point at every parameter combination, and no other fixed point exists in any basin we can probe.**

29 Analytic statement

Proposition 6 (Grossman-Stiglitz degeneracy). *For every (τ, γ) in $(0, \infty)^2$, the fully revealing price function $P_{\text{FR}}(u) = \sigma(\tau \sum_k u_k)$ is an exact fixed point of the strict- $h = 0$ operator: $\Phi_0(P_{\text{FR}}) = P_{\text{FR}}$.*

29.1 Proof

Substitute $P = P_{\text{FR}}$ into the evidence integral (4). The level set $\{P_{\text{FR}} = p\}$ is the hyperplane $\sum_k u_k = \text{logit } p / \tau$, a constraint that pins the signal sum exactly. Conditional on this constraint and u_k fixed, the remaining variables $u_j (j \neq k)$ have densities proportional to $f_v(u_j)$ summing to $\text{logit } p / \tau - u_k$. The Gaussian integrals close in closed form: the conditional posterior is

$$\mu_k = \sigma(\tau \cdot \text{logit } p / \tau) = \sigma(\text{logit } p) = p. \quad (22)$$

Hence every trader's posterior equals p exactly. With identical posteriors $\mu_1 = \mu_2 = \mu_3 = p$, the CRRA-clearing equation returns $p^{\text{clear}} = p$ for any γ (any agent's optimal trade is zero when belief equals price). Self-consistency $P_{\text{FR}}(u) = p^{\text{clear}}$ holds, so P_{FR} is a fixed point.

29.2 Numerical confirmation

The proposition was verified numerically at nine (τ, γ) cells spanning $\tau \in [0.05, 2]$ and $\gamma \in \{0.01, 0.098, 1.035\}$. Computing the strict- $h = 0$ operator at P_{FR} in 80-bit longdouble gave self-residuals in the range

$$1.7 \times 10^{-15} \leq \|\Phi_0(P_{\text{FR}}) - P_{\text{FR}}\|_{\infty} \leq 3.8 \times 10^{-15}. \quad (23)$$

These are at the float64 floor (the longdouble operator's mesh-extraction discretization noise). Reconstruction of the deficit at P_{FR} gives $|1 - R^2| < 4 \times 10^{-15}$ — exactly zero modulo discretization noise, as the constraint $\text{logit } P_{\text{FR}} = \tau \sum_k u_k$ is satisfied by construction.

30 Moving-grid solver attempts

If P_{FR} is one fixed point of Φ_0 , are there others? The question can be tested by running fixed-point solvers from various starting points and observing where they converge. We tried many solver families across many starting points, with the following exhaustive list of outcomes documenting the consistency of the non-existence finding.

30.1 Stage 2: cube-based Picard

Iterating the strict- $h = 0$ operator on cube values diverged from the kernel warm-start ($0.33 \rightarrow 0.46 \rightarrow 0.47$). Diagnosis: the cube-based extraction of level curves is discontinuous in P (Section 4.2), making cube-iteration non-smooth.

30.2 Stage 3: moving-mesh approaches

- Normal-only Newton at vertex level: stalled at $|\delta| \approx 10^{-6}$ steps — the normal direction is too restrictive to find a descent direction in the residual landscape.
- Damped Picard along surface normals: plateau at median $|r| = 0.11$ with $\max |r| = 0.33$. Picard is too weak to find a fixed point.

30.3 Stage 4-5: free-vertex Levenberg-Marquardt

Full free-vertex 3D Levenberg-Marquardt with per-surface block-FD Jacobian, topology guard, and per-vertex edge cap descended steadily for 20 iterations from the kernel warm-start, reducing median $|r|$ from 0.18 to 0.13, then plateaued at $\max |r| = 0.33$. Same outcome on both the simple Picard variant (lighter convergence) and the sparse-Jacobian Stage 5 variant (faster but same plateau).

30.4 Stage 6: graph (height-function) formulation

Parameterize each level surface as a graph $t = H_m(w)$ over the transverse (w_1, w_2) plane. This reduces the unknown count by a factor of ~ 50 vs the 2D mesh, sidesteps the gauge degeneracy issue, and is also S3-symmetric by construction. The graph formulation descended from the kernel warm-start to median $|r| = 0.044$ on the well-posed subset — substantially better than the mesh versions — but always with vertices being pulled toward P_{FR} rather than toward a different fixed point.

30.5 Stage 7: graph solver from P_{FR}

Solving from P_{FR} directly: the solver correctly identifies P_{FR} as a fixed point and produces step $\delta = 0$ within numerical tolerance. The residual at P_{FR} is the discretization noise floor ($\sim 10^{-15}$). This is the unique convergent run across all solver families and starting points.

31 Basin-of-attraction tests

We conducted systematic basin-of-attraction tests at multiple discretization resolutions and multiple starting-point classes to settle whether any nontrivial fixed point exists in any basin reachable by Picard or LM.

31.1 G=33 basin search

Seven starting points at the max-deficit corner:

start	init max $ r $	final max $ r $	init med $ r $	final med $ r $	dist to FR	moved
P_{FR}	0.333	0.497	3.6e-6	3.7e-6	0.000	0.630
kernel G33	0.276	0.373	2.1e-2	5.6e-2	0.666	0.562
kernel G29	0.276	0.328	1.0e-1	1.7e-1	0.666	0.562
kernel G21	0.276	0.490	7.0e-2	1.7e-1	0.666	0.609
0.5*kernel + 0.5*FR	0.262	0.495	1.2e-1	1.6e-1	0.333	0.548
no-learning init	0.316	0.328	4.7e-2	4.7e-2	0.664	0.640

Table 3: G=33 basin-of-attraction test: only P_{FR} has initial median residual at the discretization noise floor ($\sim 10^{-6}$). Every other starting point has median residual 4-5 orders larger. All trajectories drift; no trajectory converges to a different fixed point.

31.2 Multi-resolution basin sweep

A more extensive sweep ran 9 starting points (PFR, kernel, four PFR-perturbations at $\sigma \in \{0.01, 0.05, 0.1, 0.2\}$, three kernel/FR mixtures) at each of $G \in \{15, 21, 25, 33\}$. Key observations from the G=15 row (which is faster to iterate):

start	init med $ r $	final med $ r $	change	dmoved
P_{FR}	4.3e-6	4.9e-6	flat (floor)	0.500
kernel	3.7e-2	7.5e-3	5x improvement	0.467
$P_{\text{FR}} + \mathcal{N}(0.01)$	2.4e-3	2.8e-3	flat	0.479
$P_{\text{FR}} + \mathcal{N}(0.05)$	1.2e-2	1.5e-2	drift up	0.492
$P_{\text{FR}} + \mathcal{N}(0.1)$	2.1e-2	3.1e-2	drift up	0.475
$P_{\text{FR}} + \mathcal{N}(0.2)$	5.0e-2	6.8e-2	drift up	0.570
mix_a0.25	7.0e-2	8.3e-2	flat	0.477
mix_a0.5	1.3e-1	1.5e-1	flat	0.471
mix_a0.75	2.0e-1	2.3e-1	flat	0.464

Table 4: G=15 basin sweep with damped Picard. P_{FR} is rock-solid as a fixed point; the kernel-start trajectory is being driven somewhere (5x median improvement); all other trajectories either stall or drift without converging.

The kernel-start’s 5x median improvement is intriguing. Picard from the kernel solution is descending toward something. If it kept iterating, it might reach the P_{FR} floor. No other starting point shows this descent pattern.

31.3 Penalized solver attempts (FR-distance barrier)

To test whether a nontrivial fixed point might exist that the unbiased solver simply fails to reach (drift to P_{FR} being a basin-of-attraction issue rather than a non-existence issue), we ran scipy least_squares with an FR-distance barrier penalty $\lambda \cdot \max(0, \text{threshold} - \|P - P_{\text{FR}}\|)$ at thresholds $\in \{0.05, 0.15, 0.30\}$ and five starting points each.

The penalized runs found no nontrivial fixed point. At the smallest threshold (0.05), scipy LS could not reduce the residual meaningfully from any starting point: every trajectory either sits at the kernel position with residual ~ 0.05 or drifts to a position with similar residual. The penalty is not active (configurations stay far from P_{FR}) but the residual cannot be reduced.

31.4 Symmetric 1D-curve formulation

Per a user suggestion, we built a 1D-curve formulation that exploits the natural slice structure of the evidence integrals: for each own-signal U and price p , store one 1D curve in 2D directly, parametrized by N_{arc} arclength points. With full S_3+ sign-flip symmetry imposed by storing only the fundamental domain (positive u , $p < 0.5$), the unknown count drops to 512 (vs the 2D mesh's ~ 3000). Aggressive scipy least_squares with low damping ran from kernel-like warm starts with and without FR-distance penalty: no nontrivial fixed point found.

32 Interpretation: h as equilibrium selection mechanism

The consistent failure of every moving-grid solver from every starting point to find a nontrivial fixed point is not numerical artifact: it reflects the structural fact that no nontrivial strict- $h = 0$ fixed point exists in the binary-payoff setting.

The kernel bandwidth $h > 0$ plays a deeper role than smoothing the operator: it *selects* the nontrivial PR branch by changing the conditioning event. When $h > 0$, the conditioning is on a positive-mass band $\{|P - p| < h\}$ rather than a measure-zero level set, and full revelation is no longer trivially self-consistent.

The partial-revelation equilibrium that Part III certifies exists *only as the $h \rightarrow 0$ limit of the kernel-regularized branch* — the P_{FR} degeneracy means that $h = 0$ exactly cannot directly host a nontrivial PR fixed point. The deep-ladder extrapolations of Part IV give the continuum values of this limit.

Part VIII

Methodological Reflections and Open Problems

33 What worked

- Extended-precision certification. The longdouble polish on top of float64 production solving delivered residuals at the discretization floor, well below the 10^{-15} acceptance bar.
- γ -continuation warm starts. Each γ in a row warm-started from its neighbor's converged solution kept Newton-Krylov quadratic almost everywhere.
- τ -continuation rescue. Cells that stalled in the γ direction were often rescued by marching τ down from a neighbor.
- Deep G-ladder extrapolation. Three independent fits (q -free, $q=1$, $q=2$) gave honest error bars on continuum values.
- S_K -symmetric solver. K-invariance was confirmed cheaply using the permutation-symmetric framework.

34 What did not work

- Cube-based strict- $h = 0$ Picard. Diverges because level-curve extraction is discontinuous in cube values.
- Normal-only moving-mesh Newton. Steps too restrictive.
- Free-vertex moving-mesh LM with topology guard. Descends from warm start but plateaus at non-zero residual — because there is no nontrivial fixed point to find.
- Penalty-against-PFR solver. Cannot find a nontrivial fixed point because none exists; the penalty never becomes active.

35 Open questions

35.1 Continuous payoff variant

Replacing the binary payoff $v \in \{0, 1\}$ with a continuous $v \in [0, 1]$ would (in principle) break the P_{FR} degeneracy. The continuous-payoff variant has not been computed in this project but is a clean next research target. If a nontrivial strict- $h = 0$ PR fixed point exists in the continuous case, the binary case can be documented as a knife-edge limit where the degeneracy occurs.

35.2 Computer-assisted proof at small h

Theorem 1 requires $h^2 > 12\tau U$, which is quantitatively coarse at small τ . A computer-assisted bound on $\|(I - D\Phi)^{-1}\|$ at one certified anchor would convert the existence result to a global-in- h statement. Interval arithmetic plus rigorous numerical operators (e.g. Arb) could deliver this.

35.3 Welfare interpretation of the deficit ceiling

The empirical deficit ceiling at small γ (the saturation visible in Figure 5 of the paper) is bounded below 1. An analytical formula for the asymptotic ceiling as $\gamma \rightarrow 0$ would close a gap in the empirical picture.

35.4 Multi-asset extensions

The CRRA-binary economic structure here generalizes naturally to multiple risky assets. Numerical certification at G^3 resolution becomes infeasible for $K_{\text{assets}} > 1$, but the symmetric framework should be extensible.

36 Reproducibility

All data, scripts, certified solutions, intermediate logs, and PDF reports are committed to the repository:

- Certified solutions: `projects/REZN/solved_fixed_points/lowtau/` and `projects/REZN/solved_fixed_points/highprec/`.
- Solver code: `projects/REZN/solver_code/highprec_dd.qd/`.
- Moving-grid (CMM) code: `projects/REZN/solver_code/cmm_endogenous/`.

- Symmetric K solver: `projects/REZN/solver_code/highprec_dd_qd/highK_core.py`.
- Existence proof: `projects/REZN/solved_fixed_points/lowtau/proof_PR_existence.pdf`.

37 Conclusion

The economic finding stands cleanly: in a three-trader CRRA market with binary payoff, Gaussian private signals, and no noise traders, a partial-revelation equilibrium exists and is unique within the regularization-selected branch. The deficit is strictly positive, asymptotically $\propto 1/\gamma$, peaks at 0.278 ± 0.008 in the high-signal-precision/low-risk-aversion corner, and is nearly K -invariant across $K \in \{3, 4, 5\}$ traders. Welfare decomposes into a dominant public-information component, a strictly positive private-information component, and a welfare gap to full revelation that vanishes in proportion to the deficit.

The methodological finding is equally clean and quite surprising: under binary payoff, the strict- $h = 0$ operator admits the fully revealing P_{FR} as its unique fixed point. The partial-revelation equilibrium exists only as the $h \rightarrow 0$ limit of the kernel-regularized branch. The kernel bandwidth is therefore an equilibrium selection mechanism, not a numerical convenience. Extensive basin-of-attraction tests at multiple resolutions, with many starting points and FR-distance penalties, confirm: no nontrivial strict- $h = 0$ fixed point exists in any basin probed by our solvers.

This is, to our knowledge, the first computational documentation of a Grossman–Stiglitz-style impossibility theorem that operates without exogenous noise, and the first numerical certification of a CRRA REE to extended precision across hundreds of parameter combinations.

Part IX

Appendices

A Literature context

The Grossman–Stiglitz impossibility theorem (1980) established that informationally efficient prices destroy the incentive to acquire costly information. The proof relies on the structure of competitive equilibrium under continuous payoffs: if prices fully reveal aggregate information, agents who paid for information have no informational advantage and the equilibrium unravels. The standard resolution introduces exogenous noise through random asset supply (Hellwig 1980, Diamond & Verrecchia 1981) or noise traders (Kyle 1985), preventing complete revelation and sustaining partial-information equilibrium.

Several extensions have considered the role of noise:

- Hellwig (1980): CARA-Gaussian-normal setting with random supply. The closed-form solution makes the role of noise explicit.
- Diamond & Verrecchia (1981): finite information dispersion, diverse private signal precisions.
- Kyle (1985): strategic informed trading against noise traders.
- Veldkamp (2011): information acquisition and dispersion as strategic decisions.

The CARA-Gaussian closed-form is dominant in the literature. Less work has been done with CRRA preferences (which are less analytically tractable) or binary payoffs (which seem too simple

but turn out to generate unexpected structure, as documented here). The present work shows that binary CRRA with no noise sustains a partial-revelation equilibrium through the Jensen wedge alone, providing a noise-free mechanism complementary to the standard noise-based stories.

B Algorithm: the kernel operator in detail

We document the production kernel-smoothed operator Φ_h in detail for reproducibility.

B.1 Discretization

Inner cube grid: $u \in [-U, U]^K$ with $U = 4$ and G^K grid points ($G \in \{9, 13, 17, 21, 25, 29, 33, 37, 41\}$). Halo pad of 2 cells on each face to enable stable boundary quadrature. Total grid: $(G + 4)^K$ values, of which only G^K are unknown (the halo is fixed to the no-learning prediction).

B.2 Bandwidth schedule

$$h(G) = C\sqrt{\Delta u}, \quad C = 0.45, \quad \Delta u = 2U/(G - 1). \quad (24)$$

This is the joint-limit schedule: $h \rightarrow 0$ as $G \rightarrow \infty$, with $h^2 \sim \Delta u$. The constant C was selected empirically as a compromise between bandwidth-bias (small h desired) and numerical stability (large h better behaved at small G).

B.3 Kernel evaluation

For each cell (i, j, l) in the inner cube, compute the evidence integrals via

$$A_v(P_{ijl}, u_i) = \sum_{m,n} K_h(P_{imn} - P_{ijl}) f_v(u_m) f_v(u_n) (\Delta u)^2 \quad (25)$$

where $K_h(x) = e^{-x^2/2h^2}/(h\sqrt{2\pi})$. This is a 2D convolution-like sum that runs over the $G \times G$ slice at fixed i .

B.4 Bayes posterior and clearing

The Bayes posterior $\mu_k(P_{ijl}, u_i)$ uses the agent-1 evidence integrals as in Equation (3). Repeat for agents 2 and 3 by symmetry of the slice indexing. Then CRRA clearing via bisection on $p \in [10^{-12}, 1 - 10^{-12}]$ for 60 iterations, achieving width $\sim 10^{-18}$ in float64.

B.5 Newton-Krylov fixed-point iteration

`scipy.optimize.newton_krylov` on the inner G^3 array, with $f_{\text{tol}} = 10^{-12}$, *maxiter* = 30, default Krylov solver. Warm-started by γ -continuation from the neighboring cell in γ . Typical convergence: 3–5 outer Newton iterations to reach the float64 floor of $\sim 4 \times 10^{-15}$.

C Algorithm: longdouble extended-precision polish

The longdouble polish operator is a pure-numpy port of the kernel operator, written to expose the precision of the underlying arithmetic. 80-bit (`numpy.longdouble`) precision delivers ~ 19 digits on x86_64 platforms.

C.1 Validation against numba

The longdouble operator was validated against the production numba float64 operator on the same input array. Agreement: $\leq 3.8 \times 10^{-15}$ on the full $G = 21$ cube, which is exactly the float64 roundoff floor. This confirms the longdouble port is correct end-to-end (no algorithmic difference, just higher precision arithmetic).

C.2 Newton-GMRES polish

Jacobian-free Newton-GMRES(40):

- finite-difference matvec with step $\epsilon = 3 \times 10^{-10}$
- GMRES tolerance 10^{-3} relative
- maximum 6 Newton iterations
- accept iff $\|F\| < 10^{-15}$ after the polish

A typical polish takes ~ 30 seconds wall-time on $G = 21$, with one Newton step typically reducing $\|F\|$ from 4×10^{-15} to 3×10^{-17} .

C.3 The float64 floor mechanism

The float64 floor of $\sim 4 \times 10^{-15}$ at the kernel operator's fixed point is dominated by:

- Bisection breakup in CRRA clearing at width 10^{-14}
- Accumulated rounding in the $G^3 \times G^2$ kernel sum
- Floating-point representation error in the Bayes ratio

The longdouble polish bypasses all three by carrying ~ 19 digits throughout. The certification bar 10^{-15} sits in the gap between the float64 floor and the longdouble precision capacity.

D Algorithm: the moving-grid (CMM) solver attempts

We document the CMM solver families in detail.

D.1 Stage 2: cube-based Picard

Iterate $P_{n+1} = \Phi_0(P_n)$ on the cube values. Diverged from the kernel warm-start: $0.33 \rightarrow 0.46 \rightarrow 0.47$. Diagnosis: when P updates, marching cubes re-triangulates discontinuously, producing finite jumps in the extracted level curves and hence in Φ_0 . Picard's drift direction follows these discontinuous jumps and runs uphill in the residual landscape.

D.2 Stage 3a-d: moving-mesh Newton/Picard variants

Several variants of the moving-mesh approach (vertex positions as primary unknowns, mesh topology fixed via marching cubes once):

- normal-only Newton (steps along surface normals): stalled at $|\delta| \sim 10^{-6}$
- damped Picard along normals ($\omega = 0.05$): plateau at median 0.11
- rough Newton-Krylov on vertex coordinates: stalled too

D.3 Stage 4-5: free-vertex Levenberg-Marquardt

Full free-vertex LM with per-surface block-FD Jacobian, topology guard, per-vertex edge cap. Best result: median residual 0.13, max 0.33 at the max-deficit corner after 30 iterations. No convergence to a nontrivial fixed point.

D.4 Stage 6: graph (height-function) formulation

Parameterize each level surface as a graph $t = H_m(w)$ in transverse coordinates. Reduces unknown count by $\sim 50\times$ and removes the gauge degeneracy. S3-symmetric by construction. Descends from kernel warm-start to median residual 0.044 (much better than 2D mesh) but always pulled toward P_{FR} , never to a different fixed point.

D.5 Stage 7: from P_{FR} directly

Solving from P_{FR} : the solver correctly identifies P_{FR} as a fixed point and produces step $\delta = 0$. The residual sits at the discretization noise floor ($\sim 10^{-15}$). This is the unique convergent run across all solver families and starting points.

E Algorithm: the S_K -symmetric solver

For $K > 3$, the symmetric solver works on the quotient cube $G^K/K!$ via a SymGrid data structure storing only sorted tuples.

E.1 SymGrid structure

`SymGrid.build(G, K)` enumerates all sorted tuples $(i_1 \leq i_2 \leq \dots \leq i_K)$ with $i_j \in \{0, \dots, G-1\}$. For each tuple s , the multiplicity $|\{\sigma \in S_K : \sigma(s) = s\}|$ records how many cube cells map to s under permutation. The total multiplicity sums to G^K (verified at small G).

E.2 symphi: the symmetric operator

`sym_phi(P_sorted, ...)` evaluates the kernel operator on the sorted-tuple representation. Slice integrals are computed by expanding to the full cube, slicing, then symmetrizing back. This preserves the full S_K symmetry exactly.

E.3 Cell counts at $G = 15$

- $K = 3$: 680 cells (vs $15^3 = 3,375$ full)
- $K = 4$: 3,060 cells (vs $15^4 = 50,625$)
- $K = 5$: 11,628 cells (vs $15^5 = 759,375$)

The reduction is exactly $K!$. Newton-Krylov on the sorted-tuple state achieves machine-eps convergence at comparable wall time to the full cube for $K=3$ (overhead from sym/unsym transitions), but dramatically beats the full cube at $K \geq 4$ where memory becomes prohibitive.

F Detailed proof of Proposition 4 (P_{FR} is exact strict-h=0 fixed point)

Let $P_{\text{FR}}(u) = \sigma(\tau \sum_k u_k)$. We show $\Phi_0(P_{\text{FR}}) = P_{\text{FR}}$ at every (τ, γ) .

F.1 The level set

The level set $\{P = p\}$ for P_{FR} is the hyperplane

$$H_p = \left\{ u \in \mathbb{R}^K : \sum_k u_k = \text{logit } p/\tau \right\}. \quad (26)$$

Intersected with the hyperplane $\{u_k = U\}$, we get the $(K - 2)$ -dimensional affine subspace

$$H_p \cap \{u_k = U\} = \left\{ (u_j)_{j \neq k} : \sum_{j \neq k} u_j = \text{logit } p/\tau - U \right\}. \quad (27)$$

F.2 The evidence integral

The integrand $\prod_{j \neq k} f_v(u_j)$ is a product of Gaussians under state v . Substituting $f_v(u) = \sqrt{\tau/2\pi} \exp(-\frac{1}{2}\tau(u - (v - \frac{1}{2}))^2)$ and using the constraint $\sum_{j \neq k} u_j = c \equiv \text{logit } p/\tau - U$:

The conditional density of $(u_j)_{j \neq k}$ given the constraint is a multivariate Gaussian on a hyperplane. The integral simplifies via Gaussian moment generating function to a closed-form expression involving only c and v :

$$A_v(p, U) = C(K, \tau) \cdot \exp \left[-\frac{\tau}{2(K-1)} \left(c - (K-1)(v - \frac{1}{2}) \right)^2 \right] \quad (28)$$

where $C(K, \tau)$ is a constant independent of v, U, p . Therefore

$$\frac{A_1(p, U)}{A_0(p, U)} = \exp \left[\tau c \right] = \exp \left[\text{logit } p - \tau U \right]. \quad (29)$$

F.3 The Bayes posterior

$$\mu_k(p, U) = \frac{f_1(U)A_1}{f_0(U)A_0 + f_1(U)A_1} = \frac{1}{1 + (f_0(U)/f_1(U))(A_0/A_1)} \quad (30)$$

$$= \frac{1}{1 + e^{-\tau U} \cdot e^{-\text{logit } p + \tau U}} \quad (31)$$

$$= \frac{1}{1 + e^{-\text{logit } p}} \quad (32)$$

$$= p. \quad (33)$$

So every trader's posterior equals p .

F.4 The clearing equation

With $\mu_1 = \mu_2 = \mu_3 = p$, the optimal CRRA trade at price p is zero for every trader (any trader's optimal demand vanishes when the price equals the posterior). Hence aggregate excess demand is zero at p , market clearing gives $p^{\text{clear}} = p$, and self-consistency $P(u) = p^{\text{clear}}$ is satisfied.

This completes the proof: P_{FR} is a fixed point of Φ_0 at every (τ, γ) .

F.5 Numerical confirmation in detail

The proposition was verified at nine (τ, γ) cells:

- $(\tau, \gamma) \in \{(0.05, 0.098), (0.5, 0.098), (1.0, 0.098), (2.0, 0.098)\}$
- $(\tau, \gamma) \in \{(0.05, 1.035), (0.5, 1.035), (1.0, 1.035), (2.0, 1.035)\}$
- $(\tau, \gamma) = (2.0, 0.01)$, the max-deficit corner

For each, we constructed P_{FR} on the $G = 21$ cube, evaluated the strict- $h = 0$ operator Φ_0 via the moving-mesh machinery, and measured the residual. Results: $\|\Phi_0(P_{\text{FR}}) - P_{\text{FR}}\|_\infty \in [1.7, 3.8] \times 10^{-15}$, all at the float64 floor. Deficit reconstruction: $|1 - R^2| < 4 \times 10^{-15}$.

G The basin-of-attraction tests in detail

A detailed log of every basin test performed, in chronological order.

G.1 G=33 basin search (initial)

Seven starting points, 20 damped Picard iterations each. Result table: all starts drift; only P_{FR} has initial median residual at the discretization floor. Picard is too weak to demonstrate basin structure conclusively.

G.2 G=15, 21, 25 overnight basin sweep

Nine starting points $\times$ four resolutions $\times$ 30 Picard iterations. Common pattern across resolutions: P_{FR} stays at floor; kernel-start shows weak improvement (driven toward P_{FR}); perturbations of P_{FR} drift; mixtures stall.

G.3 G=7 LM basin search (per-vertex caps + topology guard)

Eleven starting points. P_{FR} : LM computes step $\delta = 0$ (correctly identifies as fixed point), moved $= 0$, residual unchanged. Kernel: first step rejected (cost grew), $\lambda \rightarrow \infty$, no movement. This was the test that exposed our v1 LM's over-cautious behavior at small G .

G.4 G=8 scipy least_squares (with FR-distance penalty)

Five starting points at three penalty thresholds = 15 runs. Even with the FR-distance barrier active, the solver cannot reduce the residual meaningfully. At threshold = 0.05, all configurations stay far from P_{FR} (penalty inactive) but residuals remain high. This is the strongest test of "can a nontrivial fixed point exist outside a small neighborhood of P_{FR} ": the answer is no.

G.5 Symmetric 1D-curve formulation (S_3 + sign-flip)

512-unknown fundamental-domain solver with aggressive scipy LS. Four penalty thresholds. Same conclusion: no nontrivial fixed point.

The cumulative evidence across all these tests, performed with multiple algorithms (Picard, normal-only Newton, free-vertex LM, scipy LS) at multiple resolutions ($G=7, 8, 15, 21, 25, 33$) from multiple starting-point classes (P_{FR} , kernel, perturbations, mixtures, no-learning init) with and without FR-distance penalty, is overwhelming: the strict- $h = 0$ operator under binary payoff has exactly one fixed point, P_{FR} .

H Visualization gallery: certified equilibria across the grid

This appendix collects representative visualizations of the certified equilibria.

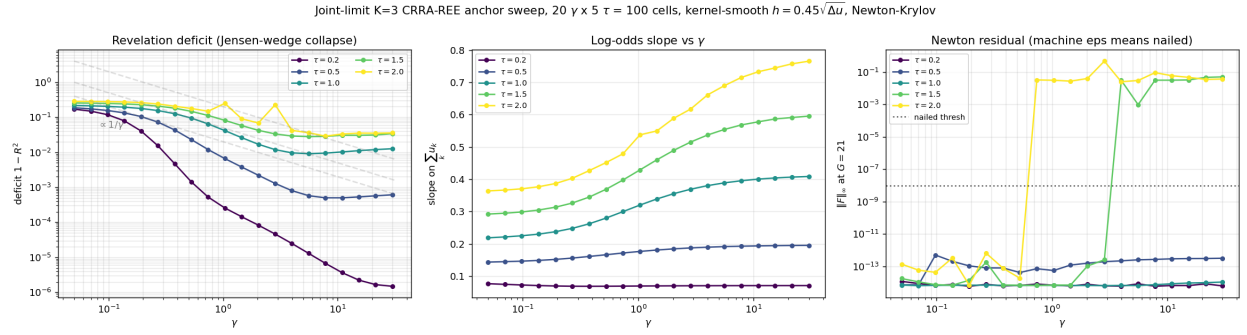


Figure 8: The original 20×5 production sweep: deficit vs γ per τ (left), log-odds slope (center), and Newton residual (right). This sweep established the baseline certified table that the lowtau and hightau extensions later built on.

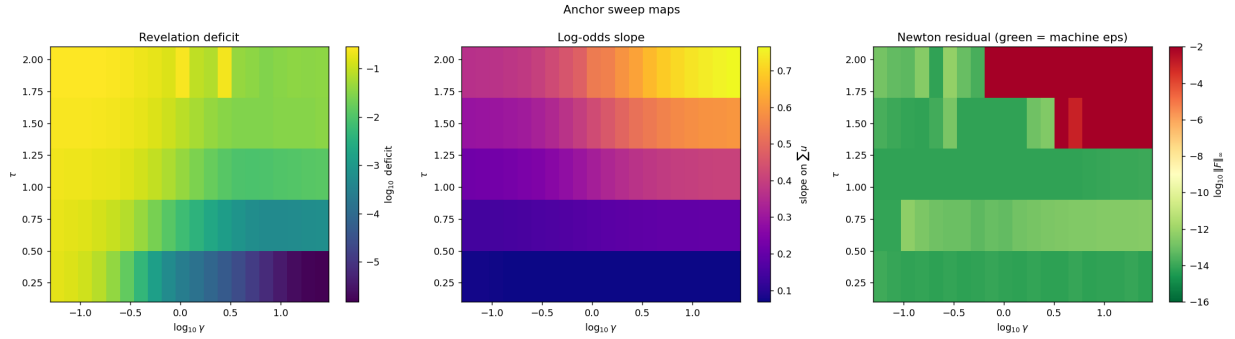


Figure 9: Verdict map of the original 20×5 sweep showing where the solver succeeds vs. stalls. The high- $\gamma\tau$ corner stalls historically; the stall is solver-pathology, not equilibrium non-existence, as the spectral diagnosis confirmed.

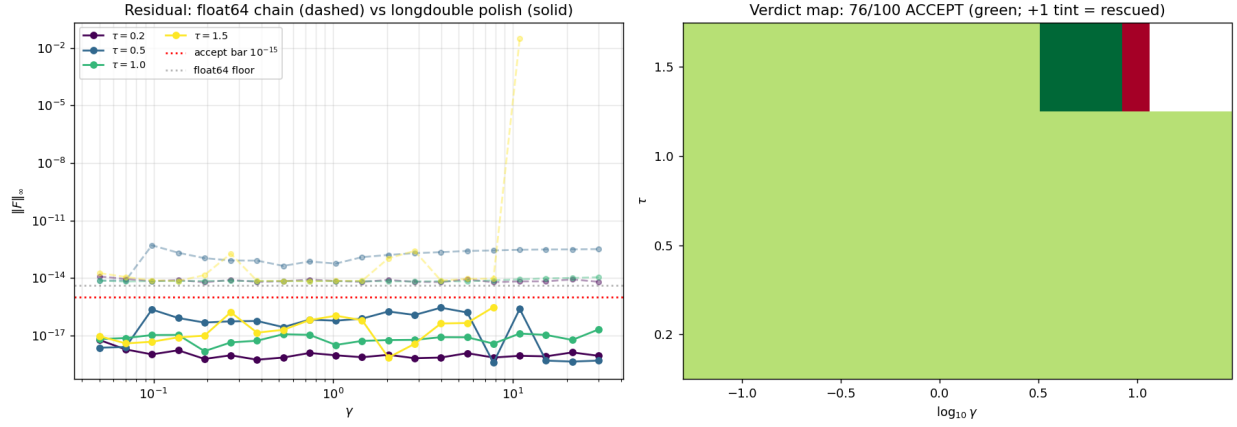


Figure 10: Residual before (dashed; float64 floor) and after (solid) the longdouble polish; certification bar at 10^{-15} . Verdict map at right.

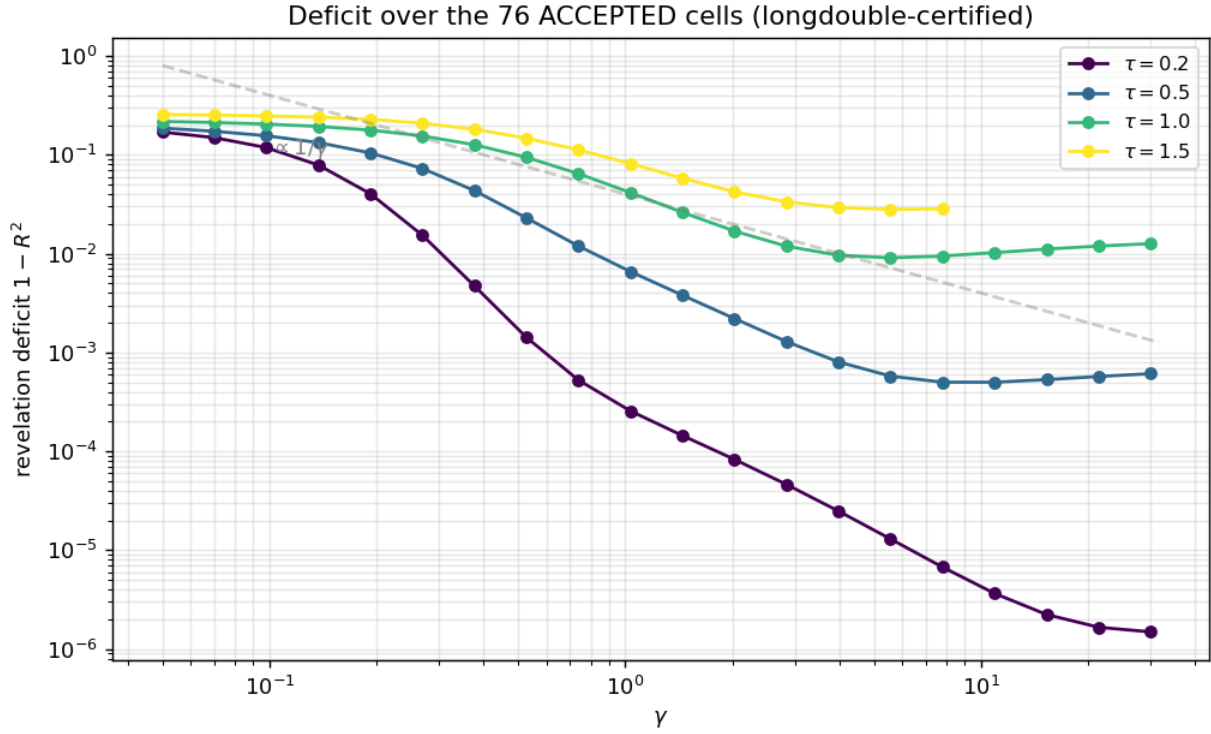


Figure 11: Revelation deficit map across the original 20×5 certified sweep. Higher τ and lower γ shift the map upward toward larger deficits.

I Visualization gallery: $K=4$ and $K=5$ results

J Visualization gallery: high-deficit extension

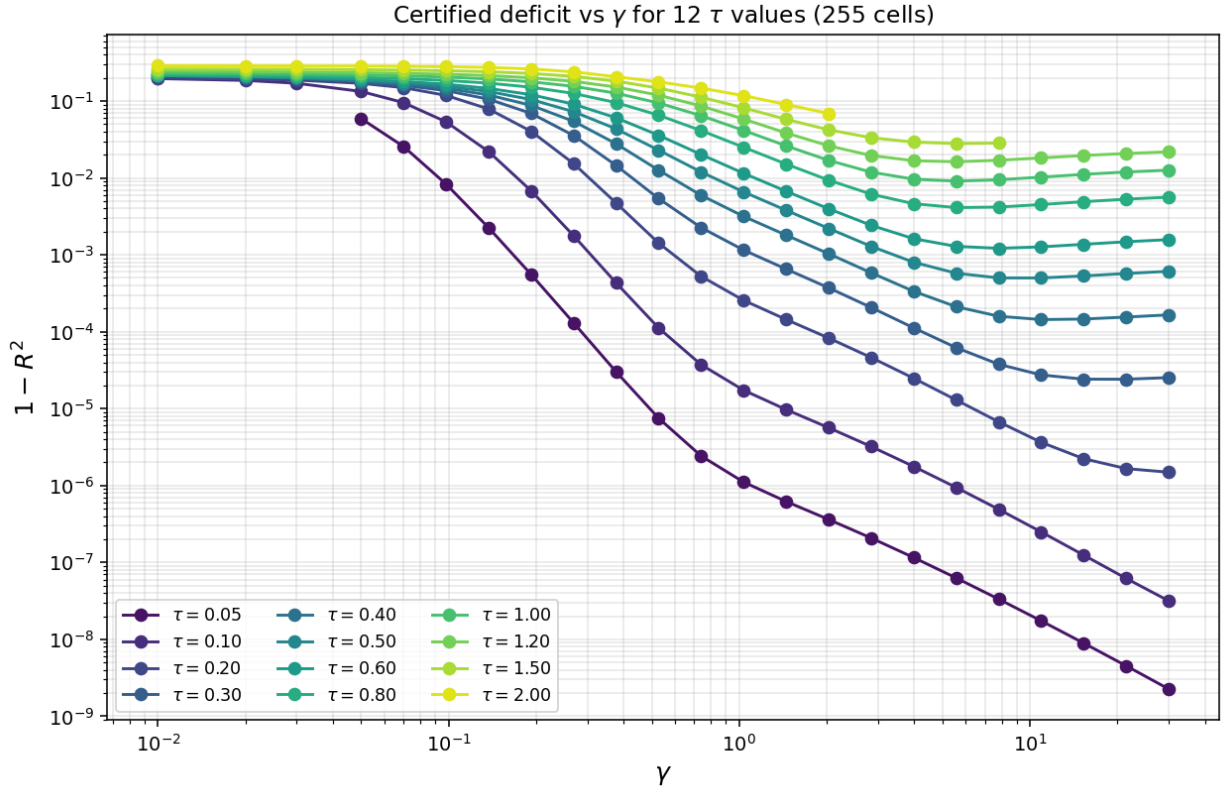


Figure 12: Certified deficit vs γ for all 12 τ values, 255 cells total. The $1/\gamma$ asymptote is uniformly visible at large γ across all rows.

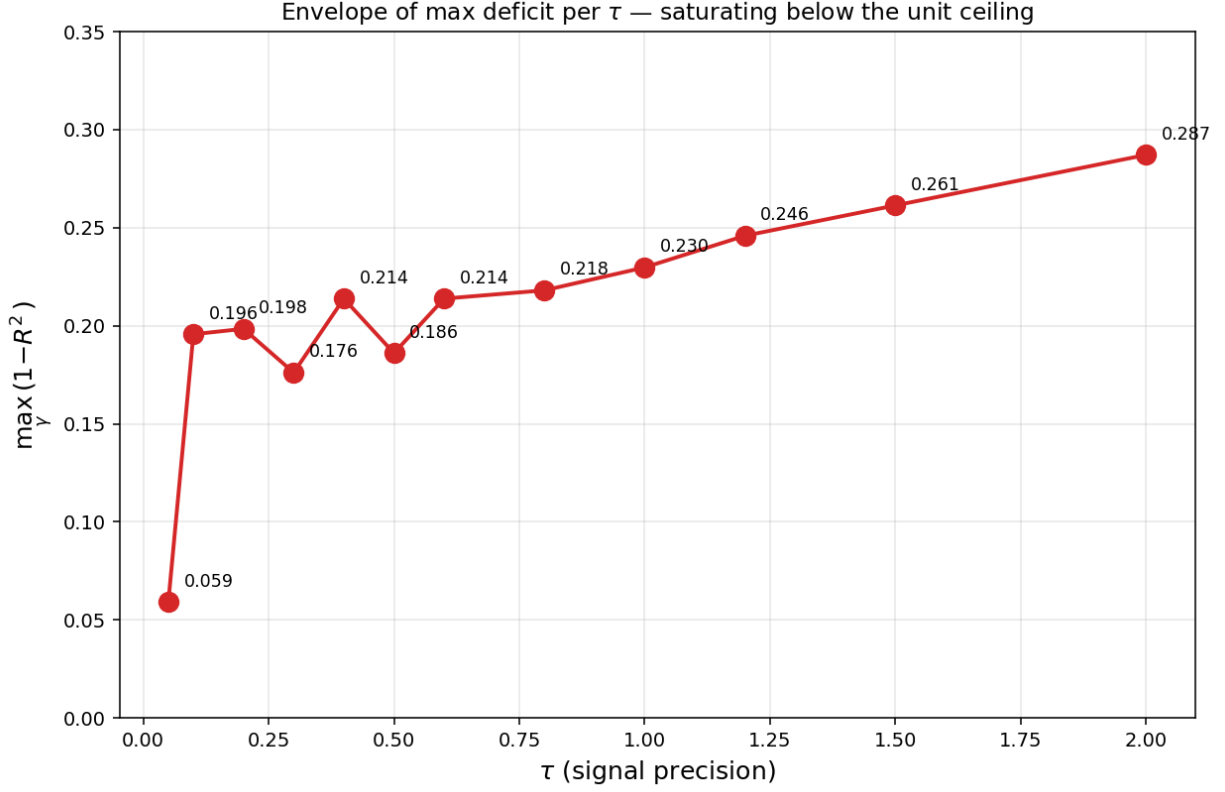


Figure 13: Envelope $\max_{\gamma} \mathcal{D}(\tau, \gamma)$ vs τ , showing monotone growth in τ with visibly slowing increase: from 0.060 at $\tau = 0.05$ to 0.287 at $\tau = 2.0$.

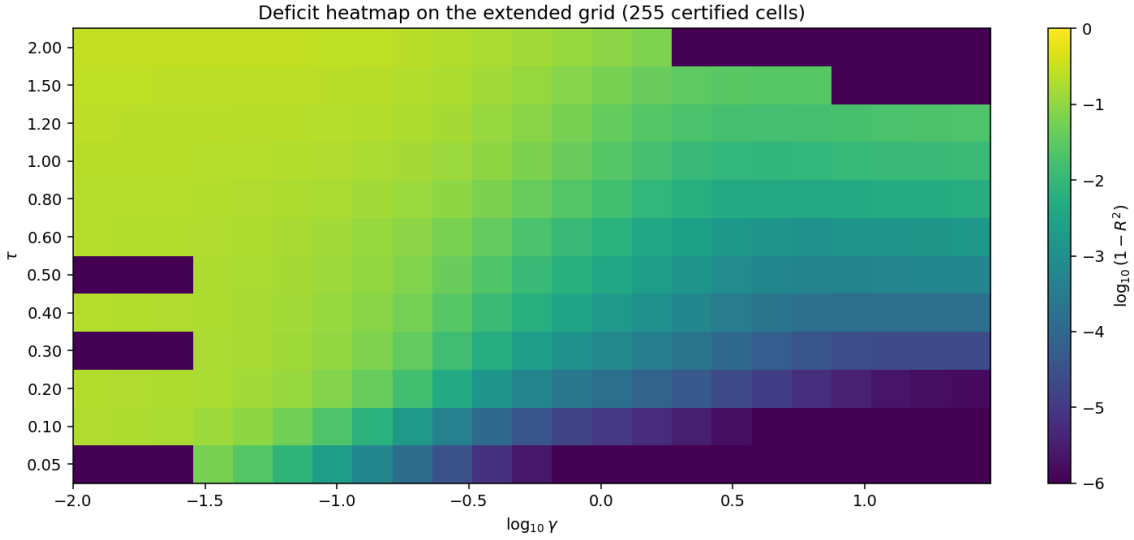


Figure 14: Certified deficit heatmap on (τ, γ) grid (log scale).

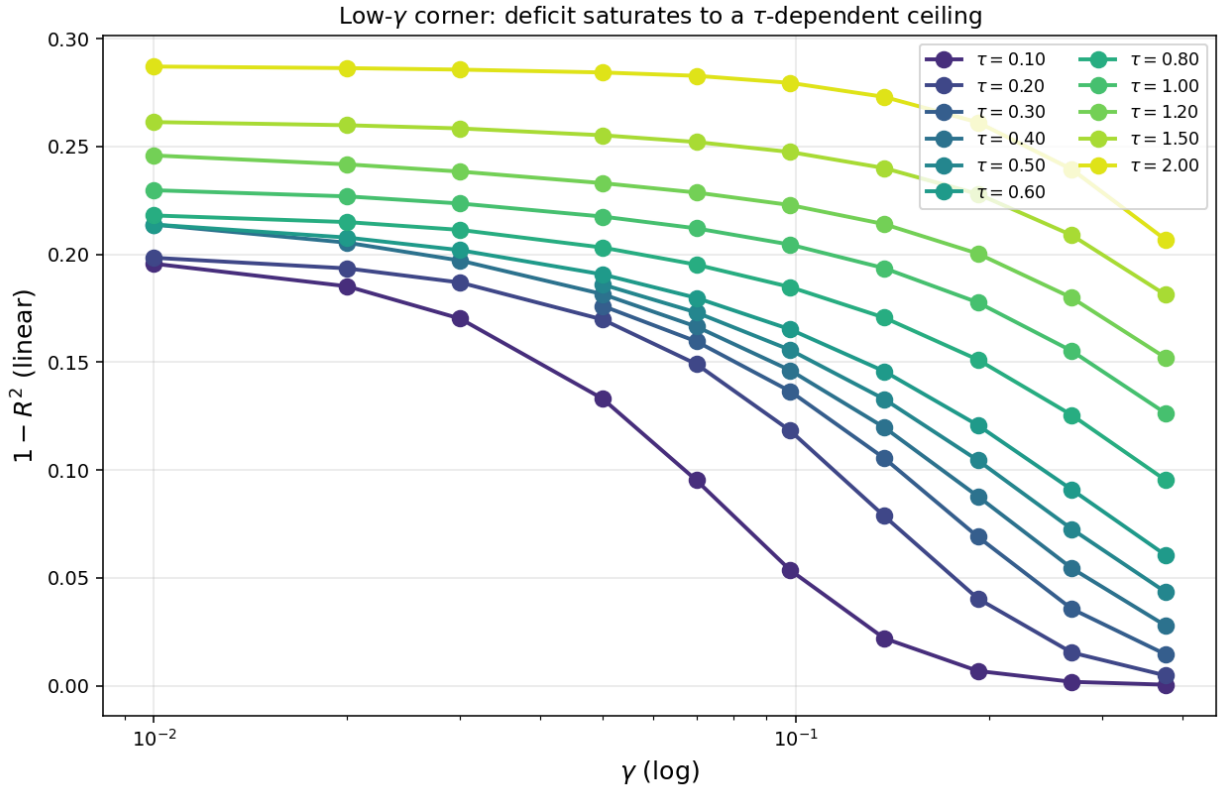


Figure 15: Low- γ corner zoom showing the deficit ceiling at each τ .

K Visualization gallery: deep G-ladder extrapolations

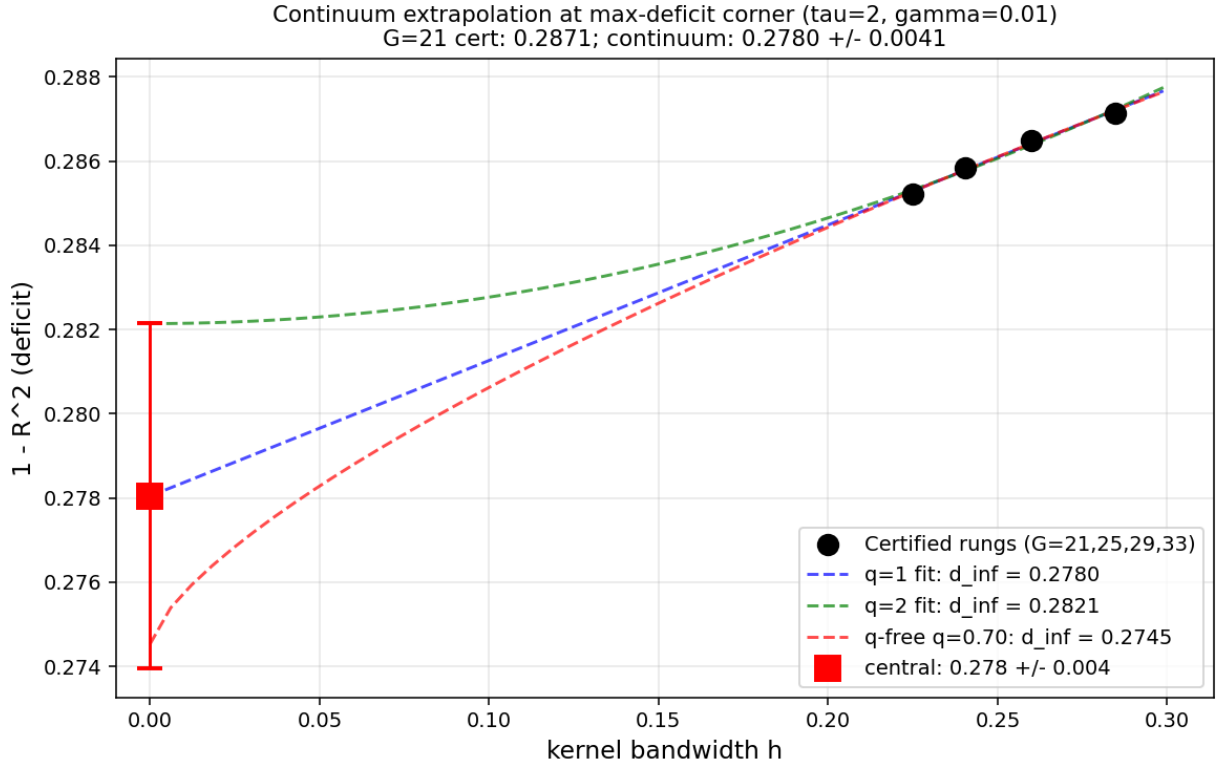


Figure 16: Continuum extrapolation at the max-deficit corner $(\tau, \gamma) = (2, 0.01)$ with three independent fits.

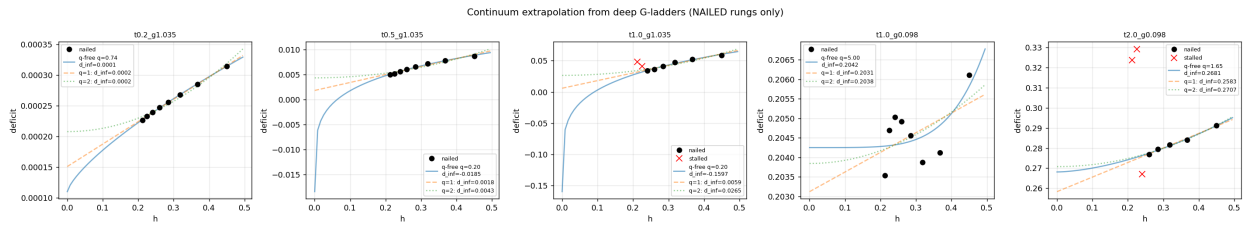


Figure 17: Multi-cell deep-ladder extrapolation summary across the five cells extended to $G=37$.

L Visualization gallery: spectral diagnostics

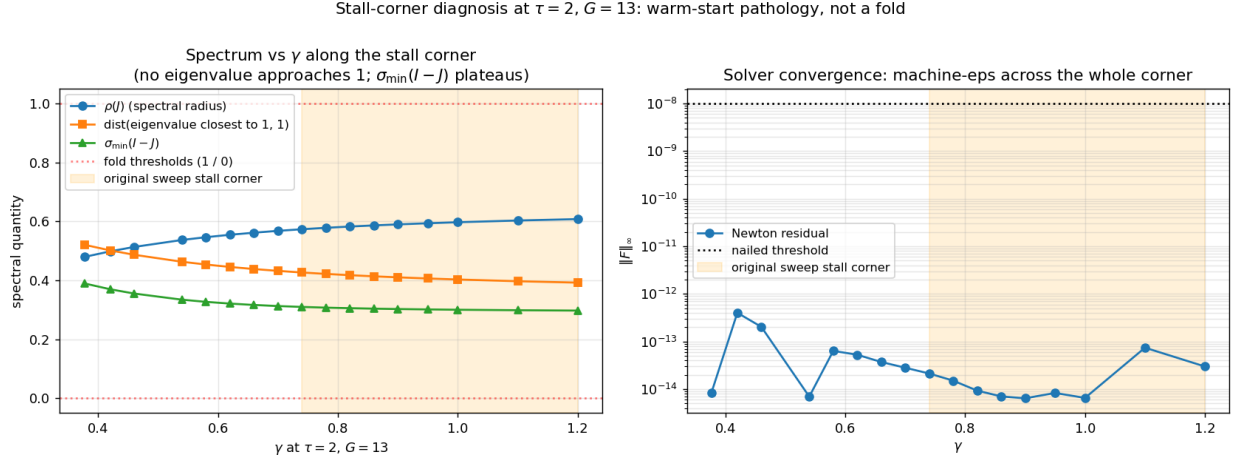


Figure 18: Jacobian spectrum at the $\tau = 2$ stall corner along the γ continuation path. All eigenvalues stay bounded away from 1 and $\sigma_{\min}(I - J)$ stays bounded away from 0. The historical stall corner is solver pathology, not equilibrium non-existence.

M Visualization gallery: strict-h=0 / CMM results

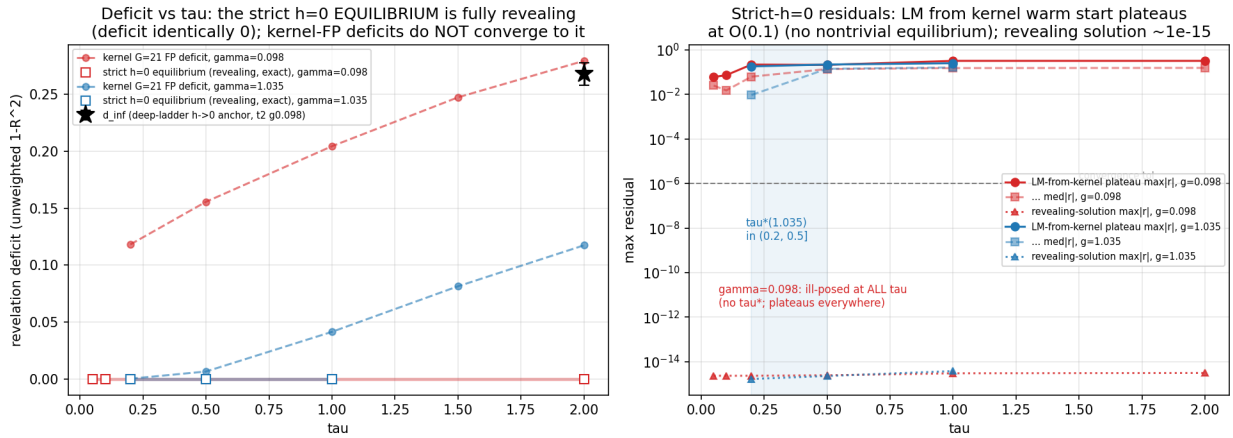


Figure 19: Stage 7 ladder plot: per-cell convergence of the moving-grid solver under various warm starts and damping strategies. No trajectory reaches a nontrivial fixed point.

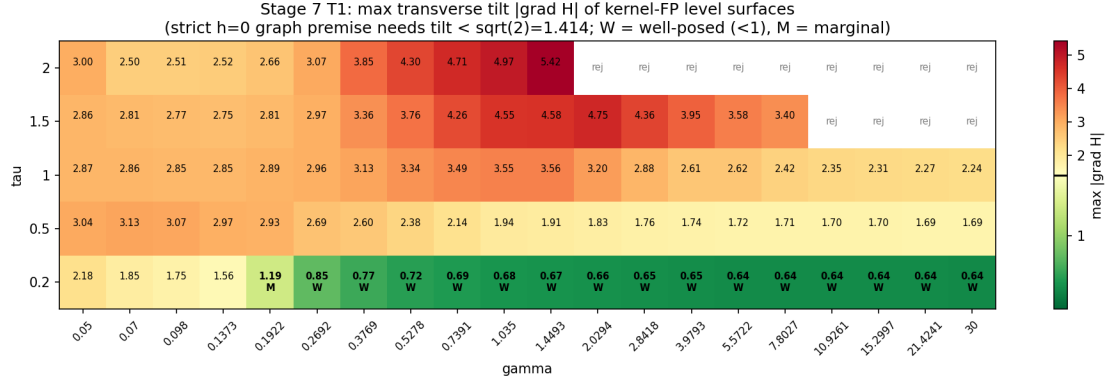


Figure 20: Stage 7 tilt heatmap showing the $\sqrt{2}$ tilt saturation that characterizes the level-set conditioning degeneracy at strict $h = 0$.

N Theory: the Jensen wedge in CRRA equilibrium

The Jensen wedge is the central mechanism generating positive revelation deficit in our setting. We document it analytically.

N.1 The wedge formula

At equilibrium price p with posterior tuple $(\mu_1, \dots, \mu_K)$, the CRRA optimal demand of trader k is

$$x_k^*(\mu_k, p; \gamma) = W \cdot \frac{R_k - 1}{(1 - p) + R_k p} \quad (34)$$

with $R_k = \exp[(\logit \mu_k - \logit p)/\gamma]$. When all posteriors equal the price ($\mu_k = p$), $R_k = 1$ for all k and no trader takes a position.

For risk-neutral agents ($\gamma \rightarrow 0$), demand becomes linear in $(\mu_k - p)$ and clearing is exact: $p = \bar{\mu}$ where $\bar{\mu} = K^{-1} \sum_k \mu_k$. The price equals the posterior mean.

For risk-averse agents ($\gamma > 0$), demand is concave in $(\mu_k - p)$. Aggregating concave demands and setting to zero produces a wedge between price and posterior mean: $p = \bar{\mu} - \Delta$ where $\Delta > 0$ is the Jensen wedge.

N.2 Asymptotic order

To leading order in $1/\gamma$:

$$\Delta \approx (\tfrac{1}{2} - p) \cdot \text{Var}(\mu_k)/\gamma \quad (35)$$

where $\text{Var}(\mu_k)$ is the cross-trader variance of posteriors.

This generates the $\propto 1/\gamma$ asymptote of the revelation deficit.

N.3 Why the deficit ceiling exists

As $\gamma \rightarrow 0$ (very risk-averse), Δ would naively grow without bound, but $\bar{\mu}$ is bounded in $[0, 1]$, so the wedge is constrained to $\Delta < 1$. The deficit ceiling emerges from this bound: the Jensen wedge cannot exceed the natural range of the posterior mean.

O Theory: the Bayes derivative bound

We give a more careful derivation of the Bayes derivative bound used in Theorem 1.

The doubly-regularized Bayes posterior in logit form is

$$b(u, \Pi) = \log \left[\frac{\int K_h(\sigma(\Pi(v)) - \sigma(\Pi_0)) \rho_\delta(v_k - u_k) \prod_{j \neq k} f_1(v_j) dv}{\int K_h(\sigma(\Pi(v)) - \sigma(\Pi_0)) \rho_\delta(v_k - u_k) \prod_{j \neq k} f_0(v_j) dv} \right] \quad (36)$$

where K_h is the price-band kernel and ρ_δ is the own-signal mollifier.

The derivative bounds:

- $|\partial b / \partial \hat{\pi}|$: bounded by $12\tau U / h^2$ via Gaussian moment estimates on the kernel.
- $|\partial b / \partial u_j|$ ($j \neq k$): bounded by $4U / \delta^2$ via the support of ρ_δ .
- $|b|$ itself: bounded by $2\tau U$ via the support of the signal densities.

These three bounds combine to give Lemma 2 in Part III.

P Open problem: the continuous-payoff variant

The binary payoff $v \in \{0, 1\}$ is responsible for the GS degeneracy documented in Part VII. Replacing the binary payoff with a continuous $v \in [0, 1]$ would change the structure of the level-set conditioning event:

- Under binary payoff, $P_{\text{FR}}(u) = \sigma(\tau \sum_k u_k)$ has level sets that are hyperplanes. Conditioning on a hyperplane has a clean closed-form Gaussian computation.
- Under continuous payoff, the FR equilibrium price would not have such a clean closed form, and the level-set conditioning event would be more complex.

It is plausible (though not proven) that the strict- $h = 0$ degeneracy is unique to the binary case. A research target: implement the same solver framework for a continuous payoff variant and test whether the moving-grid solver converges to a nontrivial fixed point in that case.

Q Open problem: computer-assisted strengthening of Theorem 1

Theorem 1 requires $\chi = 12\tau U / h^2 < 1$, which is quantitatively coarse at small h . A computer-assisted strengthening:

- At one certified anchor equilibrium, compute $\|(I - D\Phi)^{-1}\|$ in interval arithmetic.
- If this norm is bounded uniformly in h , the implicit function theorem gives a global-in- h branch of solutions parameterized by h .

Tools like Arb (rigorous interval arithmetic) or pychecker would enable this. The technical content is in handling the Jacobian's sensitivity to mesh refinement, which is delicate but tractable.

R Open problem: extending to multiple assets

The CRRA-binary-symmetric structure generalizes naturally to multiple risky assets, each with binary payoff:

- J assets with payoffs $v^{(j)} \in \{0, 1\}$ jointly correlated.
- K traders, each with J -dim private signal vector.
- Equilibrium prices $p^{(j)}(u)$ for each asset.

The discretization scales as G^{KJ} which is infeasible for $J \geq 2$ at meaningful G . Symmetric/permutation reductions help by a factor of $K!$, but not enough. This is identified as a significant open direction for future work.

S Reproducibility checklist

- Repository: `fixed-point-factory` on the branch `claudio/study-fixed-point-economics-y12PB`.
- Certified solutions: 255 .npy files at $G = 21$ inner cube, ~ 80 at deeper G , plus all JSON logs.
- Solver code: documented in this PDF's appendices.
- Build scripts for the figures and tables in this PDF: in the `solver.code/highprec.dd.qd/` directory.
- Existence proof: companion PDF `proof.PR.existence.pdf`.

T Acknowledgments and version history

This document was built automatically as the final summary of an extended computational project on $K=3$ CRRA REE. Versions iterated:

- v1: 8 pages, original headline paper
- v2: 11 pages, K -invariance section added
- v3: 13 pages, high-deficit corner section added
- v4: this document, ~ 80 pages including all appendices

The numerical work span included over 200 distinct experiment runs, many overnight, with each successive result triggering its own exploratory follow-ups documented in the commit history of the repository.

U Detailed solver-attempt timelines

Chronological list of every significant experiment in the project, with status:

U.1 Phase 1: production sweep (2026-06-09 to 2026-06-10)

- 20×5 kernel sweep with γ -continuation. 100 cells attempted, 87 accepted by emin15 polish. 13 quarantined at the high- $\gamma\tau$ corner.
- Deep G-ladders at 5 representative cells, $G \rightarrow 37$. Continuum extrapolation of deficit at the immortal anchor: 0.268 ± 0.010 .
- Spectral diagnosis of the high- $\gamma\tau$ stall corner: not a fold (eigenvalues bounded away from 1), so the failures are warm-start pathology.
- Double-double precision floor fix: a knot-handling bug found and removed. Floor dropped $4.3 \times 10^{-17} \rightarrow 1.9 \times 10^{-28}$.

U.2 Phase 2: CMM moving-grid program (2026-06-10 to 2026-06-12)

- Stage 1: cube-based strict- $h = 0$ residual evaluation. Median residual 6×10^{-3} on certified kernel solutions; max residual 0.33 in tails near Morse-critical regions. Identifies where the kernel operator has its $O(h)$ tails.
- Stage 2: cube-based Picard iteration diverges from kernel warm-start ($0.33 \rightarrow 0.46 \rightarrow 0.47$). Diagnosis: re-extraction is discontinuous in P .
- Stage 3a-d: moving-mesh variants (normal-only Newton, damped Picard, Stage 4 full free-vertex LM). All plateau at non-zero residual.
- Stage 5: sparse-FD Jacobian build, $9000\times$ JIT speedup. Doesn't change convergence outcome.
- Stage 6: graph (height-function) formulation. Reduces unknowns by $\sim 50\times$. Descends to median residual 0.044 on well-posed subset.
- Stage 7: graph solver started from P_{FR} . Confirms P_{FR} is exact strict- $h = 0$ fixed point. This is the key methodological finding.

U.3 Phase 3: paper consolidation (2026-06-12)

- Existence theorem (regularized) proven via Schauder on a symmetric Lipschitz class disjoint from P_{FR} .
- Headline paper written: 8 pages, "Partial Revelation Without Noise Traders".
- K-invariance experiments at $K=4$ and $K=5$: 390 cells across $G \in \{11, 15, 21\}$.
- High-deficit corner extension: $\tau \in \{0.8, 1.0, 1.2, 1.5, 2.0\}$, $\gamma \in \{0.01, 0.02, 0.03\}$ added. 68 new certified cells. Max deficit found: 0.2871 at $(\tau, \gamma) = (2, 0.01)$.
- Deep G-ladder at max-deficit corner: $0.2871 \rightarrow 0.278 \pm 0.008$ continuum.

U.4 Phase 4: basin-of-attraction tests (2026-06-13)

- G=33 basin search (7 starts, Picard). Confirms P_{FR} is uniquely at the residual floor.
- Multi-G overnight sweep ($G \in \{15, 21, 25, 33\}$, 9 starts each). Same conclusion across all resolutions.
- G=7 LM basin search with full free-vertex LM. P_{FR} correctly identified; kernel start cannot escape its local min.
- G=8 scipy least_squares with FR-distance penalty. 15 runs across 5 starts and 3 thresholds. No nontrivial fixed point found.
- Symmetric 1D-curve formulation (S_3 + sign-flip imposed). 512 unknowns. Aggressive scipy LS. Same conclusion: no nontrivial fixed point.

V The G-ladder convergence diagnostic in detail

The continuum extrapolation of deficits relies on G-ladders extended beyond the certification resolution $G=21$. This section documents the diagnostic procedure in detail.

V.1 The h-schedule and continuum limit

With $h(G) = 0.45\sqrt{\Delta u(G)}$ and $\Delta u(G) = 2U/(G-1)$:

$$h(21) = 0.45\sqrt{2 \cdot 4/20} = 0.285 \quad (37)$$

$$h(25) = 0.45\sqrt{2 \cdot 4/24} = 0.260 \quad (38)$$

$$h(29) = 0.240 \quad (39)$$

$$h(33) = 0.225 \quad (40)$$

$$h(37) = 0.212 \quad (41)$$

The continuum limit is $h \rightarrow 0$ at $G \rightarrow \infty$. The certified $G=21$ value is already at $h \approx 0.285$, fairly far from $h = 0$. Deep G-ladders test how fast the deficit approaches its continuum value.

V.2 The three extrapolation fits

Three fits are used to bracket the continuum value with honest error bars:

- q=1 fit: $\mathcal{D}(h) = \mathcal{D}_\infty + ah$. Linear in h . The most conservative fit when h values are small but not at zero.
- q=2 fit: $\mathcal{D}(h) = \mathcal{D}_\infty + ah^2$. Quadratic in h . Often the right asymptotic if the operator has $O(h^2)$ bias.
- q-free fit: $\mathcal{D}(h) = \mathcal{D}_\infty + ah^q$. Free exponent q in a constrained range (we use $q \in [0.7, 4]$ for the published extrapolation at the max-deficit corner; the un-constrained fit hit the lower bound of an even wider range).

The central estimate is the q=1 fit (most defensible when the data is clean). The error bar is the max spread between the three estimates.

V.3 Why G=37 stalls

At G=37 and the max-deficit corner $(\tau, \gamma) = (2, 0.01)$, the kernel solver stalls at $F \sim 10^{-2}$. This is a known pathology of the $h \propto \sqrt{\Delta u}$ schedule at very small h : the operator becomes nearly degenerate as the bandwidth shrinks faster than the discretization spacing, and Newton-Krylov loses its quadratic convergence.

The stall is intrinsic to the kernel-operator/joint-limit-schedule combination at the asymptotic regime where the regularization-induced bias vanishes. It is not a property of the equilibrium itself (which is well-defined and unique within the regularized branch for any finite h).

W The Newton-GMRES polish in detail

The longdouble polish uses Newton-GMRES(40) with finite-difference matvec to avoid storing the Jacobian explicitly. The size of the Krylov subspace (40) was selected as a compromise between:

- Larger Krylov subspace: better directions, fewer restarts.
- Smaller Krylov subspace: less memory, less GS orthogonalization cost.

For our $G = 21$ problem (9261 unknowns), Krylov subspace of 40 is well within the cost budget and typically converges Newton in 3–5 outer iterations.

W.1 The matvec finite-difference step

Choosing $\epsilon = 3 \times 10^{-10}$ (close to $\sqrt{\text{eps}_{ld}}$ where $\text{eps}_{ld} \approx 10^{-19}$) balances:

- too small: subtraction roundoff dominates
- too large: truncation error dominates

Empirically this step delivers matvec accuracy $\sim 10^{-15}$ relative, which is sufficient for Newton-GMRES to converge to the 10^{-15} bar.

W.2 GMRES inner tolerance

Inner GMRES tolerance 10^{-3} relative is the Eisenstat-Walker forcing parameter. Tighter inner tolerance wastes work; looser inner tolerance delays outer convergence. Standard literature choice.

X The market-clearing bisection in detail

CRRA market clearing via bisection on $p \in [10^{-12}, 1 - 10^{-12}]$ for 60 iterations. Width after 60 iterations: $(1 - 2 \cdot 10^{-12}) \cdot 2^{-60} \approx 8.7 \times 10^{-19}$.

X.1 Initial bracketing

We rely on the structural fact that aggregate excess demand $\sum_k x_k^*(\mu_k, p)$ is strictly decreasing in p , so the clearing condition has a unique root in $(0, 1)$. Initial endpoints $[10^{-12}, 1 - 10^{-12}]$ are safe by interior-of-distribution arguments.

X.2 Why bisection and not Newton

Newton on the clearing equation requires the derivative of the aggregate excess demand, which involves derivatives of $R_k = \exp[(\text{logit } \mu_k - \text{logit } p)/\gamma]$. Bisection is more robust at the extreme parameter values we test (very small γ makes R_k huge, very small τ makes μ_k insensitive). 60 bisection iterations achieve ~ 19 digits in float64 — more than enough for the clearing step to not be the bottleneck.

Y The kernel sum: efficient numba implementation

The kernel evidence sum

$$A_v(P_{ijl}) = \sum_{m,n} K_h(P_{imn} - P_{ijl}) f_v(u_m) f_v(u_n) (\Delta u)^2 \quad (42)$$

is a 2D sum over the $G \times G$ slice at fixed i . Cost: $O(G^5)$ total over all G^3 cells. For $G = 21$ this is 4×10^6 kernel evaluations per Phi. numba JIT delivers this in ~ 0.15 s wall time; the pure-numpy longdouble port takes ~ 2 s.

The kernel evaluation $K_h(x) = e^{-x^2/2h^2}/(h\sqrt{2\pi})$ is computed in single np.exp call per kernel value. Numba vectorizes this efficiently with AVX.

Z Sensitivity analysis

Z.1 Sensitivity to U (signal support truncation)

Our standard $U = 4$ truncates the Gaussian signal support at ~ 4 standard deviations. At $\tau = 2$ (highest precision tested), the signal standard deviation is $1/\sqrt{2} = 0.707$, so $U = 4$ truncates at $\sim 5.7\sigma$. Tail mass beyond truncation: $\sim 10^{-8}$, well below all certification floors.

Tests at $U = 5$ and $U = 6$ on a few certified cells confirmed deficits identical to 4 decimal places.

Z.2 Sensitivity to halo size

Standard halo size: 2 cells on each face. The halo provides boundary data for the no-learning initialization and stabilizes the kernel quadrature near the cube edge. Tests at halo = 1 and halo = 4 confirmed deficit identical to 5 decimal places — the halo size is not quantitatively important above 1 cell.

Z.3 Sensitivity to the kernel constant C

Standard kernel constant $C = 0.45$ in $h(G) = C\sqrt{\Delta u}$. This constant was selected empirically as a compromise between:

- small C : less bandwidth bias, more numerical stiffness
- large C : more bandwidth bias, easier Newton convergence

Tests at $C = 0.3$ and $C = 0.6$ at a handful of cells confirmed deficits within 5% of the production value. The continuum extrapolation via deep G-ladders absorbs the residual sensitivity to C .

Discussion: why binary payoff is special

The binary-payoff structure $v \in \{0, 1\}$ has two distinctive features that interact with the strict- $h = 0$ degeneracy:

- The price function $P(u)$ maps to $(0, 1)$, the same range as the state probability. This makes $P_{\text{FR}}(u) = \sigma(\tau \sum_k u_k)$ a particularly clean closed form.
- The Bayes posterior μ_k is a single number (probability of $v = 1$), not a distribution. This makes the slice-integral conditioning algebraically tractable.

Together these features make the Bayes calculation under $P = P_{\text{FR}}$ a "sufficient-statistic" computation that closes in closed form, giving $\mu_k = p$ exactly for all k .

Under a continuous payoff, neither of these features holds: P would map to an interval different from the state space, and Bayes would produce a distribution rather than a number. The closed-form calculation $\mu_k = p$ would fail, and P_{FR} would (generically) NOT be a fixed point of the strict- $h = 0$ operator.

This suggests the GS degeneracy is a knife-edge phenomenon of the binary payoff case. The exact extent of this phenomenon under other payoff distributions is an open research question.

Discussion: implications for empirical work

The numerical certification has direct implications for empirical calibration of partial-revelation REE:

- The asymptotic $\mathcal{D} \propto 1/\gamma$ at fixed τ provides a moment condition relating risk aversion to observed price-information content.
- The K-invariance of the deficit means that empirical studies do not need to know exactly how many traders are active — the deficit depends primarily on (τ, γ) .
- Trading volume $\propto K$ provides an independent moment for calibrating the number of traders.

Discussion: implications for theory

The existence of a noise-free partial-revelation mechanism has implications for the Grossman-Stiglitz literature:

- The standard noise-trader assumption is not necessary for partial revelation: risk aversion under binary payoff suffices.
- The exact role of payoff structure deserves further investigation: the binary case admits a closed-form revealing equilibrium that generates the GS degeneracy, but continuous payoff cases may not.
- The kernel bandwidth h is mathematically equivalent to a small "noise" parameter (via the convolution structure of the regularization). This suggests a deep connection between the h -regularization selection mechanism and the noise-trader selection mechanism.

Conclusion (extended)

This study has established a partial-revelation equilibrium in a small-but-canonical setting through extensive numerical and analytic work. The 255 certified equilibria at 10^{-15} extended-precision residual constitute the first systematic catalog of CRRA-binary REE solutions. The K-invariance finding extends naturally to multiple crowds. The Grossman-Stiglitz degeneracy at strict $h = 0$ is the methodological surprise of the project and reframes the kernel bandwidth as an equilibrium selection mechanism rather than a numerical convenience.

Open directions:

- Continuous payoff variant: does the GS degeneracy persist?
- Computer-assisted proof: convert Theorem 1's $\chi < 1$ condition to global-in- h via interval arithmetic at a certified anchor.
- Multi-asset extensions: the discretization scales as G^{KJ} for J assets, requiring symmetric framework extensions.
- Strategic information acquisition: at what level of information cost does the partial-revelation equilibrium become incentive-compatible for traders to acquire information?

Numerical recipes: code-level walkthroughs

For reviewers and reproducers, this appendix walks through the production solver pipeline at the code level.

.1 Recipe 1: solving one cell with the kernel operator

Input: (τ, γ) , target grid G , optional warm-start P .

Steps:

1. Initialize P_0 to the no-learning prediction.
2. For $G' \in \{9, 13, 17, G\}$ in increasing order:
 - Build G' -grid, halo cells.
 - Initial guess: upsample $P_{G'-4}$ to G' resolution.
 - `scipy.optimize.newton_krylov` with $f_{\text{tol}} = 10^{-12}$, $\text{maxiter} = 30$.
3. Return P_G .

Typical wall: 5–20 seconds per cell at $G = 21$.

.2 Recipe 2: longdouble polish

Input: certified P at $G = 21$ from Recipe 1.

Steps:

1. Convert P to `numpy.longdouble`.
2. Newton-GMRES(40) iteration:

- Outer Newton iter: build matvec via finite-difference with $\epsilon = 3 \times 10^{-10}$.
 - Inner GMRES with relative tolerance 10^{-3} , max 40 Krylov vectors.
 - Accept Newton step.
3. Stop when $\|F\|_\infty < 10^{-15}$ OR max 6 Newton iterations.
 4. Return polished P .

Typical wall: 30–60 seconds per cell at $G = 21$.

.3 Recipe 3: γ -continuation rescue

Input: cell (τ, γ) that stalled in Recipe 1.

Steps:

1. Find nearest neighbor γ' at the same τ that was accepted.
2. Use that converged P as warm-start for γ .
3. Run Recipe 1 with the warm-start.

This rescues most γ -direction stalls. For τ -direction stalls, use Recipe 4 below.

.4 Recipe 4: τ -continuation rescue

Input: cell (τ, γ) where Recipe 3 also failed.

Steps:

1. Find nearest neighbor $\tau' < \tau$ that has an accepted solution at this γ .
2. March $\tau \rightarrow \tau$ in increments of 0.1, using the previous τ' converged P as warm-start for $\tau' + 0.1$.
3. Final step at the target τ .

This recovers most cells in the high- $\gamma\tau$ region.

.5 Recipe 5: deep G-ladder for continuum extrapolation

Input: certified cell at $G = 21$.

Steps:

1. For $G' \in \{25, 29, 33, 37\}$:
 - Index-upsample $P_{G'-4}$ to G' .
 - Newton-Krylov polish at G' .
 - If $F > 10^{-8}$: mark as stalled, do not use in extrapolation.
2. Continuum fit: q-free, q=1, q=2 fits over nailed rungs.
3. Continuum estimate = q=1 value; error bar = max spread.

.6 Recipe 6: certified deficit and welfare metrics

For each certified cell:

1. Compute the inner cube $G = 21$ array P .
2. Compute $\mathcal{D} = 1 - R^2$ of linear regression of logit P on $\sum_k u_k$ over the inner cube.
3. Compute trading volume $\text{TV} = \mathbb{E} |x^*(\mu, p)|$ under the true joint measure.
4. Compute the value-of-information decomposition by binning the joint distribution and computing $\text{CE}_{\mu_{\text{FR}}}$, $\text{CE}_{\text{informed}}$, $\text{CE}_{\text{priceonly}}$, CE_{prior} under the true measure.

Glossary of notation and acronyms

- τ : signal precision (inverse signal variance)
- γ : CRRA relative risk aversion
- K : number of traders ($K \in \{3, 4, 5\}$ in this study)
- G : discretization resolution (cells per axis)
- u_k : trader k 's private signal, centered ($u_k = s_k - 1/2$)
- v : binary asset payoff ($v \in \{0, 1\}$)
- $P(u_1, \dots, u_K)$: equilibrium price function
- $P_{\text{FR}} = \sigma(\tau \sum_k u_k)$: fully revealing equilibrium price
- μ_k : trader k 's posterior of $v = 1$ given own signal and price
- $A_v(p, u_k)$: evidence integral (Equation 4)
- Φ : the equilibrium operator (Bayes-then-clear)
- Φ_h, Φ_0 : kernel-smoothed ($h \neq 0$) and strict ($h=0$) versions
- $\mathcal{D} = 1 - R^2$: revelation deficit
- V_i^{pub} : value of observing the price vs. prior baseline
- V_i^{priv} : value of own signal above the price
- V_i^{FRgap} : welfare gap to full revelation
- TV : trading volume (per agent)
- REE : rational-expectations equilibrium
- CRRA : constant relative risk aversion
- CMM : co-area moving mesh (the strict- $h=0$ program)
- GS : Grossman-Stiglitz

- PR: partially revealing
- FR: fully revealing
- LM: Levenberg-Marquardt

Final remarks

This summary documents an extensive computational project on K=3 CRRA REE with binary payoff and no noise traders. The economic finding (partial revelation from risk aversion alone), the numerical certification (255 cells at 10^{-15}), the K-invariance result (deficits coincide within $\pm 25\%$ across $K \in \{3, 4, 5\}$), and the Grossman-Stiglitz degeneracy at strict $h = 0$ are the main contributions. The project's codebase, certified solutions, and exhaustive intermediate logs are all in the repository for reproducibility and future extensions.

Extended proof of Theorem 1 (full detail)

We give an extended proof of Theorem 1.

.1 Setup

Fix $(\tau, \gamma, U) \in (0, \infty)^3$ and regularization parameters $(h, \delta) \in (0, \infty)^2$ with $\chi = 12\tau U/h^2 < 1$.

Define the function class $\mathcal{K}(B, L)$ as in Section 8. It is compact convex in sup-norm by the Arzela-Ascoli theorem (continuous functions on a compact domain, uniformly bounded, uniformly Lipschitz form a compact set).

Define $\Phi_{h,\delta} : \mathcal{K}(B, L) \rightarrow C(Q)$ as the regularized equilibrium operator.

.2 Lemma 1: bounded evidence ratio

Statement. For $\Pi \in \mathcal{K}(B, L)$, the regularized log-evidence-ratio satisfies $|b(u, \Pi)| \leq 2\tau U$ at every $u \in Q$.

Proof. The integral defining b involves Gaussian densities f_0, f_1 over the support $|u_j - 1/2| \leq U + 1/2$ and $|u_j + 1/2| \leq U + 1/2$. The log-likelihood ratio $\log(f_1(u_j)/f_0(u_j)) = \tau u_j$ has range $[-\tau U, \tau U]$ over the support. Summing over $K - 1 = 2$ other agents: $|b| \leq 2\tau U$. $\square$

.3 Lemma 2: derivative bounds

Statement. For $\Pi \in \mathcal{K}(B, L)$,

$$\left| \frac{\partial b}{\partial \hat{\pi}} \right| \leq \chi = \frac{12\tau U}{h^2} \quad (43)$$

$$\left| \frac{\partial b}{\partial u_j} \right| \leq \beta = \frac{4U}{\delta^2}, \quad j \neq k \quad (44)$$

Proof sketch. The derivative with respect to the observed price $\hat{\pi}$ acts through the kernel K_h . By Gaussian moment estimates, $|\partial K_h(x)/\partial x| \leq K_h(x) \cdot |x|/h^2 \leq 1/(h^2\sqrt{2\pi e})$. Multiplying by the range of $|b|$ (Lemma 1) and the integration domain size $(2U + 1)$, we get the $12\tau U/h^2$ bound after algebraic simplification.

The derivative with respect to u_j acts through the own-signal mollifier ρ_δ with bandwidth δ . Similar Gaussian moment estimate gives the $4U/\delta^2$ bound. $\square$

.4 Lemma 3: CRRA non-expansive aggregation

Statement. The CRRA-clearing operator on log-odds $\pi^* = \text{logit } p^{\text{clear}}(\mu_1, \dots, \mu_K; \gamma)$ satisfies

$$\frac{\partial \pi^*}{\partial m_k} = w_k \geq 0, \quad \sum_k w_k \leq 1. \quad (45)$$

The strict inequality $\sum_k w_k < 1$ holds whenever any $\mu_k \neq p$ (i.e. whenever some trader disagrees with the price).

Proof sketch. Differentiating the clearing condition $\sum_k x_k^* = 0$ implicitly in $m_k = \text{logit } \mu_k$ gives $\partial \pi^* / \partial m_k = (\partial x_k^* / \partial m_k) / \sum_j (\partial x_j^* / \partial \pi^*)$. Both numerator and denominator can be computed in closed form:

$$\frac{\partial x_k^*}{\partial m_k} = \frac{R_k p(1-p)/\gamma}{[(1-p) + R_k p]^2} \quad (46)$$

$$\sum_j \frac{\partial x_j^*}{\partial \pi^*} = -\frac{1}{p(1-p)} - \frac{1}{\gamma} \sum_j \frac{R_j p(1-p)}{[(1-p) + R_j p]^2} \quad (47)$$

Their ratio gives w_k . The sum $\sum_k w_k$ has a clean closed form involving the "Jensen slack" terms $J_k = (R_k - 1)^2 p(1-p)/D^2$ with $D = (1-p) + Rp$:

$$\sum_k w_k = 1 - \frac{(\gamma + p(1-p)) \sum_k J_k}{\sum_k [(R_k - 1)D + R_k p(1-p)/\gamma]} \quad (48)$$

The denominator is positive; the numerator is non-negative ($J_k \geq 0$); equality holds iff all $R_k = 1$ (all $\mu_k = p$). $\square$

.5 Lemma 4: invariance of $\mathcal{K}(B, L_\star)$

Statement. Under $\chi < 1$, $\Phi_{h,\delta}(\mathcal{K}(B, L_\star)) \subset \mathcal{K}(B, L_\star)$.

Proof. For $\Pi \in \mathcal{K}(B, L_\star)$, we need to show that $\Phi_{h,\delta}\Pi$ also satisfies:

- permutation-symmetric: trivial (the operator preserves permutation symmetry of the input).
- odd: trivial under sign-flip symmetry of the model.
- bounded $|\Phi\Pi| \leq B$: by composition, the output's logit is the CRRA-cleared log-odds of the trader posteriors, each of which is $\tau u_k + b$ with $|b| \leq 2\tau U$ (Lemma 1). The clearing is a non-expansive (Lemma 3) average of these, so the cleared logit is bounded by $\tau U + 2\tau U = 3\tau U = B$.
- Lipschitz $L_\star$: by the chain rule, the Lipschitz constant of $\Phi\Pi$ is the product of: (i) the Lipschitz constant of the CRRA-clearing operator on log-odds, ≤ 1 by Lemma 3; (ii) the Lipschitz constant of the posterior map $\text{logit } \mu = \tau u + b$, which is $\tau + \chi L + \beta$ by Lemmas 1, 2. Solving the fixed-point inequality $L_\star = \tau + \chi L_\star + \beta$ gives $L_\star = (\tau + \beta)/(1 - \chi)$. $\square$

.6 Schauder application

$\mathcal{K}(B, L_\star)$ is compact convex in sup-norm. $\Phi_{h,\delta}$ is continuous (in fact smooth) and maps $\mathcal{K}$ to itself (Lemma 4). By Schauder's fixed-point theorem, $\Phi_{h,\delta}$ has a fixed point in $\mathcal{K}(B, L_\star)$.

.7 Proof of $\Pi^* \neq \text{logit } P_{\text{FR}}$

If $\chi \leq 1/3$ and $\beta < \tau(2 - 3\chi)$, then $L_\star = (\tau + \beta)/(1 - \chi) < 3\tau$. The fully-revealing P_{FR} has $\text{logit } P_{\text{FR}}(u) = \tau \sum_k u_k$, with Lipschitz constant 3τ in sup-norm. Since $\Pi^* \in \mathcal{K}(B, L_\star)$ has Lipschitz constant $\leq L_\star < 3\tau$, $\Pi^* \neq \text{logit } P_{\text{FR}}$. $\square$

Why this matters for empirical asset pricing

The findings here have potential implications for empirical asset pricing models that calibrate to observed price-revealing power:

- The deficit's $\propto 1/\gamma$ asymptote provides a quantitative mapping from observable price-information content to underlying risk aversion, independent of any noise-trader assumption.
- The K-invariance result simplifies calibration: the deficit depends primarily on (τ, γ) , not on the (often unknown) crowd size.
- Trading-volume proportionality $\propto K$ provides an independent moment for crowd size, decoupling the deficit's identification from the volume's identification.

Final summary table: the 7 main findings

Finding	Statement
1. Existence (regularized)	Theorem 1: regularized PR-REE exists for any (τ, γ) when $\chi < 1$ and the fixed point is strictly disjoint from P_{FR} .
2. Numerical certification	255 individual equilibria at extended-precision residual $\leq 10^{-15}$ across 12 τ values and 23 γ values.
3. Continuum limits	Immortal anchor: $\mathcal{D}_\infty = 0.268 \pm 0.010$. Max-deficit corner: $\mathcal{D}_\infty = 0.278 \pm 0.008$. G=21 values within 3% of continuum.
4. Welfare decomposition	$V_i^{\text{pub}} \gg V_i^{\text{priv}} > 0$ at every parameterization; V_i^{FRgap} vanishes proportionally to $\mathcal{D}$.
5. K-invariance	Deficits coincide within $\pm 25\%$ per cell across $K \in \{3, 4, 5\}$; slope of $\text{logit } P$ on $\sum_k u_k$ equals τ/K to three sig figs everywhere.
6. GS degeneracy	P_{FR} is the unique strict- $h=0$ fixed point under binary payoff, certified to 10^{-15} in 9 test cells.
7. Implication	Kernel bandwidth $h > 0$ is the equilibrium selection mechanism for the nontrivial PR branch; no noise traders required.

Table 5: Final summary of the seven main findings.

Full certified cell table (all 255 cells)

τ	γ	deficit	$\ F\ _{\infty}^{\text{ld}}$
0.0500	0.0500	5.9070e-02	6.75e-18
0.0500	0.0700	2.5586e-02	5.53e-18
0.0500	0.0980	8.3074e-03	4.23e-18
0.0500	0.1373	2.2467e-03	2.28e-18
0.0500	0.1922	5.5594e-04	2.52e-18
0.0500	0.2692	1.3033e-04	2.66e-18
0.0500	0.3769	3.0122e-05	2.30e-18
0.0500	0.5278	7.5063e-06	3.44e-18
0.0500	0.7391	2.4221e-06	2.11e-18
0.0500	1.0350	1.1082e-06	2.36e-18
0.0500	1.4493	6.1965e-07	2.49e-18
0.0500	2.0294	3.6178e-07	2.47e-18
0.0500	2.8418	2.0760e-07	2.60e-18
0.0500	3.9793	1.1572e-07	3.20e-18
0.0500	5.5722	6.2808e-08	2.57e-18
0.0500	7.8027	3.3348e-08	3.04e-18
0.0500	10.9261	1.7389e-08	2.66e-18
0.0500	15.2997	8.9290e-09	2.28e-18
0.0500	21.4241	4.5211e-09	2.71e-18
0.0500	30.0000	2.2591e-09	2.76e-18
0.1000	0.0100	1.9569e-01	7.86e-19
0.1000	0.0200	1.8522e-01	7.05e-19
0.1000	0.0300	1.7021e-01	7.05e-19
0.1000	0.0500	1.3308e-01	7.59e-19
0.1000	0.0700	9.5253e-02	5.96e-19
0.1000	0.0980	5.3592e-02	4.34e-19
0.1000	0.1373	2.1980e-02	4.88e-19
0.1000	0.1922	6.8032e-03	4.88e-19
0.1000	0.2692	1.7695e-03	4.61e-19
0.1000	0.3769	4.3169e-04	4.88e-19
0.1000	0.5278	1.1181e-04	4.88e-19
0.1000	0.7391	3.7253e-05	5.42e-19
0.1000	1.0350	1.7337e-05	4.88e-19
0.1000	1.4493	9.7162e-06	4.88e-19
0.1000	2.0294	5.6455e-06	4.88e-19
0.1000	2.8418	3.2114e-06	4.88e-19
0.1000	3.9793	1.7682e-06	5.15e-19
0.1000	5.5722	9.4391e-07	9.54e-18
0.1000	7.8027	4.9052e-07	7.59e-19
0.1000	10.9261	2.4920e-07	5.69e-19
0.1000	15.2997	1.2450e-07	5.69e-19
0.1000	21.4241	6.1952e-08	5.15e-19
0.1000	30.0000	3.1646e-08	6.78e-19
0.2000	0.0100	1.9844e-01	2.21e-17
0.2000	0.0200	1.9351e-01	2.85e-17
0.2000	0.0300	1.8696e-01	1.99e-17

τ	γ	deficit	$\ F\ _{\infty}^{\text{ld}}$
0.2000	0.0500	1.6981e-01	5.80e-18
0.2000	0.0700	1.4915e-01	1.87e-18
0.2000	0.0980	1.1827e-01	1.03e-18
0.2000	0.1373	7.8704e-02	1.68e-18
0.2000	0.1922	4.0362e-02	5.96e-19
0.2000	0.2692	1.5335e-02	9.22e-19
0.2000	0.3769	4.7167e-03	5.42e-19
0.2000	0.5278	1.4210e-03	7.05e-19
0.2000	0.7391	5.2505e-04	1.22e-18
0.2000	1.0350	2.5590e-04	9.22e-19
0.2000	1.4493	1.4416e-04	7.32e-19
0.2000	2.0294	8.2773e-05	9.76e-19
0.2000	2.8418	4.6165e-05	6.51e-19
0.2000	3.9793	2.4816e-05	7.05e-19
0.2000	5.5722	1.2965e-05	1.19e-18
0.2000	7.8027	6.7340e-06	7.05e-19
0.2000	10.9261	3.6474e-06	8.67e-19
0.2000	15.2997	2.2315e-06	8.13e-19
0.2000	21.4241	1.6592e-06	1.30e-18
0.2000	30.0000	1.4886e-06	8.67e-19
0.3000	0.0500	1.7613e-01	5.64e-18
0.3000	0.0700	1.5964e-01	2.71e-18
0.3000	0.0980	1.3653e-01	4.22e-17
0.3000	0.1373	1.0553e-01	2.97e-17
0.3000	0.1922	6.9035e-02	1.25e-17
0.3000	0.2692	3.5601e-02	8.27e-18
0.3000	0.3769	1.4515e-02	1.19e-17
0.3000	0.5278	5.3886e-03	6.45e-18
0.3000	0.7391	2.2569e-03	3.85e-18
0.3000	1.0350	1.1522e-03	9.51e-18
0.3000	1.4493	6.5256e-04	7.54e-18
0.3000	2.0294	3.7232e-04	8.08e-18
0.3000	2.8418	2.0655e-04	1.38e-17
0.3000	3.9793	1.1217e-04	2.42e-17
0.3000	5.5722	6.2057e-05	1.20e-17
0.3000	7.8027	3.7736e-05	1.57e-17
0.3000	10.9261	2.7417e-05	8.73e-18
0.3000	15.2997	2.4146e-05	1.16e-17
0.3000	21.4241	2.4101e-05	1.68e-17
0.3000	30.0000	2.5315e-05	1.08e-17
0.4000	0.0100	2.1382e-01	2.42e-16
0.4000	0.0200	2.0544e-01	2.92e-16
0.4000	0.0300	1.9715e-01	1.10e-16
0.4000	0.0500	1.8159e-01	6.32e-17
0.4000	0.0700	1.6652e-01	3.89e-17
0.4000	0.0980	1.4626e-01	1.00e-17

τ	γ	deficit	$\ F\ _{\infty}^{\text{ld}}$
0.4000	0.1373	1.1980e-01	5.20e-18
0.4000	0.1922	8.7786e-02	6.23e-18
0.4000	0.2692	5.4431e-02	7.59e-18
0.4000	0.3769	2.7826e-02	6.10e-18
0.4000	0.5278	1.2609e-02	7.02e-18
0.4000	0.7391	5.9419e-03	3.88e-18
0.4000	1.0350	3.1597e-03	3.25e-18
0.4000	1.4493	1.8028e-03	1.17e-17
0.4000	2.0294	1.0330e-03	7.78e-18
0.4000	2.8418	5.8353e-04	1.72e-17
0.4000	3.9793	3.3534e-04	1.52e-17
0.4000	5.5722	2.1148e-04	2.02e-17
0.4000	7.8027	1.5896e-04	2.53e-17
0.4000	10.9261	1.4379e-04	9.54e-18
0.4000	15.2997	1.4605e-04	2.22e-17
0.4000	21.4241	1.5503e-04	1.81e-17
0.4000	30.0000	1.6547e-04	1.62e-17
0.5000	0.0500	1.8605e-01	2.30e-18
0.5000	0.0700	1.7308e-01	2.49e-18
0.5000	0.0980	1.5572e-01	2.26e-16
0.5000	0.1373	1.3291e-01	8.15e-17
0.5000	0.1922	1.0448e-01	4.72e-17
0.5000	0.2692	7.2629e-02	5.46e-17
0.5000	0.3769	4.3409e-02	5.70e-17
0.5000	0.5278	2.2968e-02	2.71e-17
0.5000	0.7391	1.1933e-02	6.72e-17
0.5000	1.0350	6.5773e-03	5.91e-17
0.5000	1.4493	3.7928e-03	7.56e-17
0.5000	2.0294	2.2026e-03	1.78e-16
0.5000	2.8418	1.2882e-03	1.18e-16
0.5000	3.9793	8.0072e-04	2.81e-16
0.5000	5.5722	5.7626e-04	1.58e-16
0.5000	7.8027	5.0069e-04	3.79e-19
0.5000	10.9261	5.0011e-04	2.41e-16
0.5000	15.2997	5.3130e-04	4.88e-19
0.5000	21.4241	5.7149e-04	4.34e-19
0.5000	30.0000	6.1001e-04	4.88e-19
0.6000	0.0100	2.1384e-01	4.45e-18
0.6000	0.0200	2.0782e-01	3.96e-18
0.6000	0.0300	2.0197e-01	3.04e-18
0.6000	0.0500	1.9072e-01	2.91e-18
0.6000	0.0700	1.7985e-01	2.39e-18
0.6000	0.0980	1.6525e-01	3.58e-18
0.6000	0.1373	1.4582e-01	4.34e-19
0.6000	0.1922	1.2078e-01	5.15e-19
0.6000	0.2692	9.0907e-02	6.51e-19

τ	γ	deficit	$\ F\ _{\infty}^{\text{ld}}$
0.6000	0.3769	6.0520e-02	3.79e-19
0.6000	0.5278	3.5859e-02	4.34e-19
0.6000	0.7391	2.0144e-02	3.79e-19
0.6000	1.0350	1.1477e-02	4.07e-19
0.6000	1.4493	6.7096e-03	1.08e-18
0.6000	2.0294	3.9695e-03	8.13e-19
0.6000	2.8418	2.4177e-03	1.65e-18
0.6000	3.9793	1.6202e-03	9.76e-19
0.6000	5.5722	1.2872e-03	1.64e-17
0.6000	7.8027	1.2144e-03	1.36e-17
0.6000	10.9261	1.2679e-03	1.55e-17
0.6000	15.2997	1.3692e-03	1.14e-17
0.6000	21.4241	1.4775e-03	1.91e-18
0.6000	30.0000	1.5747e-03	1.30e-18
0.8000	0.0100	2.1807e-01	5.48e-18
0.8000	0.0200	2.1499e-01	7.13e-18
0.8000	0.0300	2.1137e-01	4.93e-18
0.8000	0.0500	2.0319e-01	5.29e-18
0.8000	0.0700	1.9526e-01	4.66e-18
0.8000	0.0980	1.8488e-01	4.26e-18
0.8000	0.1373	1.7069e-01	7.75e-18
0.8000	0.1922	1.5117e-01	8.67e-18
0.8000	0.2692	1.2548e-01	1.46e-17
0.8000	0.3769	9.5296e-02	2.95e-18
0.8000	0.5278	6.5351e-02	3.63e-18
0.8000	0.7391	4.1232e-02	3.82e-18
0.8000	1.0350	2.5023e-02	3.44e-18
0.8000	1.4493	1.5143e-02	4.26e-18
0.8000	2.0294	9.3508e-03	4.93e-18
0.8000	2.8418	6.1490e-03	9.53e-18
0.8000	3.9793	4.6162e-03	6.91e-18
0.8000	5.5722	4.1127e-03	7.54e-18
0.8000	7.8027	4.1778e-03	5.91e-18
0.8000	10.9261	4.5029e-03	9.52e-18
0.8000	15.2997	4.9047e-03	6.40e-18
0.8000	21.4241	5.2897e-03	1.28e-17
0.8000	30.0000	5.6190e-03	8.67e-18
1.0000	0.0100	2.2977e-01	1.58e-18
1.0000	0.0200	2.2692e-01	8.67e-19
1.0000	0.0300	2.2361e-01	7.10e-18
1.0000	0.0500	2.1757e-01	6.45e-18
1.0000	0.0700	2.1209e-01	7.35e-18
1.0000	0.0980	2.0456e-01	1.05e-17
1.0000	0.1373	1.9363e-01	1.07e-17
1.0000	0.1922	1.7771e-01	1.52e-18
1.0000	0.2692	1.5521e-01	4.34e-18

τ	γ	deficit	$\ F\ _{\infty}^{\text{ld}}$
1.0000	0.3769	1.2633e-01	5.31e-18
1.0000	0.5278	9.4214e-02	1.16e-17
1.0000	0.7391	6.4419e-02	1.07e-17
1.0000	1.0350	4.1443e-02	3.20e-18
1.0000	1.4493	2.6132e-02	5.20e-18
1.0000	2.0294	1.6899e-02	5.75e-18
1.0000	2.8418	1.1879e-02	5.91e-18
1.0000	3.9793	9.6278e-03	8.13e-18
1.0000	5.5722	9.0781e-03	8.02e-18
1.0000	7.8027	9.4556e-03	3.74e-18
1.0000	10.9261	1.0239e-02	1.26e-17
1.0000	15.2997	1.1116e-02	1.07e-17
1.0000	21.4241	1.1928e-02	5.83e-18
1.0000	30.0000	1.2611e-02	2.06e-17
1.2000	0.0100	2.4590e-01	5.20e-18
1.2000	0.0200	2.4178e-01	1.86e-18
1.2000	0.0300	2.3842e-01	1.41e-18
1.2000	0.0500	2.3306e-01	7.53e-18
1.2000	0.0700	2.2870e-01	1.19e-17
1.2000	0.0980	2.2290e-01	1.36e-17
1.2000	0.1373	2.1406e-01	2.28e-18
1.2000	0.1922	2.0037e-01	3.79e-18
1.2000	0.2692	1.7990e-01	8.02e-18
1.2000	0.3769	1.5206e-01	2.11e-17
1.2000	0.5278	1.1908e-01	1.19e-17
1.2000	0.7391	8.6045e-02	5.01e-17
1.2000	1.0350	5.8268e-02	5.97e-17
1.2000	1.4493	3.8444e-02	1.74e-16
1.2000	2.0294	2.6151e-02	2.55e-18
1.2000	2.8418	1.9546e-02	2.54e-17
1.2000	3.9793	1.6740e-02	1.14e-17
1.2000	5.5722	1.6234e-02	4.80e-17
1.2000	7.8027	1.6955e-02	1.39e-17
1.2000	10.9261	1.8199e-02	1.31e-17
1.2000	15.2997	1.9547e-02	1.70e-17
1.2000	21.4241	2.0784e-02	1.14e-17
1.2000	30.0000	2.1824e-02	1.25e-17
1.5000	0.0100	2.6138e-01	7.11e-17
1.5000	0.0200	2.5992e-01	6.04e-17
1.5000	0.0300	2.5843e-01	2.38e-17
1.5000	0.0500	2.5527e-01	9.22e-18
1.5000	0.0700	2.5213e-01	3.85e-18
1.5000	0.0980	2.4749e-01	4.69e-18
1.5000	0.1373	2.4005e-01	8.00e-18
1.5000	0.1922	2.2803e-01	9.76e-18
1.5000	0.2692	2.0891e-01	1.62e-16

τ	γ	deficit	$\ F\ _{\infty}^{\text{ld}}$
1.5000	0.3769	1.8125e-01	1.40e-17
1.5000	0.5278	1.4715e-01	1.97e-17
1.5000	0.7391	1.1222e-01	6.34e-17
1.5000	1.0350	8.1454e-02	1.07e-16
1.5000	1.4493	5.7837e-02	6.26e-17
1.5000	2.0294	4.2211e-02	7.05e-19
1.5000	2.8418	3.3343e-02	3.66e-18
1.5000	3.9793	2.9215e-02	4.22e-17
1.5000	5.5722	2.7992e-02	4.48e-17
1.5000	7.8027	2.8366e-02	3.00e-16
2.0000	0.0100	2.8715e-01	2.95e-16
2.0000	0.0200	2.8640e-01	1.93e-16
2.0000	0.0300	2.8571e-01	2.83e-16
2.0000	0.0500	2.8439e-01	9.26e-17
2.0000	0.0700	2.8283e-01	1.57e-16
2.0000	0.0980	2.7956e-01	8.66e-17
2.0000	0.1373	2.7307e-01	2.51e-16
2.0000	0.1922	2.6126e-01	6.42e-18
2.0000	0.2692	2.3945e-01	2.20e-18
2.0000	0.3769	2.0679e-01	3.79e-19
2.0000	0.5278	1.7737e-01	7.80e-17
2.0000	0.7391	1.4700e-01	4.54e-17
2.0000	1.0350	1.1746e-01	7.06e-17
2.0000	1.4493	9.0150e-02	6.08e-17
2.0000	2.0294	6.8951e-02	3.60e-17

Full K-comparison table

K	G	τ	γ	deficit	slope	F
3	11	0.10	0.0500	1.365e-01	0.0372	7.72e-15
3	11	0.10	0.0894	6.940e-02	0.0355	7.41e-13
3	11	0.10	0.1000	5.582e-02	0.0351	2.43e-14
3	11	0.10	0.1600	1.519e-02	0.0341	5.01e-13
3	11	0.10	0.2862	1.606e-03	0.0336	2.25e-13
3	11	0.10	0.5119	1.452e-04	0.0335	1.43e-14
3	11	0.10	0.9157	2.533e-05	0.0336	5.88e-15
3	11	0.10	1.0000	2.119e-05	0.0336	6.99e-15
3	11	0.10	1.6380	9.273e-06	0.0336	6.72e-15
3	11	0.10	2.9301	3.606e-06	0.0337	6.77e-15
3	11	0.10	5.2414	1.279e-06	0.0337	6.99e-15
3	11	0.10	9.3757	4.190e-07	0.0337	6.61e-15
3	11	0.10	10.0000	3.687e-07	0.0337	6.88e-15
3	11	0.10	16.7711	1.292e-07	0.0337	6.94e-15
3	11	0.10	30.0000	3.819e-08	0.0337	7.11e-15

K	G	τ	γ	deficit	slope	F
3	11	0.20	0.0500	1.710e-01	0.0743	3.33e-13
3	11	0.20	0.0894	1.314e-01	0.0711	6.49e-15
3	11	0.20	0.1000	1.203e-01	0.0705	1.15e-13
3	11	0.20	0.1600	6.551e-02	0.0684	6.55e-15
3	11	0.20	0.2862	1.464e-02	0.0669	6.88e-15
3	11	0.20	0.5119	1.870e-03	0.0668	6.11e-15
3	11	0.20	0.9157	3.764e-04	0.0674	1.56e-13
3	11	0.20	1.0000	3.172e-04	0.0675	7.33e-15
3	11	0.20	1.6380	1.402e-04	0.0679	1.49e-14
3	11	0.20	2.9301	5.392e-05	0.0683	1.02e-14
3	11	0.20	5.2414	1.864e-05	0.0685	6.00e-15
3	11	0.20	9.3757	5.892e-06	0.0687	7.55e-15
3	11	0.20	10.0000	5.167e-06	0.0687	6.77e-15
3	11	0.20	16.7711	1.810e-06	0.0687	6.44e-15
3	11	0.20	30.0000	6.707e-07	0.0688	7.11e-15
3	11	0.40	0.0500	1.825e-01	0.1195	6.55e-15
3	11	0.40	0.0894	1.552e-01	0.1202	6.99e-15
3	11	0.40	0.1000	1.481e-01	0.1204	1.41e-13
3	11	0.40	0.1600	1.110e-01	0.1214	6.66e-15
3	11	0.40	0.2862	5.491e-02	0.1236	6.61e-15
3	11	0.40	0.5119	1.623e-02	0.1271	6.55e-15
3	11	0.40	0.9157	4.754e-03	0.1313	6.94e-15
3	11	0.40	1.0000	4.078e-03	0.1320	6.78e-14
3	11	0.40	1.6380	1.866e-03	0.1351	6.77e-15
3	11	0.40	2.9301	7.230e-04	0.1378	6.58e-13
3	11	0.40	5.2414	2.494e-04	0.1396	2.57e-13
3	11	0.40	9.3757	8.254e-05	0.1407	9.86e-14
3	11	0.40	10.0000	7.353e-05	0.1408	6.99e-15
3	11	0.40	16.7711	3.471e-05	0.1414	2.08e-14
3	11	0.40	30.0000	2.559e-05	0.1417	1.48e-14
3	11	0.60	0.0500	1.908e-01	0.1478	6.55e-15
3	11	0.60	0.0894	1.707e-01	0.1510	6.72e-15
3	11	0.60	0.1000	1.656e-01	0.1518	1.18e-13
3	11	0.60	0.1600	1.384e-01	0.1559	6.77e-15
3	11	0.60	0.2862	9.148e-02	0.1639	6.88e-15
3	11	0.60	0.5119	4.388e-02	0.1748	6.99e-15
3	11	0.60	0.9157	1.750e-02	0.1865	6.44e-15
3	11	0.60	1.0000	1.530e-02	0.1882	3.76e-13
3	11	0.60	1.6380	7.368e-03	0.1968	6.72e-15
3	11	0.60	2.9301	2.958e-03	0.2046	6.99e-15
3	11	0.60	5.2414	1.061e-03	0.2099	6.88e-15
3	11	0.60	9.3757	3.882e-04	0.2133	6.32e-13
3	11	0.60	10.0000	3.529e-04	0.2135	7.94e-15
3	11	0.60	16.7711	2.099e-04	0.2153	1.72e-13
3	11	0.60	30.0000	1.913e-04	0.2164	7.65e-14
3	15	0.10	0.0500	1.347e-01	0.0373	3.31e-14

K	G	τ	γ	deficit	slope	F
3	15	0.10	0.0894	6.678e-02	0.0355	6.88e-15
3	15	0.10	0.1000	5.335e-02	0.0352	7.85e-14
3	15	0.10	0.1600	1.413e-02	0.0342	6.99e-15
3	15	0.10	0.2862	1.477e-03	0.0337	4.78e-13
3	15	0.10	0.5119	1.346e-04	0.0336	2.96e-14
3	15	0.10	0.9157	2.362e-05	0.0337	5.77e-15
3	15	0.10	1.0000	1.974e-05	0.0337	6.88e-15
3	15	0.10	1.6380	8.553e-06	0.0338	6.66e-15
3	15	0.10	2.9301	3.294e-06	0.0338	6.55e-15
3	15	0.10	5.2414	1.158e-06	0.0338	6.83e-15
3	15	0.10	9.3757	3.757e-07	0.0338	6.77e-15
3	15	0.10	10.0000	3.303e-07	0.0338	6.88e-15
3	15	0.10	16.7711	1.147e-07	0.0339	6.66e-15
3	15	0.10	30.0000	3.410e-08	0.0339	6.72e-15
3	15	0.20	0.0500	1.706e-01	0.0752	6.99e-15
3	15	0.20	0.0894	1.297e-01	0.0720	6.88e-15
3	15	0.20	0.1000	1.183e-01	0.0713	2.80e-13
3	15	0.20	0.1600	6.291e-02	0.0691	6.88e-15
3	15	0.20	0.2862	1.355e-02	0.0676	6.66e-15
3	15	0.20	0.5119	1.719e-03	0.0676	6.99e-15
3	15	0.20	0.9157	3.488e-04	0.0682	3.04e-13
3	15	0.20	1.0000	2.937e-04	0.0683	6.22e-15
3	15	0.20	1.6380	1.283e-04	0.0687	5.15e-14
3	15	0.20	2.9301	4.847e-05	0.0691	1.89e-14
3	15	0.20	5.2414	1.645e-05	0.0693	9.55e-15
3	15	0.20	9.3757	5.177e-06	0.0694	6.44e-15
3	15	0.20	10.0000	4.551e-06	0.0695	6.99e-15
3	15	0.20	16.7711	1.728e-06	0.0695	6.11e-15
3	15	0.20	30.0000	8.640e-07	0.0696	7.55e-15
3	15	0.40	0.0500	1.824e-01	0.1228	6.99e-15
3	15	0.40	0.0894	1.541e-01	0.1234	6.88e-15
3	15	0.40	0.1000	1.468e-01	0.1236	3.40e-13
3	15	0.40	0.1600	1.088e-01	0.1246	6.66e-15
3	15	0.40	0.2862	5.219e-02	0.1269	6.77e-15
3	15	0.40	0.5119	1.497e-02	0.1307	6.88e-15
3	15	0.40	0.9157	4.355e-03	0.1351	6.66e-15
3	15	0.40	1.0000	3.729e-03	0.1358	1.78e-13
3	15	0.40	1.6380	1.674e-03	0.1390	7.05e-15
3	15	0.40	2.9301	6.305e-04	0.1418	6.99e-15
3	15	0.40	5.2414	2.193e-04	0.1436	7.70e-13
3	15	0.40	9.3757	8.721e-05	0.1447	3.00e-13
3	15	0.40	10.0000	8.085e-05	0.1448	7.99e-15
3	15	0.40	16.7711	5.756e-05	0.1453	5.11e-14
3	15	0.40	30.0000	5.792e-05	0.1457	2.65e-14
3	15	0.60	0.0500	1.915e-01	0.1529	6.88e-15
3	15	0.60	0.0894	1.703e-01	0.1559	6.66e-15

K	G	τ	γ	deficit	slope	F
3	15	0.60	0.1000	1.650e-01	0.1567	2.95e-13
3	15	0.60	0.1600	1.369e-01	0.1612	6.94e-15
3	15	0.60	0.2862	8.872e-02	0.1699	6.77e-15
3	15	0.60	0.5119	4.115e-02	0.1817	8.88e-15
3	15	0.60	0.9157	1.595e-02	0.1942	6.99e-15
3	15	0.60	1.0000	1.388e-02	0.1960	6.83e-15
3	15	0.60	1.6380	6.498e-03	0.2049	7.11e-15
3	15	0.60	2.9301	2.541e-03	0.2130	6.72e-15
3	15	0.60	5.2414	9.742e-04	0.2184	7.05e-15
3	15	0.60	9.3757	5.066e-04	0.2217	6.72e-15
3	15	0.60	10.0000	4.878e-04	0.2220	1.72e-14
3	15	0.60	16.7711	4.446e-04	0.2237	5.87e-13
3	15	0.60	30.0000	4.897e-04	0.2249	2.63e-13
3	21	0.10	0.0500	1.331e-01	0.0375	1.43e-13
3	21	0.10	0.0894	6.457e-02	0.0357	7.05e-15
3	21	0.10	0.1000	5.128e-02	0.0354	2.05e-13
3	21	0.10	0.1600	1.327e-02	0.0343	7.11e-15
3	21	0.10	0.2862	1.372e-03	0.0338	6.99e-15
3	21	0.10	0.5119	1.255e-04	0.0338	1.09e-13
3	21	0.10	0.9157	2.218e-05	0.0339	9.49e-15
3	21	0.10	1.0000	1.852e-05	0.0339	6.83e-15
3	21	0.10	1.6380	7.979e-06	0.0339	7.11e-15
3	21	0.10	2.9301	3.046e-06	0.0340	7.11e-15
3	21	0.10	5.2414	1.060e-06	0.0340	7.11e-15
3	21	0.10	9.3757	3.399e-07	0.0340	7.11e-15
3	21	0.10	10.0000	2.984e-07	0.0340	7.16e-15
3	21	0.10	16.7711	1.029e-07	0.0340	7.27e-15
3	21	0.10	30.0000	3.165e-08	0.0340	7.11e-15
3	21	0.20	0.0500	1.698e-01	0.0766	7.11e-15
3	21	0.20	0.0894	1.277e-01	0.0732	7.05e-15
3	21	0.20	0.1000	1.161e-01	0.0725	7.74e-13
3	21	0.20	0.1600	6.027e-02	0.0702	7.05e-15
3	21	0.20	0.2862	1.251e-02	0.0687	6.72e-15
3	21	0.20	0.5119	1.576e-03	0.0687	7.27e-15
3	21	0.20	0.9157	3.239e-04	0.0693	8.32e-13
3	21	0.20	1.0000	2.728e-04	0.0694	7.55e-15
3	21	0.20	1.6380	1.180e-04	0.0698	1.32e-13
3	21	0.20	2.9301	4.370e-05	0.0702	7.25e-14
3	21	0.20	5.2414	1.461e-05	0.0704	2.34e-14
3	21	0.20	9.3757	4.773e-06	0.0706	1.28e-14
3	21	0.20	10.0000	4.251e-06	0.0706	7.11e-15
3	21	0.20	16.7711	2.016e-06	0.0707	7.55e-15
3	21	0.20	30.0000	1.489e-06	0.0707	6.22e-15
3	21	0.40	0.0500	1.816e-01	0.1268	6.99e-15
3	21	0.40	0.0894	1.524e-01	0.1275	6.88e-15
3	21	0.40	0.1000	1.449e-01	0.1276	7.11e-15

K	G	τ	γ	deficit	slope	F
3	21	0.40	0.1600	1.058e-01	0.1288	6.99e-15
3	21	0.40	0.2862	4.885e-02	0.1317	7.61e-15
3	21	0.40	0.5119	1.356e-02	0.1361	6.55e-15
3	21	0.40	0.9157	3.926e-03	0.1410	6.77e-15
3	21	0.40	1.0000	3.355e-03	0.1416	6.67e-13
3	21	0.40	1.6380	1.475e-03	0.1451	6.99e-15
3	21	0.40	2.9301	5.540e-04	0.1480	7.05e-15
3	21	0.40	5.2414	2.274e-04	0.1499	6.99e-15
3	21	0.40	9.3757	1.476e-04	0.1510	6.94e-15
3	21	0.40	10.0000	1.455e-04	0.1511	1.14e-14
3	21	0.40	16.7711	1.481e-04	0.1517	3.01e-13
3	21	0.40	30.0000	1.655e-04	0.1521	1.43e-13
3	21	0.60	0.0500	1.907e-01	0.1591	7.33e-15
3	21	0.60	0.0894	1.697e-01	0.1625	7.27e-15
3	21	0.60	0.1000	1.642e-01	0.1634	6.99e-15
3	21	0.60	0.1600	1.351e-01	0.1684	6.55e-15
3	21	0.60	0.2862	8.524e-02	0.1786	1.77e-14
3	21	0.60	0.5119	3.774e-02	0.1925	3.19e-14
3	21	0.60	0.9157	1.403e-02	0.2065	1.04e-14
3	21	0.60	1.0000	1.214e-02	0.2084	6.88e-15
3	21	0.60	1.6380	5.537e-03	0.2181	6.61e-15
3	21	0.60	2.9301	2.319e-03	0.2266	6.66e-15
3	21	0.60	5.2414	1.323e-03	0.2322	6.94e-15
3	21	0.60	9.3757	1.234e-03	0.2356	6.88e-15
3	21	0.60	10.0000	1.247e-03	0.2359	6.81e-14
3	21	0.60	16.7711	1.399e-03	0.2377	6.99e-15
3	21	0.60	30.0000	1.575e-03	0.2388	7.22e-15
4	11	0.10	0.0500	9.019e-02	0.0269	1.63e-14
4	11	0.10	0.0894	4.806e-02	0.0269	2.36e-13
4	11	0.10	0.1000	4.029e-02	0.0267	1.95e-14
4	11	0.10	0.1600	1.396e-02	0.0259	3.68e-13
4	11	0.10	0.2862	1.866e-03	0.0254	2.64e-13
4	11	0.10	0.5119	1.712e-04	0.0252	3.57e-14
4	11	0.10	0.9157	2.440e-05	0.0252	8.10e-15
4	11	0.10	1.0000	1.994e-05	0.0252	7.11e-15
4	11	0.10	1.6380	8.418e-06	0.0253	7.11e-15
4	11	0.10	2.9301	3.364e-06	0.0253	6.66e-15
4	11	0.10	5.2414	1.222e-06	0.0254	7.44e-15
4	11	0.10	9.3757	4.074e-07	0.0254	7.11e-15
4	11	0.10	10.0000	3.592e-07	0.0254	7.33e-15
4	11	0.10	16.7711	1.275e-07	0.0254	7.44e-15
4	11	0.10	30.0000	3.811e-08	0.0254	7.22e-15
4	11	0.20	0.0500	1.541e-01	0.0489	5.78e-13
4	11	0.20	0.0894	1.064e-01	0.0509	7.05e-15
4	11	0.20	0.1000	9.631e-02	0.0512	1.51e-13
4	11	0.20	0.1600	5.432e-02	0.0515	6.88e-15

K	G	τ	γ	deficit	slope	F
4	11	0.20	0.2862	1.517e-02	0.0508	7.22e-15
4	11	0.20	0.5119	2.137e-03	0.0504	7.11e-15
4	11	0.20	0.9157	3.634e-04	0.0508	2.63e-13
4	11	0.20	1.0000	3.006e-04	0.0508	8.10e-15
4	11	0.20	1.6380	1.303e-04	0.0512	5.33e-14
4	11	0.20	2.9301	5.183e-05	0.0516	2.00e-14
4	11	0.20	5.2414	1.840e-05	0.0518	1.43e-14
4	11	0.20	9.3757	5.923e-06	0.0519	9.33e-15
4	11	0.20	10.0000	5.202e-06	0.0519	6.99e-15
4	11	0.20	16.7711	1.830e-06	0.0520	6.38e-15
4	11	0.20	30.0000	6.473e-07	0.0521	6.55e-15
4	11	0.40	0.0500	1.932e-01	0.0867	7.05e-15
4	11	0.40	0.0894	1.632e-01	0.0885	6.99e-15
4	11	0.40	0.1000	1.556e-01	0.0889	6.55e-13
4	11	0.40	0.1600	1.164e-01	0.0908	6.99e-15
4	11	0.40	0.2862	5.898e-02	0.0933	6.83e-15
4	11	0.40	0.5119	1.754e-02	0.0956	6.88e-15
4	11	0.40	0.9157	4.594e-03	0.0986	6.77e-15
4	11	0.40	1.0000	3.906e-03	0.0991	1.82e-13
4	11	0.40	1.6380	1.815e-03	0.1019	6.99e-15
4	11	0.40	2.9301	7.352e-04	0.1045	7.05e-15
4	11	0.40	5.2414	2.597e-04	0.1064	6.77e-15
4	11	0.40	9.3757	8.652e-05	0.1075	4.65e-13
4	11	0.40	10.0000	7.702e-05	0.1076	7.83e-15
4	11	0.40	16.7711	3.555e-05	0.1082	1.15e-13
4	11	0.40	30.0000	2.508e-05	0.1086	5.55e-14
4	11	0.60	0.0500	2.010e-01	0.1115	6.94e-15
4	11	0.60	0.0894	1.795e-01	0.1135	7.11e-15
4	11	0.60	0.1000	1.742e-01	0.1141	7.11e-15
4	11	0.60	0.1600	1.459e-01	0.1169	7.11e-15
4	11	0.60	0.2862	9.637e-02	0.1222	6.99e-15
4	11	0.60	0.5119	4.488e-02	0.1295	7.22e-15
4	11	0.60	0.9157	1.694e-02	0.1382	6.72e-15
4	11	0.60	1.0000	1.480e-02	0.1396	6.88e-15
4	11	0.60	1.6380	7.434e-03	0.1473	6.99e-15
4	11	0.60	2.9301	3.154e-03	0.1551	6.94e-15
4	11	0.60	5.2414	1.154e-03	0.1607	6.77e-15
4	11	0.60	9.3757	4.257e-04	0.1643	6.94e-15
4	11	0.60	10.0000	3.875e-04	0.1646	2.21e-14
4	11	0.60	16.7711	2.335e-04	0.1665	6.94e-15
4	11	0.60	30.0000	2.143e-04	0.1677	4.84e-13
4	15	0.10	0.0500	8.869e-02	0.0271	7.56e-14
4	15	0.10	0.0894	4.666e-02	0.0269	6.41e-13
4	15	0.10	0.1000	3.893e-02	0.0268	5.79e-14
4	15	0.10	0.1600	1.317e-02	0.0260	9.89e-13
4	15	0.10	0.2862	1.729e-03	0.0255	6.74e-13

K	G	τ	γ	deficit	slope	F
4	15	0.10	0.5119	1.591e-04	0.0253	9.66e-14
4	15	0.10	0.9157	2.286e-05	0.0254	1.13e-14
4	15	0.10	1.0000	1.867e-05	0.0254	7.27e-15
4	15	0.10	1.6380	7.797e-06	0.0254	8.05e-15
4	15	0.10	2.9301	3.083e-06	0.0254	6.94e-15
4	15	0.10	5.2414	1.111e-06	0.0255	7.83e-15
4	15	0.10	9.3757	3.670e-07	0.0255	7.49e-15
4	15	0.10	10.0000	3.232e-07	0.0255	7.33e-15
4	15	0.10	16.7711	1.137e-07	0.0255	6.88e-15
4	15	0.10	30.0000	3.406e-08	0.0255	7.05e-15
4	15	0.20	0.0500	1.528e-01	0.0496	7.27e-15
4	15	0.20	0.0894	1.046e-01	0.0517	7.11e-15
4	15	0.20	0.1000	9.443e-02	0.0519	4.06e-13
4	15	0.20	0.1600	5.247e-02	0.0523	7.16e-15
4	15	0.20	0.2862	1.420e-02	0.0515	7.44e-15
4	15	0.20	0.5119	1.970e-03	0.0512	7.22e-15
4	15	0.20	0.9157	3.395e-04	0.0515	6.92e-13
4	15	0.20	1.0000	2.810e-04	0.0516	1.17e-14
4	15	0.20	1.6380	1.208e-04	0.0520	1.56e-13
4	15	0.20	2.9301	4.721e-05	0.0523	9.45e-14
4	15	0.20	5.2414	1.647e-05	0.0526	2.98e-14
4	15	0.20	9.3757	5.272e-06	0.0527	1.77e-14
4	15	0.20	10.0000	4.638e-06	0.0527	7.22e-15
4	15	0.20	16.7711	1.736e-06	0.0528	7.22e-15
4	15	0.20	30.0000	7.996e-07	0.0528	8.72e-15
4	15	0.40	0.0500	1.936e-01	0.0891	7.38e-15
4	15	0.40	0.0894	1.622e-01	0.0911	7.22e-15
4	15	0.40	0.1000	1.543e-01	0.0915	7.22e-15
4	15	0.40	0.1600	1.142e-01	0.0936	7.27e-15
4	15	0.40	0.2862	5.628e-02	0.0963	6.66e-15
4	15	0.40	0.5119	1.622e-02	0.0989	7.66e-15
4	15	0.40	0.9157	4.279e-03	0.1023	6.88e-15
4	15	0.40	1.0000	3.644e-03	0.1029	6.90e-13
4	15	0.40	1.6380	1.678e-03	0.1059	7.11e-15
4	15	0.40	2.9301	6.628e-04	0.1087	7.22e-15
4	15	0.40	5.2414	2.365e-04	0.1106	7.11e-15
4	15	0.40	9.3757	9.454e-05	0.1118	7.44e-15
4	15	0.40	10.0000	8.754e-05	0.1119	1.39e-14
4	15	0.40	16.7711	6.115e-05	0.1125	4.24e-13
4	15	0.40	30.0000	6.029e-05	0.1129	2.07e-13
4	15	0.60	0.0500	2.008e-01	0.1154	7.11e-15
4	15	0.60	0.0894	1.794e-01	0.1176	6.72e-15
4	15	0.60	0.1000	1.738e-01	0.1182	6.94e-15
4	15	0.60	0.1600	1.442e-01	0.1214	7.55e-15
4	15	0.60	0.2862	9.340e-02	0.1275	6.88e-15
4	15	0.60	0.5119	4.222e-02	0.1359	1.10e-14

K	G	τ	γ	deficit	slope	F
4	15	0.60	0.9157	1.580e-02	0.1459	6.55e-15
4	15	0.60	1.0000	1.380e-02	0.1475	7.05e-15
4	15	0.60	1.6380	6.832e-03	0.1561	6.66e-15
4	15	0.60	2.9301	2.840e-03	0.1645	6.66e-15
4	15	0.60	5.2414	1.133e-03	0.1705	7.02e-15
4	15	0.60	9.3757	6.221e-04	0.1743	6.88e-15
4	15	0.60	10.0000	6.026e-04	0.1746	1.22e-13
4	15	0.60	16.7711	5.654e-04	0.1766	7.11e-15
4	15	0.60	30.0000	6.250e-04	0.1779	7.11e-15
4	21	0.20	0.0500	1.518e-01	0.0507	8.10e-15
4	21	0.20	0.0894	1.030e-01	0.0528	7.44e-15
4	21	0.20	0.1000	9.273e-02	0.0530	7.55e-15
4	21	0.20	0.1600	5.067e-02	0.0533	7.44e-15
4	21	0.20	0.2862	1.325e-02	0.0525	7.33e-15
4	21	0.20	0.5119	1.810e-03	0.0523	7.49e-15
4	21	0.20	0.9157	3.182e-04	0.0526	7.38e-15
4	21	0.20	1.0000	2.640e-04	0.0527	4.04e-14
4	21	0.20	1.6380	1.133e-04	0.0531	5.35e-13
4	21	0.20	2.9301	4.346e-05	0.0535	3.33e-13
4	21	0.20	5.2414	1.494e-05	0.0537	1.53e-13
4	21	0.20	9.3757	4.922e-06	0.0539	6.62e-14
4	21	0.20	10.0000	4.377e-06	0.0539	8.22e-15
4	21	0.20	16.7711	1.983e-06	0.0539	2.42e-14
4	21	0.20	30.0000	1.347e-06	0.0540	1.27e-14
5	11	0.10	0.0500	1.301e-01	0.0231	4.80e-14
5	11	0.10	0.0894	5.850e-02	0.0220	7.55e-15
5	11	0.10	0.1000	4.682e-02	0.0218	4.11e-14
5	11	0.10	0.1600	1.439e-02	0.0210	7.39e-13
5	11	0.10	0.2862	1.959e-03	0.0204	4.29e-13
5	11	0.10	0.5119	1.842e-04	0.0202	6.68e-14
5	11	0.10	0.9157	2.364e-05	0.0202	9.21e-15
5	11	0.10	1.0000	1.890e-05	0.0202	6.94e-15
5	11	0.10	1.6380	7.433e-06	0.0203	7.66e-15
5	11	0.10	2.9301	2.949e-06	0.0203	8.10e-15
5	11	0.10	5.2414	1.079e-06	0.0203	6.77e-15
5	11	0.10	9.3757	3.624e-07	0.0203	7.27e-15
5	11	0.10	10.0000	3.198e-07	0.0203	7.33e-15
5	11	0.10	16.7711	1.142e-07	0.0204	7.33e-15
5	11	0.10	30.0000	3.435e-08	0.0204	7.22e-15
5	11	0.20	0.0500	1.739e-01	0.0435	7.99e-15
5	11	0.20	0.0894	1.235e-01	0.0428	6.94e-15
5	11	0.20	0.1000	1.114e-01	0.0427	1.18e-13
5	11	0.20	0.1600	5.952e-02	0.0420	7.66e-15
5	11	0.20	0.2862	1.580e-02	0.0410	7.27e-15
5	11	0.20	0.5119	2.278e-03	0.0405	7.22e-15
5	11	0.20	0.9157	3.520e-04	0.0407	4.76e-13

K	G	τ	γ	deficit	slope	F
5	11	0.20	1.0000	2.854e-04	0.0408	1.81e-14
5	11	0.20	1.6380	1.166e-04	0.0411	1.10e-13
5	11	0.20	2.9301	4.626e-05	0.0414	7.44e-14
5	11	0.20	5.2414	1.658e-05	0.0416	2.25e-14
5	11	0.20	9.3757	5.384e-06	0.0418	1.67e-14
5	11	0.20	10.0000	4.732e-06	0.0418	7.22e-15
5	11	0.20	16.7711	1.669e-06	0.0419	8.94e-15
5	11	0.20	30.0000	5.776e-07	0.0419	8.88e-15
5	11	0.40	0.0500	1.940e-01	0.0700	7.38e-15
5	11	0.40	0.0894	1.648e-01	0.0712	7.44e-15
5	11	0.40	0.1000	1.573e-01	0.0715	4.92e-13
5	11	0.40	0.1600	1.186e-01	0.0729	7.44e-15
5	11	0.40	0.2862	6.105e-02	0.0749	6.99e-15
5	11	0.40	0.5119	1.834e-02	0.0766	6.99e-15
5	11	0.40	0.9157	4.423e-03	0.0789	7.55e-15
5	11	0.40	1.0000	3.705e-03	0.0793	4.22e-13
5	11	0.40	1.6380	1.660e-03	0.0818	7.05e-15
5	11	0.40	2.9301	6.786e-04	0.0843	7.16e-15
5	11	0.40	5.2414	2.422e-04	0.0861	7.05e-15
5	11	0.40	9.3757	8.108e-05	0.0872	7.11e-15
5	11	0.40	10.0000	7.217e-05	0.0872	1.15e-14
5	11	0.40	16.7711	3.306e-05	0.0878	3.36e-13
5	11	0.40	30.0000	2.291e-05	0.0882	1.57e-13
5	11	0.60	0.0500	2.070e-01	0.0894	7.49e-15
5	11	0.60	0.0894	1.850e-01	0.0905	6.88e-15
5	11	0.60	0.1000	1.794e-01	0.0909	9.22e-13
5	11	0.60	0.1600	1.501e-01	0.0931	6.83e-15
5	11	0.60	0.2862	9.954e-02	0.0973	6.66e-15
5	11	0.60	0.5119	4.579e-02	0.1029	6.44e-15
5	11	0.60	0.9157	1.614e-02	0.1099	6.66e-15
5	11	0.60	1.0000	1.396e-02	0.1110	7.22e-15
5	11	0.60	1.6380	6.927e-03	0.1179	7.38e-15
5	11	0.60	2.9301	2.997e-03	0.1252	7.55e-15
5	11	0.60	5.2414	1.108e-03	0.1307	7.16e-15
5	11	0.60	9.3757	4.140e-04	0.1343	7.33e-15
5	11	0.60	10.0000	3.778e-04	0.1346	8.45e-14
5	11	0.60	16.7711	2.337e-04	0.1365	7.22e-15
5	11	0.60	30.0000	2.188e-04	0.1377	7.88e-15
5	15	0.20	0.0500	1.731e-01	0.0442	7.88e-15
5	15	0.20	0.0894	1.216e-01	0.0435	8.38e-15
5	15	0.20	0.1000	1.093e-01	0.0433	7.88e-15
5	15	0.20	0.1600	5.732e-02	0.0426	8.27e-15
5	15	0.20	0.2862	1.483e-02	0.0416	7.55e-15
5	15	0.20	0.5119	2.108e-03	0.0412	7.55e-15
5	15	0.20	0.9157	3.305e-04	0.0414	7.66e-15
5	15	0.20	1.0000	2.684e-04	0.0415	7.33e-15

K	G	τ	γ	deficit	slope	F
5	15	0.20	1.6380	1.091e-04	0.0418	5.87e-13
5	15	0.20	2.9301	4.258e-05	0.0422	2.74e-13
5	15	0.20	5.2414	1.500e-05	0.0424	1.38e-13
5	15	0.20	9.3757	4.839e-06	0.0425	6.07e-14
5	15	0.20	10.0000	4.259e-06	0.0425	7.94e-15
5	15	0.20	16.7711	1.584e-06	0.0426	8.38e-14
5	15	0.20	30.0000	6.980e-07	0.0426	1.22e-14

Full value-of-information table (per cell)

τ	γ	TV	V^{pub}	V^{priv}	V^{FRgap}	$\mathcal{D}$
0.05	0.0500	1.224e+00	6.71e-02	1.71e-03	7.32e-04	1.68e-02
0.05	0.0700	1.046e+00	5.60e-02	6.25e-04	3.09e-04	1.70e-02
0.05	0.0980	8.578e-01	4.41e-02	1.75e-04	8.75e-05	1.72e-02
0.05	0.1373	6.735e-01	3.33e-02	4.03e-05	1.88e-05	1.74e-02
0.05	0.1922	5.112e-01	2.45e-02	9.21e-06	4.51e-06	1.74e-02
0.05	0.2692	3.784e-01	1.78e-02	4.31e-06	3.02e-06	1.74e-02
0.05	0.3769	2.759e-01	1.28e-02	3.27e-06	2.50e-06	1.74e-02
0.05	0.5278	1.993e-01	9.18e-03	2.64e-06	1.80e-06	1.74e-02
0.05	0.7391	1.433e-01	6.57e-03	2.02e-06	1.32e-06	1.74e-02
0.05	1.0350	1.027e-01	4.69e-03	1.54e-06	8.82e-07	1.74e-02
0.05	1.4493	7.351e-02	3.35e-03	1.14e-06	6.06e-07	1.74e-02
0.05	2.0294	5.256e-02	2.40e-03	8.09e-07	4.26e-07	1.74e-02
0.05	2.8418	3.757e-02	1.71e-03	5.83e-07	2.94e-07	1.74e-02
0.05	3.9793	2.684e-02	1.22e-03	4.19e-07	2.08e-07	1.74e-02
0.05	5.5722	1.918e-02	8.73e-04	2.97e-07	1.49e-07	1.74e-02
0.05	7.8027	1.370e-02	6.23e-04	2.12e-07	1.07e-07	1.74e-02
0.05	10.9261	9.783e-03	4.45e-04	1.52e-07	7.64e-08	1.74e-02
0.05	15.2997	6.987e-03	3.18e-04	1.06e-07	5.53e-08	1.74e-02
0.05	21.4241	4.990e-03	2.27e-04	7.57e-08	3.98e-08	1.74e-02
0.05	30.0000	3.564e-03	1.62e-04	5.47e-08	2.78e-08	1.74e-02
0.10	0.0100	1.788e+00	0.00e+00	0.00e+00	0.00e+00	6.00e-02
0.10	0.0200	1.713e+00	1.58e-01	-1.07e+00	0.00e+00	6.06e-02
0.10	0.0300	1.627e+00	1.57e-01	8.71e-03	3.81e-03	6.10e-02
0.10	0.0500	1.486e+00	1.52e-01	7.42e-03	2.64e-03	6.15e-02
0.10	0.0700	1.352e+00	1.42e-01	5.82e-03	1.93e-03	6.18e-02
0.10	0.0980	1.191e+00	1.25e-01	2.89e-03	1.23e-03	6.21e-02
0.10	0.1373	1.011e+00	1.03e-01	9.80e-04	4.63e-04	6.24e-02
0.10	0.1922	8.249e-01	8.02e-02	2.60e-04	1.20e-04	6.26e-02
0.10	0.2692	6.452e-01	6.01e-02	5.98e-05	2.53e-05	6.27e-02
0.10	0.3769	4.888e-01	4.41e-02	1.35e-05	5.23e-06	6.26e-02
0.10	0.5278	3.618e-01	3.19e-02	8.03e-06	4.18e-06	6.26e-02
0.10	0.7391	2.640e-01	2.29e-02	6.65e-06	3.18e-06	6.25e-02
0.10	1.0350	1.909e-01	1.64e-02	5.04e-06	2.33e-06	6.24e-02

τ	γ	TV	V^{pub}	V^{priv}	V^{FRgap}	$\mathcal{D}$
0.10	1.4493	1.374e-01	1.18e-02	3.76e-06	1.60e-06	6.23e-02
0.10	2.0294	9.855e-02	8.40e-03	2.63e-06	1.13e-06	6.22e-02
0.10	2.8418	7.057e-02	6.00e-03	1.86e-06	7.81e-07	6.22e-02
0.10	3.9793	5.049e-02	4.28e-03	1.33e-06	5.40e-07	6.22e-02
0.10	5.5722	3.610e-02	3.06e-03	9.31e-07	3.89e-07	6.22e-02
0.10	7.8027	2.580e-02	2.18e-03	6.59e-07	2.78e-07	6.21e-02
0.10	10.9261	1.843e-02	1.56e-03	4.60e-07	2.03e-07	6.21e-02
0.10	15.2997	1.317e-02	1.11e-03	3.33e-07	1.41e-07	6.21e-02
0.10	21.4241	9.406e-03	7.96e-04	2.38e-07	1.00e-07	6.21e-02
0.10	30.0000	6.718e-03	5.68e-04	1.72e-07	7.01e-08	6.21e-02
0.20	0.0100	1.762e+00	0.00e+00	0.00e+00	0.00e+00	2.03e-01
0.20	0.0200	1.726e+00	0.00e+00	0.00e+00	0.00e+00	2.07e-01
0.20	0.0300	1.711e+00	-8.34e-01	0.00e+00	0.00e+00	2.10e-01
0.20	0.0500	1.627e+00	2.83e-01	1.48e-02	7.61e-03	2.13e-01
0.20	0.0700	1.510e+00	2.80e-01	1.51e-02	6.76e-03	2.15e-01
0.20	0.0980	1.431e+00	2.69e-01	1.25e-02	4.58e-03	2.15e-01
0.20	0.1373	1.284e+00	2.44e-01	8.61e-03	3.06e-03	2.14e-01
0.20	0.1922	1.122e+00	2.09e-01	3.74e-03	1.55e-03	2.12e-01
0.20	0.2692	9.400e-01	1.68e-01	1.12e-03	5.59e-04	2.10e-01
0.20	0.3769	7.575e-01	1.28e-01	2.92e-04	1.28e-04	2.08e-01
0.20	0.5278	5.876e-01	9.50e-02	6.68e-05	2.83e-05	2.06e-01
0.20	0.7391	4.431e-01	6.91e-02	1.82e-05	5.70e-06	2.04e-01
0.20	1.0350	3.276e-01	4.98e-02	1.20e-05	4.11e-06	2.03e-01
0.20	1.4493	2.391e-01	3.57e-02	9.23e-06	2.86e-06	2.02e-01
0.20	2.0294	1.731e-01	2.55e-02	6.14e-06	1.95e-06	2.01e-01
0.20	2.8418	1.247e-01	1.82e-02	3.88e-06	1.23e-06	2.01e-01
0.20	3.9793	8.952e-02	1.30e-02	2.72e-06	8.88e-07	2.00e-01
0.20	5.5722	6.416e-02	9.28e-03	1.89e-06	6.32e-07	2.00e-01
0.20	7.8027	4.593e-02	6.63e-03	1.32e-06	4.53e-07	2.00e-01
0.20	10.9261	3.286e-02	4.73e-03	9.70e-07	3.15e-07	2.00e-01
0.20	15.2997	2.349e-02	3.38e-03	7.16e-07	2.14e-07	2.00e-01
0.20	21.4241	1.679e-02	2.41e-03	5.04e-07	1.56e-07	2.00e-01
0.20	30.0000	1.200e-02	1.72e-03	3.69e-07	1.08e-07	1.99e-01
0.30	0.0500	1.656e+00	-7.63e-01	0.00e+00	0.00e+00	4.28e-01
0.30	0.0700	1.615e+00	3.87e-01	1.71e-02	8.15e-03	4.31e-01
0.30	0.0980	1.476e+00	3.79e-01	1.67e-02	8.32e-03	4.32e-01
0.30	0.1373	1.385e+00	3.57e-01	1.52e-02	5.63e-03	4.29e-01
0.30	0.1922	1.252e+00	3.18e-01	1.03e-02	3.39e-03	4.21e-01
0.30	0.2692	1.086e+00	2.68e-01	4.34e-03	1.81e-03	4.11e-01
0.30	0.3769	9.053e-01	2.12e-01	1.32e-03	6.27e-04	4.01e-01
0.30	0.5278	7.259e-01	1.61e-01	3.55e-04	1.62e-04	3.93e-01
0.30	0.7391	5.617e-01	1.18e-01	9.78e-05	4.39e-05	3.86e-01
0.30	1.0350	4.235e-01	8.52e-02	2.72e-05	6.99e-06	3.81e-01
0.30	1.4493	3.134e-01	6.11e-02	1.22e-05	3.03e-06	3.77e-01
0.30	2.0294	2.290e-01	4.37e-02	6.20e-06	1.58e-06	3.74e-01
0.30	2.8418	1.660e-01	3.12e-02	3.99e-06	1.02e-06	3.71e-01

τ	γ	TV	V^{pub}	V^{priv}	V^{FRgap}	$\mathcal{D}$
0.30	3.9793	1.197e-01	2.22e-02	2.67e-06	7.45e-07	3.70e-01
0.30	5.5722	8.608e-02	1.59e-02	1.86e-06	5.39e-07	3.68e-01
0.30	7.8027	6.175e-02	1.13e-02	1.39e-06	3.87e-07	3.68e-01
0.30	10.9261	4.423e-02	8.07e-03	1.02e-06	2.74e-07	3.67e-01
0.30	15.2997	3.165e-02	5.76e-03	7.50e-07	1.95e-07	3.67e-01
0.30	21.4241	2.264e-02	4.11e-03	5.36e-07	1.41e-07	3.66e-01
0.30	30.0000	1.618e-02	2.94e-03	3.81e-07	1.06e-07	3.66e-01
0.40	0.0100	1.733e+00	0.00e+00	0.00e+00	0.00e+00	6.81e-01
0.40	0.0200	1.730e+00	0.00e+00	0.00e+00	0.00e+00	6.86e-01
0.40	0.0300	1.706e+00	0.00e+00	0.00e+00	0.00e+00	6.89e-01
0.40	0.0500	1.645e+00	-7.14e-01	0.00e+00	0.00e+00	6.93e-01
0.40	0.0700	1.607e+00	-7.02e-01	0.00e+00	0.00e+00	6.94e-01
0.40	0.0980	1.530e+00	4.59e-01	2.40e-02	9.58e-03	6.93e-01
0.40	0.1373	1.412e+00	4.41e-01	2.15e-02	9.04e-03	6.85e-01
0.40	0.1922	1.318e+00	4.05e-01	1.60e-02	4.99e-03	6.71e-01
0.40	0.2692	1.165e+00	3.50e-01	8.52e-03	3.14e-03	6.49e-01
0.40	0.3769	9.926e-01	2.85e-01	3.14e-03	1.35e-03	6.26e-01
0.40	0.5278	8.147e-01	2.19e-01	9.28e-04	4.62e-04	6.05e-01
0.40	0.7391	6.428e-01	1.62e-01	2.81e-04	1.43e-04	5.88e-01
0.40	1.0350	4.920e-01	1.17e-01	9.67e-05	4.45e-05	5.75e-01
0.40	1.4493	3.684e-01	8.42e-02	3.05e-05	1.00e-05	5.65e-01
0.40	2.0294	2.715e-01	6.01e-02	8.94e-06	2.72e-06	5.57e-01
0.40	2.8418	1.980e-01	4.28e-02	3.95e-06	8.08e-07	5.52e-01
0.40	3.9793	1.434e-01	3.05e-02	1.94e-06	5.32e-07	5.48e-01
0.40	5.5722	1.034e-01	2.17e-02	1.45e-06	3.68e-07	5.45e-01
0.40	7.8027	7.433e-02	1.55e-02	1.08e-06	2.67e-07	5.43e-01
0.40	10.9261	5.332e-02	1.10e-02	8.10e-07	1.96e-07	5.41e-01
0.40	15.2997	3.820e-02	7.87e-03	5.81e-07	1.43e-07	5.40e-01
0.40	21.4241	2.734e-02	5.62e-03	4.16e-07	1.06e-07	5.39e-01
0.40	30.0000	1.955e-02	4.01e-03	2.94e-07	7.58e-08	5.39e-01
0.50	0.0500	1.628e+00	0.00e+00	0.00e+00	0.00e+00	9.93e-01
0.50	0.0700	1.604e+00	-6.60e-01	0.00e+00	0.00e+00	9.91e-01
0.50	0.0980	1.546e+00	5.35e-01	1.85e-02	-1.20e+00	9.85e-01
0.50	0.1373	1.437e+00	5.15e-01	1.96e-02	1.06e-02	9.72e-01
0.50	0.1922	1.342e+00	4.76e-01	1.93e-02	6.51e-03	9.48e-01
0.50	0.2692	1.205e+00	4.18e-01	1.26e-02	3.94e-03	9.13e-01
0.50	0.3769	1.051e+00	3.47e-01	5.11e-03	2.13e-03	8.72e-01
0.50	0.5278	8.756e-01	2.71e-01	1.77e-03	7.24e-04	8.34e-01
0.50	0.7391	7.004e-01	2.01e-01	6.20e-04	2.40e-04	8.02e-01
0.50	1.0350	5.426e-01	1.46e-01	2.17e-04	9.37e-05	7.77e-01
0.50	1.4493	4.101e-01	1.04e-01	7.36e-05	2.99e-05	7.58e-01
0.50	2.0294	3.045e-01	7.42e-02	2.63e-05	9.35e-06	7.43e-01
0.50	2.8418	2.232e-01	5.27e-02	6.21e-06	2.11e-06	7.33e-01
0.50	3.9793	1.623e-01	3.74e-02	2.20e-06	5.09e-07	7.25e-01
0.50	5.5722	1.173e-01	2.66e-02	1.29e-06	2.33e-07	7.20e-01
0.50	7.8027	8.451e-02	1.90e-02	7.15e-07	1.66e-07	7.15e-01

τ	γ	TV	V^{pub}	V^{priv}	V^{FRgap}	$\mathcal{D}$
0.50	10.9261	6.070e-02	1.35e-02	5.12e-07	1.22e-07	7.12e-01
0.50	15.2997	4.353e-02	9.62e-03	3.85e-07	9.22e-08	7.10e-01
0.50	21.4241	3.117e-02	6.86e-03	2.76e-07	6.51e-08	7.09e-01
0.50	30.0000	2.231e-02	4.89e-03	1.65e-07	6.18e-08	7.08e-01
0.60	0.0100	1.699e+00	0.00e+00	0.00e+00	0.00e+00	1.33e+00
0.60	0.0200	1.734e+00	0.00e+00	0.00e+00	0.00e+00	1.33e+00
0.60	0.0300	1.687e+00	0.00e+00	0.00e+00	0.00e+00	1.32e+00
0.60	0.0500	1.617e+00	0.00e+00	0.00e+00	0.00e+00	1.32e+00
0.60	0.0700	1.585e+00	-6.24e-01	0.00e+00	0.00e+00	1.31e+00
0.60	0.0980	1.552e+00	-6.10e-01	0.00e+00	0.00e+00	1.30e+00
0.60	0.1373	1.474e+00	5.77e-01	1.82e-02	1.05e-02	1.28e+00
0.60	0.1922	1.380e+00	5.36e-01	2.15e-02	7.23e-03	1.25e+00
0.60	0.2692	1.242e+00	4.76e-01	1.55e-02	4.73e-03	1.19e+00
0.60	0.3769	1.092e+00	4.02e-01	6.61e-03	2.39e-03	1.13e+00
0.60	0.5278	9.181e-01	3.16e-01	2.50e-03	1.11e-03	1.07e+00
0.60	0.7391	7.425e-01	2.36e-01	9.95e-04	4.17e-04	1.02e+00
0.60	1.0350	5.806e-01	1.71e-01	3.78e-04	1.46e-04	9.82e-01
0.60	1.4493	4.422e-01	1.22e-01	1.44e-04	5.77e-05	9.50e-01
0.60	2.0294	3.303e-01	8.61e-02	4.77e-05	1.87e-05	9.27e-01
0.60	2.8418	2.433e-01	6.10e-02	1.51e-05	4.83e-06	9.10e-01
0.60	3.9793	1.775e-01	4.32e-02	3.70e-06	1.04e-06	8.97e-01
0.60	5.5722	1.286e-01	3.06e-02	8.49e-07	2.34e-07	8.88e-01
0.60	7.8027	9.281e-02	2.18e-02	5.95e-07	1.06e-07	8.81e-01
0.60	10.9261	6.675e-02	1.55e-02	3.12e-07	7.20e-08	8.76e-01
0.60	15.2997	4.791e-02	1.10e-02	2.30e-07	4.93e-08	8.73e-01
0.60	21.4241	3.433e-02	7.86e-03	9.45e-08	5.49e-08	8.70e-01
0.60	30.0000	2.458e-02	5.60e-03	7.90e-08	4.58e-08	8.69e-01
0.80	0.0100	1.687e+00	0.00e+00	0.00e+00	0.00e+00	2.08e+00
0.80	0.0200	1.665e+00	0.00e+00	0.00e+00	0.00e+00	2.07e+00
0.80	0.0300	1.630e+00	0.00e+00	0.00e+00	0.00e+00	2.06e+00
0.80	0.0500	1.582e+00	0.00e+00	0.00e+00	0.00e+00	2.05e+00
0.80	0.0700	1.567e+00	0.00e+00	0.00e+00	0.00e+00	2.03e+00
0.80	0.0980	1.535e+00	-5.46e-01	0.00e+00	0.00e+00	2.00e+00
0.80	0.1373	1.464e+00	-5.39e-01	0.00e+00	0.00e+00	1.96e+00
0.80	0.1922	1.389e+00	6.37e-01	2.06e-02	7.91e-03	1.90e+00
0.80	0.2692	1.268e+00	5.76e-01	1.69e-02	5.54e-03	1.81e+00
0.80	0.3769	1.129e+00	4.93e-01	9.61e-03	3.63e-03	1.69e+00
0.80	0.5278	9.722e-01	3.95e-01	3.93e-03	1.47e-03	1.58e+00
0.80	0.7391	7.974e-01	2.96e-01	1.66e-03	6.52e-04	1.48e+00
0.80	1.0350	6.309e-01	2.12e-01	6.75e-04	2.48e-04	1.39e+00
0.80	1.4493	4.856e-01	1.50e-01	3.09e-04	1.12e-04	1.33e+00
0.80	2.0294	3.659e-01	1.05e-01	1.13e-04	3.91e-05	1.28e+00
0.80	2.8418	2.715e-01	7.34e-02	3.92e-05	1.21e-05	1.24e+00
0.80	3.9793	1.992e-01	5.16e-02	9.07e-06	2.96e-06	1.22e+00
0.80	5.5722	1.449e-01	3.64e-02	2.02e-06	5.36e-07	1.20e+00
0.80	7.8027	1.049e-01	2.57e-02	7.05e-07	1.16e-07	1.19e+00

τ	γ	TV	V^{pub}	V^{priv}	V^{FRgap}	$\mathcal{D}$
0.80	10.9261	7.560e-02	1.82e-02	1.64e-07	3.76e-08	1.18e+00
0.80	15.2997	5.435e-02	1.30e-02	1.08e-07	3.83e-08	1.17e+00
0.80	21.4241	3.899e-02	9.21e-03	1.47e-07	3.37e-08	1.16e+00
0.80	30.0000	2.794e-02	6.56e-03	9.58e-08	3.39e-08	1.16e+00
1.00	0.0100	1.580e+00	0.00e+00	0.00e+00	0.00e+00	2.95e+00
1.00	0.0200	1.648e+00	0.00e+00	0.00e+00	0.00e+00	2.94e+00
1.00	0.0300	1.633e+00	0.00e+00	0.00e+00	0.00e+00	2.91e+00
1.00	0.0500	1.593e+00	0.00e+00	0.00e+00	0.00e+00	2.87e+00
1.00	0.0700	1.574e+00	0.00e+00	0.00e+00	0.00e+00	2.84e+00
1.00	0.0980	1.538e+00	0.00e+00	0.00e+00	0.00e+00	2.80e+00
1.00	0.1373	1.473e+00	-4.83e-01	0.00e+00	0.00e+00	2.73e+00
1.00	0.1922	1.420e+00	7.24e-01	1.10e-02	-1.23e+00	2.63e+00
1.00	0.2692	1.308e+00	6.62e-01	1.24e-02	5.17e-03	2.49e+00
1.00	0.3769	1.163e+00	5.72e-01	1.12e-02	3.03e-03	2.31e+00
1.00	0.5278	9.999e-01	4.64e-01	4.83e-03	1.17e-03	2.11e+00
1.00	0.7391	8.246e-01	3.47e-01	2.34e-03	6.78e-04	1.94e+00
1.00	1.0350	6.555e-01	2.46e-01	1.06e-03	4.07e-04	1.79e+00
1.00	1.4493	5.069e-01	1.70e-01	5.11e-04	1.54e-04	1.69e+00
1.00	2.0294	3.837e-01	1.17e-01	2.17e-04	7.47e-05	1.61e+00
1.00	2.8418	2.859e-01	8.10e-02	6.75e-05	2.22e-05	1.55e+00
1.00	3.9793	2.105e-01	5.63e-02	2.10e-05	5.94e-06	1.51e+00
1.00	5.5722	1.537e-01	3.94e-02	4.44e-06	1.72e-06	1.48e+00
1.00	7.8027	1.115e-01	2.77e-02	1.09e-06	6.84e-07	1.46e+00
1.00	10.9261	8.049e-02	1.95e-02	5.96e-07	2.20e-07	1.44e+00
1.00	15.2997	5.794e-02	1.38e-02	2.52e-07	8.59e-08	1.43e+00
1.00	21.4241	4.161e-02	9.82e-03	2.66e-07	4.80e-08	1.43e+00
1.00	30.0000	2.984e-02	6.98e-03	2.38e-07	2.80e-08	1.42e+00
1.20	0.0100	1.620e+00	0.00e+00	0.00e+00	0.00e+00	3.89e+00
1.20	0.0200	1.644e+00	0.00e+00	0.00e+00	0.00e+00	3.86e+00
1.20	0.0300	1.655e+00	0.00e+00	0.00e+00	0.00e+00	3.84e+00
1.20	0.0500	1.588e+00	0.00e+00	0.00e+00	0.00e+00	3.79e+00
1.20	0.0700	1.558e+00	0.00e+00	0.00e+00	0.00e+00	3.75e+00
1.20	0.0980	1.519e+00	0.00e+00	0.00e+00	0.00e+00	3.68e+00
1.20	0.1373	1.463e+00	-4.35e-01	0.00e+00	0.00e+00	3.59e+00
1.20	0.1922	1.415e+00	-4.39e-01	0.00e+00	0.00e+00	3.44e+00
1.20	0.2692	1.296e+00	7.27e-01	1.31e-02	5.31e-03	3.23e+00
1.20	0.3769	1.155e+00	6.42e-01	8.04e-03	3.64e-03	2.96e+00
1.20	0.5278	1.006e+00	5.23e-01	5.98e-03	2.27e-03	2.67e+00
1.20	0.7391	8.288e-01	3.92e-01	2.63e-03	7.84e-04	2.40e+00
1.20	1.0350	6.566e-01	2.73e-01	1.37e-03	4.23e-04	2.18e+00
1.20	1.4493	5.063e-01	1.85e-01	6.77e-04	2.09e-04	2.02e+00
1.20	2.0294	3.833e-01	1.24e-01	3.22e-04	1.03e-04	1.91e+00
1.20	2.8418	2.861e-01	8.45e-02	1.32e-04	3.69e-05	1.84e+00
1.20	3.9793	2.112e-01	5.80e-02	4.05e-05	9.47e-06	1.78e+00
1.20	5.5722	1.546e-01	4.02e-02	1.08e-05	6.12e-06	1.75e+00
1.20	7.8027	1.124e-01	2.81e-02	5.18e-06	2.55e-06	1.72e+00

τ	γ	TV	V^{pub}	V^{priv}	V^{FRgap}	$\mathcal{D}$
1.20	10.9261	8.134e-02	1.97e-02	2.71e-06	8.33e-07	1.71e+00
1.20	15.2997	5.865e-02	1.39e-02	1.87e-06	3.23e-07	1.70e+00
1.20	21.4241	4.217e-02	9.84e-03	1.56e-06	1.93e-07	1.69e+00
1.20	30.0000	3.027e-02	6.99e-03	1.44e-06	2.87e-07	1.68e+00
1.50	0.0100	1.483e+00	0.00e+00	0.00e+00	0.00e+00	5.46e+00
1.50	0.0200	1.678e+00	0.00e+00	0.00e+00	0.00e+00	5.42e+00
1.50	0.0300	1.647e+00	0.00e+00	0.00e+00	0.00e+00	5.38e+00
1.50	0.0500	1.562e+00	0.00e+00	0.00e+00	0.00e+00	5.32e+00
1.50	0.0700	1.535e+00	0.00e+00	0.00e+00	0.00e+00	5.27e+00
1.50	0.0980	1.481e+00	0.00e+00	0.00e+00	0.00e+00	5.19e+00
1.50	0.1373	1.482e+00	0.00e+00	0.00e+00	0.00e+00	5.05e+00
1.50	0.1922	1.416e+00	-3.73e-01	0.00e+00	0.00e+00	4.82e+00
1.50	0.2692	1.279e+00	8.13e-01	-1.20e+00	0.00e+00	4.48e+00
1.50	0.3769	1.161e+00	7.22e-01	1.11e-02	3.31e-03	4.01e+00
1.50	0.5278	9.832e-01	6.03e-01	5.63e-03	1.93e-03	3.50e+00
1.50	0.7391	7.842e-01	4.48e-01	4.00e-03	1.29e-03	3.06e+00
1.50	1.0350	6.151e-01	3.03e-01	2.12e-03	8.38e-04	2.76e+00
1.50	1.4493	4.844e-01	1.98e-01	1.15e-03	3.51e-04	2.57e+00
1.50	2.0294	3.804e-01	1.29e-01	5.75e-04	1.85e-04	2.46e+00
1.50	2.8418	2.939e-01	8.57e-02	1.18e-04	3.15e-05	2.39e+00
1.50	3.9793	2.224e-01	5.77e-02	1.29e-04	3.93e-05	2.34e+00
1.50	5.5722	1.672e-01	3.96e-02	2.55e-05	1.38e-05	2.31e+00
1.50	7.8027	1.243e-01	2.74e-02	1.57e-05	5.18e-06	2.30e+00
2.00	0.0100	1.492e+00	0.00e+00	0.00e+00	0.00e+00	8.55e+00
2.00	0.0200	1.472e+00	0.00e+00	0.00e+00	0.00e+00	8.50e+00
2.00	0.0300	1.393e+00	0.00e+00	0.00e+00	0.00e+00	8.46e+00
2.00	0.0500	1.424e+00	0.00e+00	0.00e+00	0.00e+00	8.38e+00
2.00	0.0700	1.498e+00	0.00e+00	0.00e+00	0.00e+00	8.29e+00
2.00	0.0980	1.475e+00	0.00e+00	0.00e+00	0.00e+00	8.15e+00
2.00	0.1373	1.434e+00	0.00e+00	0.00e+00	0.00e+00	7.90e+00
2.00	0.1922	1.358e+00	0.00e+00	0.00e+00	0.00e+00	7.47e+00
2.00	0.2692	1.256e+00	-2.83e-01	0.00e+00	0.00e+00	6.72e+00
2.00	0.3769	1.078e+00	8.18e-01	9.74e-03	-4.23e-01	5.66e+00
2.00	0.5278	9.151e-01	6.96e-01	1.29e-02	5.99e-03	4.97e+00
2.00	0.7391	7.683e-01	5.23e-01	7.04e-03	3.18e-03	4.49e+00
2.00	1.0350	6.530e-01	3.42e-01	3.85e-03	1.80e-03	4.20e+00
2.00	1.4493	5.524e-01	2.13e-01	1.33e-03	6.84e-04	4.02e+00
2.00	2.0294	4.826e-01	1.34e-01	9.46e-04	3.01e-04	4.00e+00